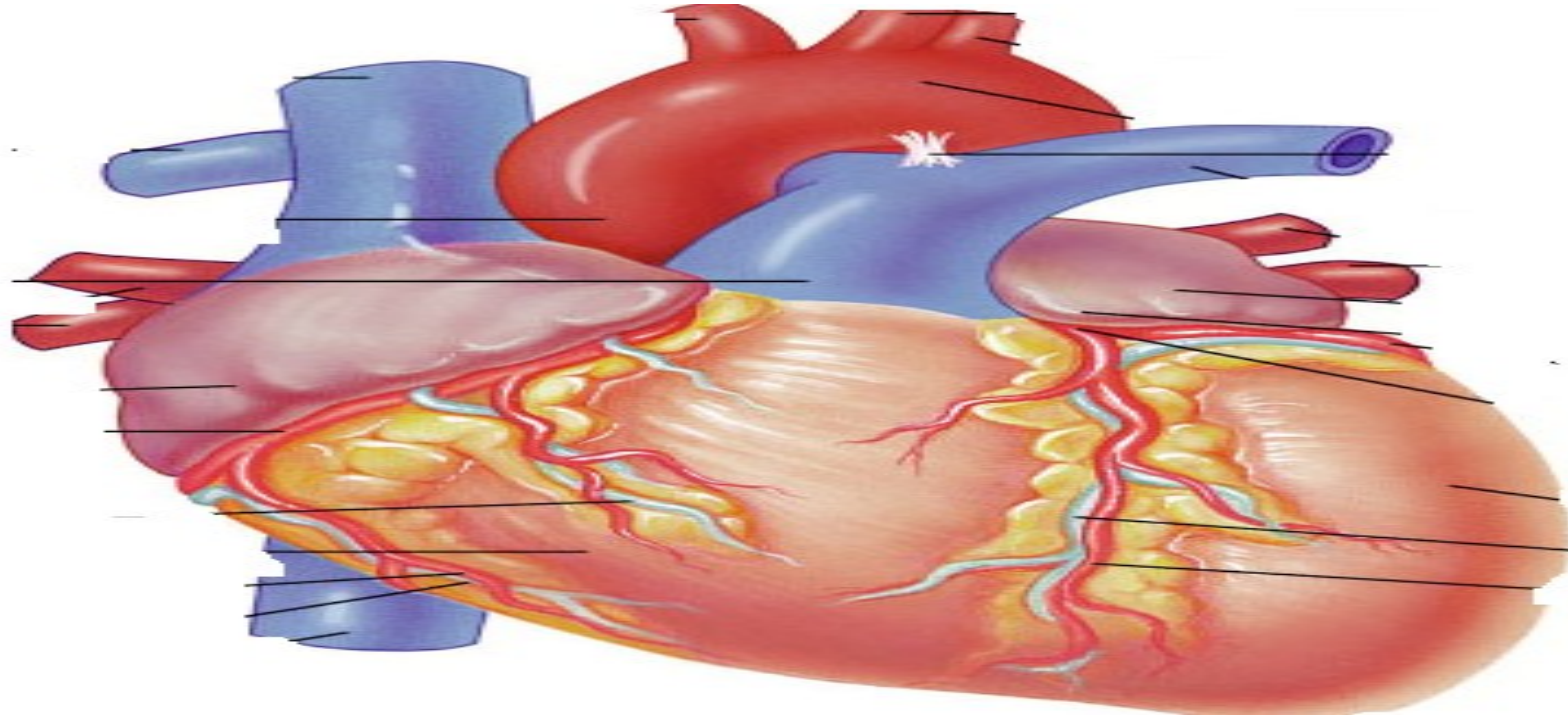


PHYSIOLOGY OF THE CARDIOVASCULAR SYSTEM



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Learning Objectives

- ④ At the end of teaching/learning activities, students are expected to:
 - ▶ Identify functional structure of the cardiovascular system
 - ▶ Characterize histology of the heart muscle cells
 - ▶ Describe pumping function of the heart chambers
 - ▶ Explain cardiac output and its regulation
 - ▶ Explain blood pressure and its regulation
 - ▶ Describe capillary fluid exchange

Historical Perspective

- The vital nature of the CVS was apparent to the ancients, but its actual function was a mystery.
- **Galen** (130–201 A.D.) a Greek physician and writer(Father of Experimental Physiology) made the first attempt to explain the function of the heart and the blood.
- **Key Aspects of Galen's "Ebb and Flow" Theory**
 - **Creation in the Liver:** Galen believed the liver was the center of the venous system and continuously produced new blood from food.
 - **Centrifugal Movement:** Blood was thought to flow outward from the liver and heart to the rest of the body, where it was completely consumed by the tissues as nourishment.
 - **Bidirectional Flow:** Instead of circulating in a closed loop, blood was believed to move back and forth—or ebb and flow—within the vessels, similar to the tides.
 - **Two Separate Systems:** Galen argued for two separate, open-ended systems rather than a single unified circulation:
 - **Venous System:** Carried nutrient-rich blood from the liver to the tissues.
 - **Arterial System:** Carried "vital spirits" (pneuma) and blood, mixed with air from the lungs, to the rest of the body.
- This erroneous concept remained a cornerstone of medical thought for the next 14 centuries.

Historical Perspective...cont'd

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- This erroneous concept remained a cornerstone of medical understanding for over 1,300 years until it was disproven by **William Harvey in 1628**
- Physiology started as an experimental science in 1628 **when William Harvey**, an English physician and anatomist, demonstrated that:
 - the CVS forms a circle in which blood was pumped out of the heart (the central pump) continuously through one set of vessels and returned to the heart through another set, i.e. blood flows in a completely closed circulation.
- This gave rise to a modern era of understanding and investigation of cardiovascular physiology.

General Functions of the CVS

Transportation of Materials: The blood serves as the transport medium, carrying:

- **Oxygen** from the lungs to cells.
- **Nutrients** (e.g., glucose, amino acids) from the digestive tract to cells.
- **Waste Products** (e.g., carbon dioxide, urea) to the kidneys and lungs for excretion.
- **Hormones** from endocrine glands to target organs.

Protection against Disease and Blood Loss:

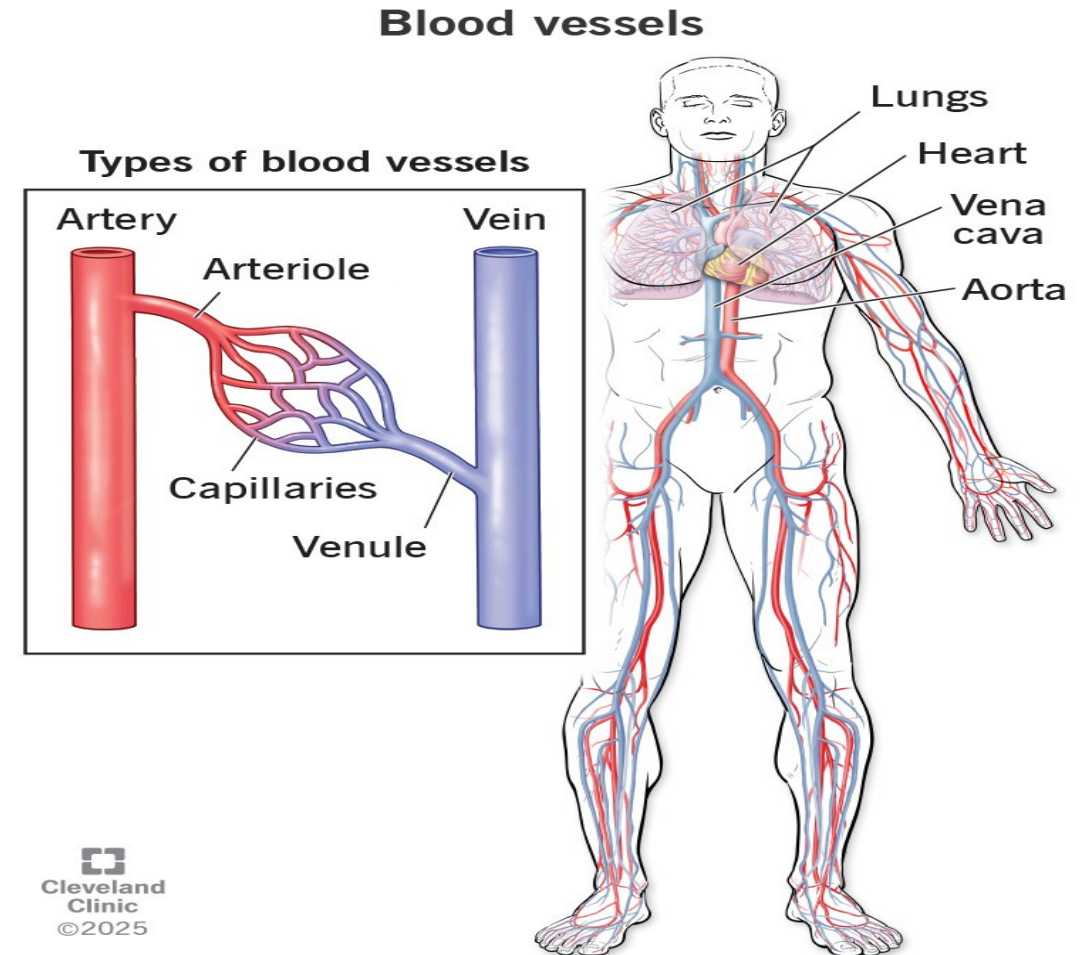
- **Immune response:** White blood cells travel through the blood to fight infection.
- **Clotting:** Platelets and clotting factors in the blood prevent excessive bleeding at injury sites.

Regulation of Body Environment (Homeostasis):

- **Temperature Regulation:** The system redistributes heat by adjusting blood flow to the skin, using vasodilation (widening) to lose heat and vasoconstriction (narrowing) to retain heat.
- **pH Stability:** Blood carries buffers that maintain a stable pH for body cells.
- **Fluid Balance:** Proper blood circulation helps maintain the necessary balance of fluids between blood and tissue spaces

Components of the CVS

- **The Heart:** central pump
- **Systemic arteries:** designed to carry blood under high pressure out to the tissue beds
- **Arterioles & pre capillary sphincters:**
 - act as control valves to regulate local flow
- **Capillaries:** one cell layer thick
 - exchange between tissue (cells) & blood
- **Venules:**
 - collect blood from capillaries
- **Systemic veins:**
 - return blood to heart/dynamic storage



Types of Circulation

- Blood Circulation is an evolutionary consequence of body size.
- Isolated small cells and small organisms with no circulatory system meet metabolic needs by simple processes of diffusion and convection of solutes.
- In lower animals and vertebrates the circulation may be:
 - ❑ Open or closed circulation
 - ❑ Single or double circulation

Open Circulatory System

Mechanism: A heart pumps hemolymph through short vessels into open spaces called sinuses (hemocoel). The fluid directly bathes tissues, allowing oxygen and nutrient exchange.

Examples: Insects (cockroaches, grasshoppers), mollusks (snails, clams), and arthropods (lobsters, crabs).

Closed Circulatory System

Mechanism: Blood is pumped by the heart, flowing exclusively through a continuous network of vessels (arteries, veins, and capillaries) to reach tissues.

Examples: Humans, fish, mammals, reptiles, birds, and annelids (earthworms).

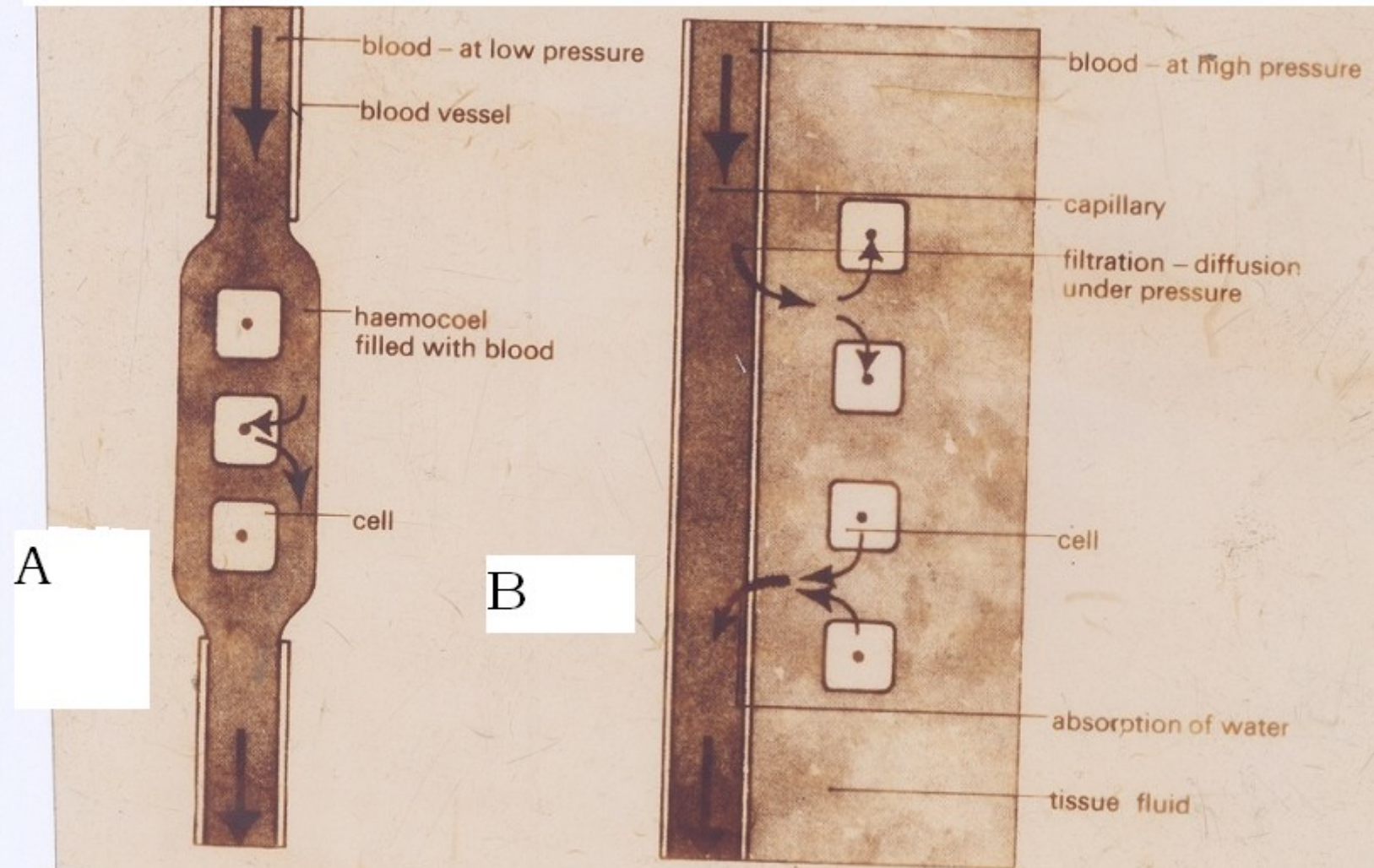


Fig..Open (A) and closed(B) circulations

Comparison of Circulatory Circuits

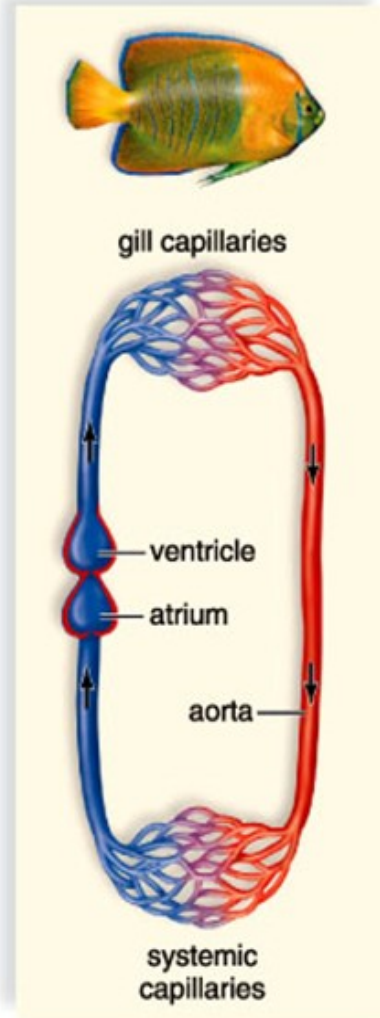
Single Circulation

In single circulation, blood passes through the heart only once during a complete circuit of the body. It is characterized by a low-pressure system commonly found in fish.

Heart Structure: Usually a two-chambered heart (one atrium and one ventricle)

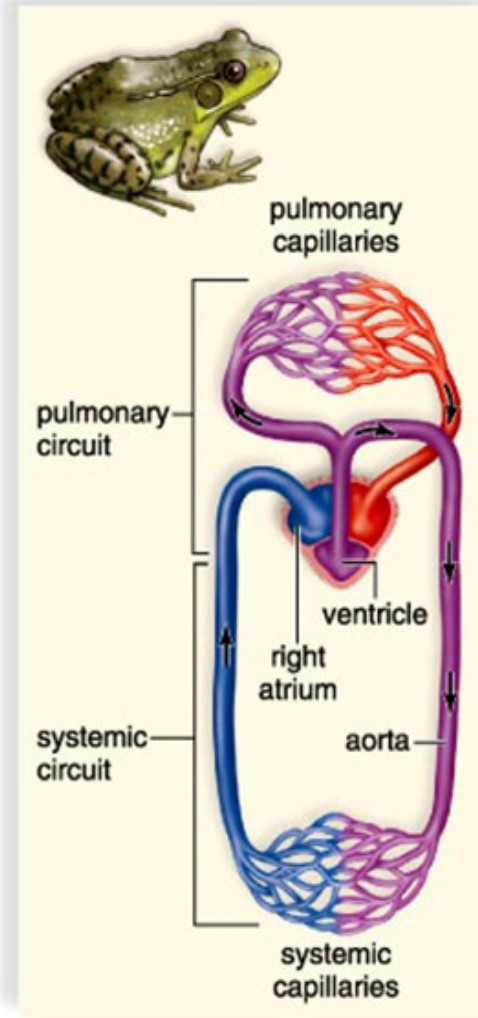
Characteristics: Blood is oxygenated in the gills and then travels directly to body tissues without returning to the heart, leading to lower pressure and efficiency

single

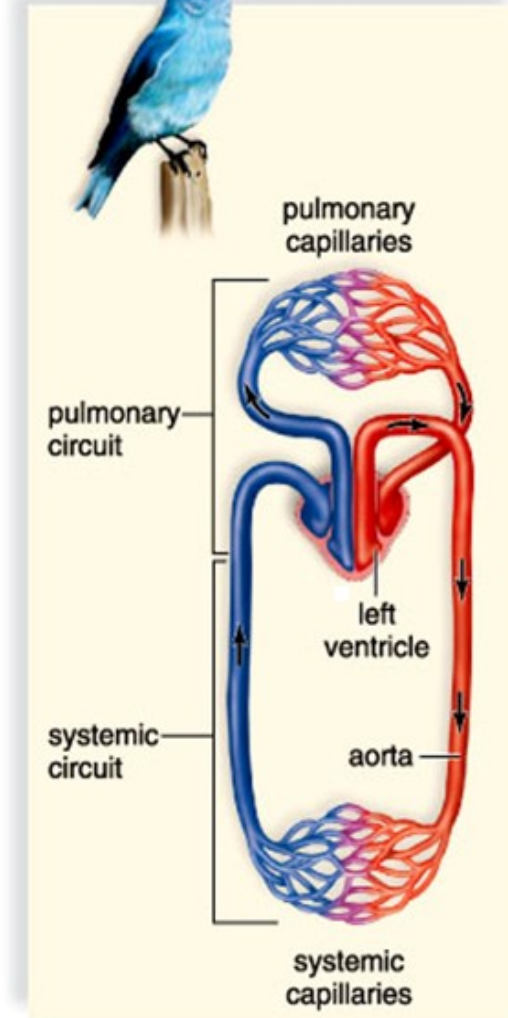


a. Fishes

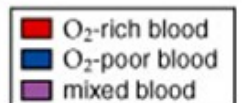
Double



b. Amphibians and most reptiles



c. Some reptiles; birds and mammals



Division of the Circulation

The human circulatory system operates through three primary circuits: pulmonary.

1. Systemic circulation (heart-body-heart)
2. Pulmonary circulation (heart-lungs-heart)
3. Coronary (heart muscle blood supply)

- **Systemic circulation:**

Starts in the LV → Aorta → Systemic arteries → Systemic capillaries → Veins → SVC & IVC → ends in RA

- **Pulmonary circulation:**

Starts in the RV → Pulmonary trunk → Pulmonary arteries → Pulmonary capillaries → Pulmonary veins → ends in the LA.

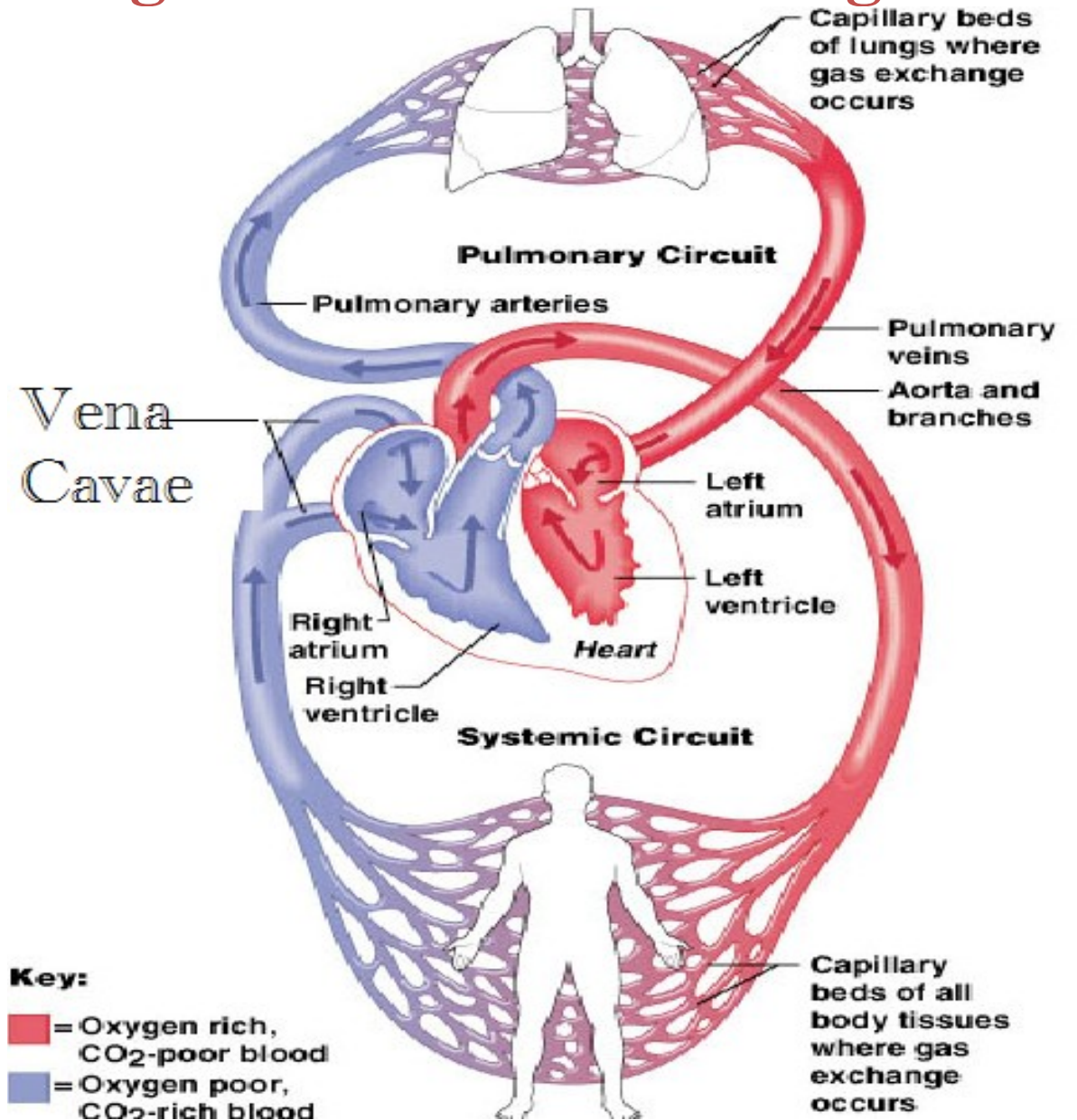
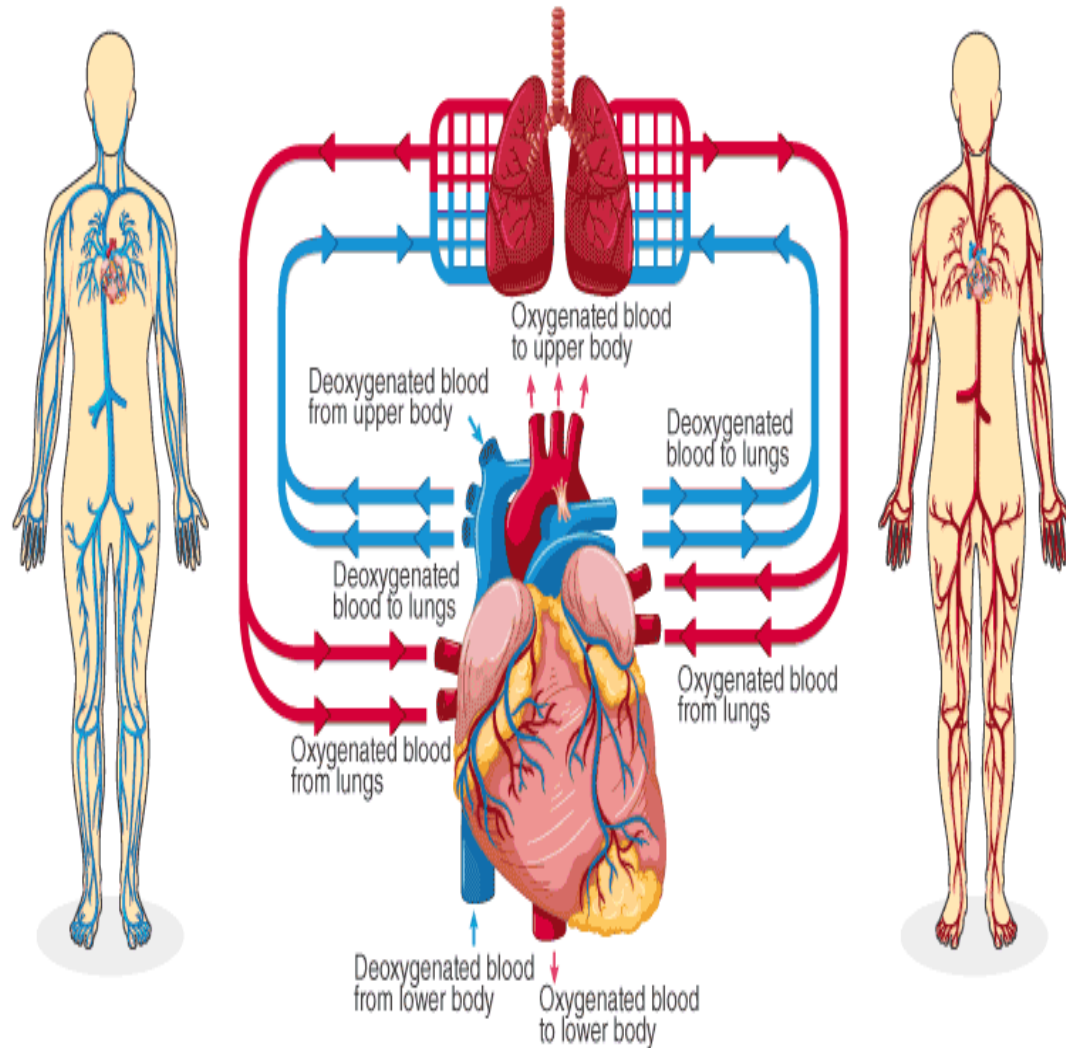
LA=left atrium

LV=left ventricle, SVC=superior vena cava

IVC=inferior vena cava

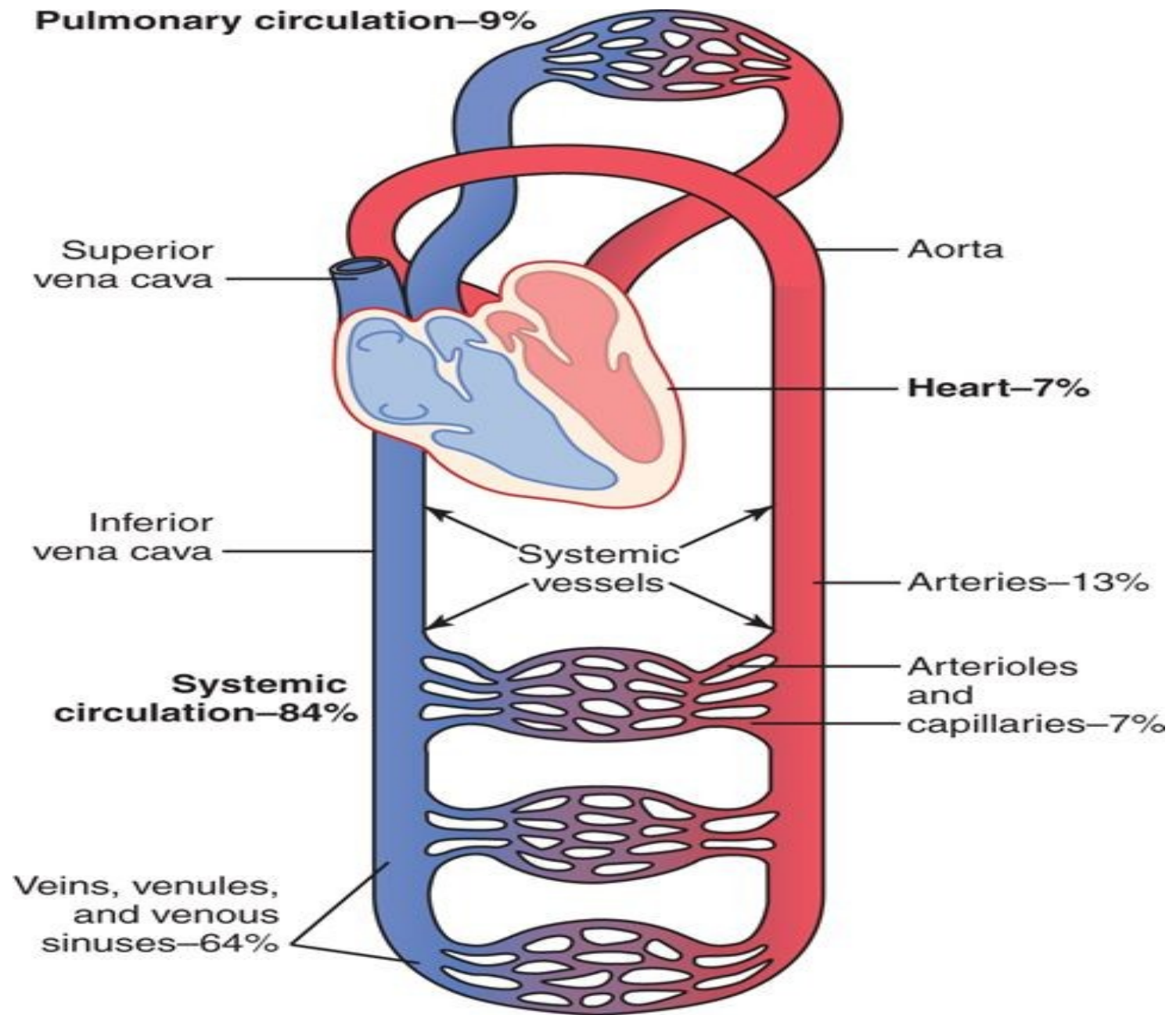
Pathway of Blood Through the Heart and Lungs

HUMAN CIRCULATORY SYSTEM TRANSPORTATION



Distribution of Blood

1. Veins, venules, venous sinuses=64%
2. Pulmonary Circulation=9%
3. Heart=7%
4. Capillaries=5%
5. Arteries=13-15%



The Heart- Key Facts

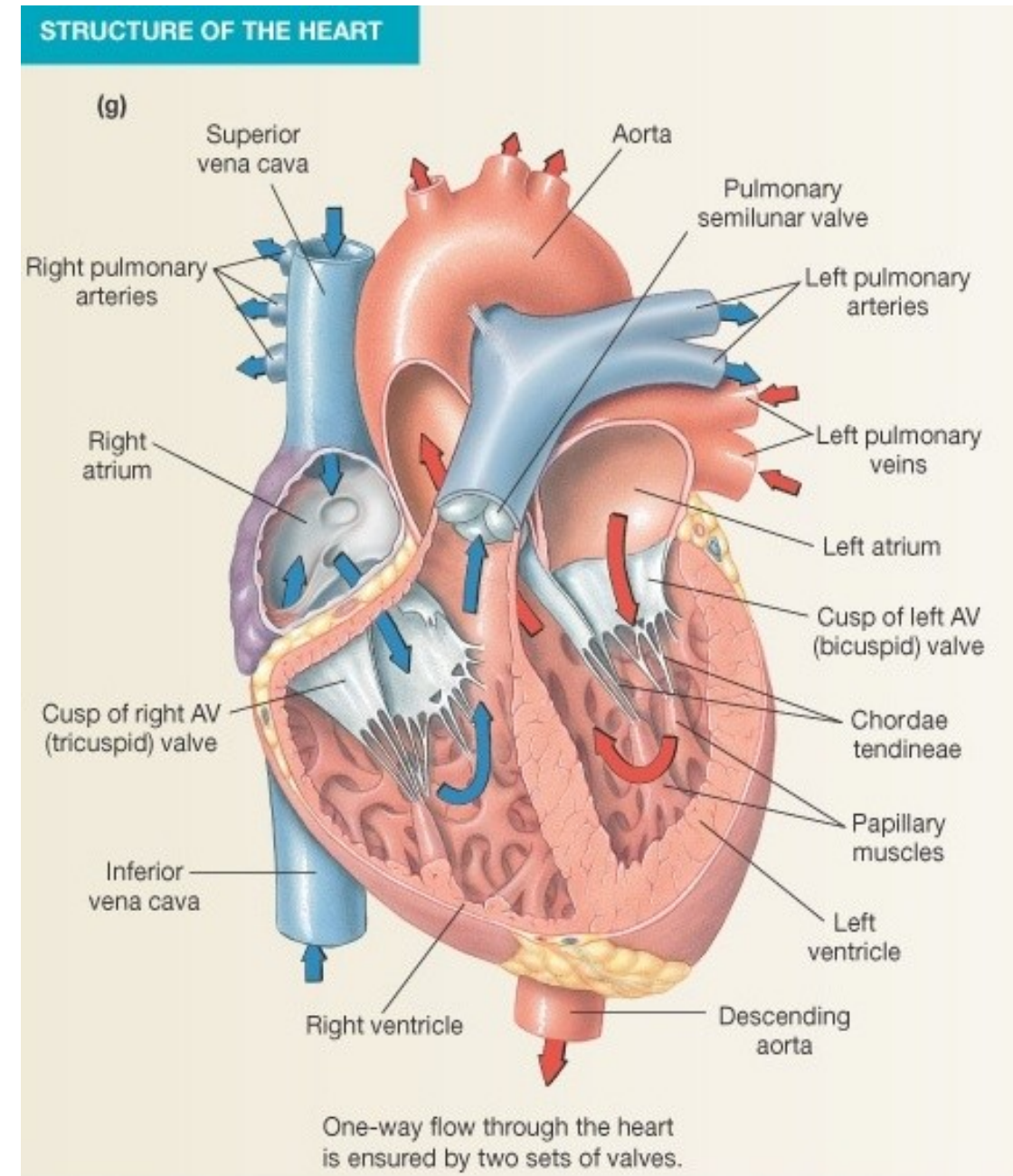
- The human heart begins to beat during intrauterine life around **3 weeks (approximately day 22-23) after fertilization.**
- It is the first functional organ to form, necessary for delivering oxygen and nutrients as the embryo grows

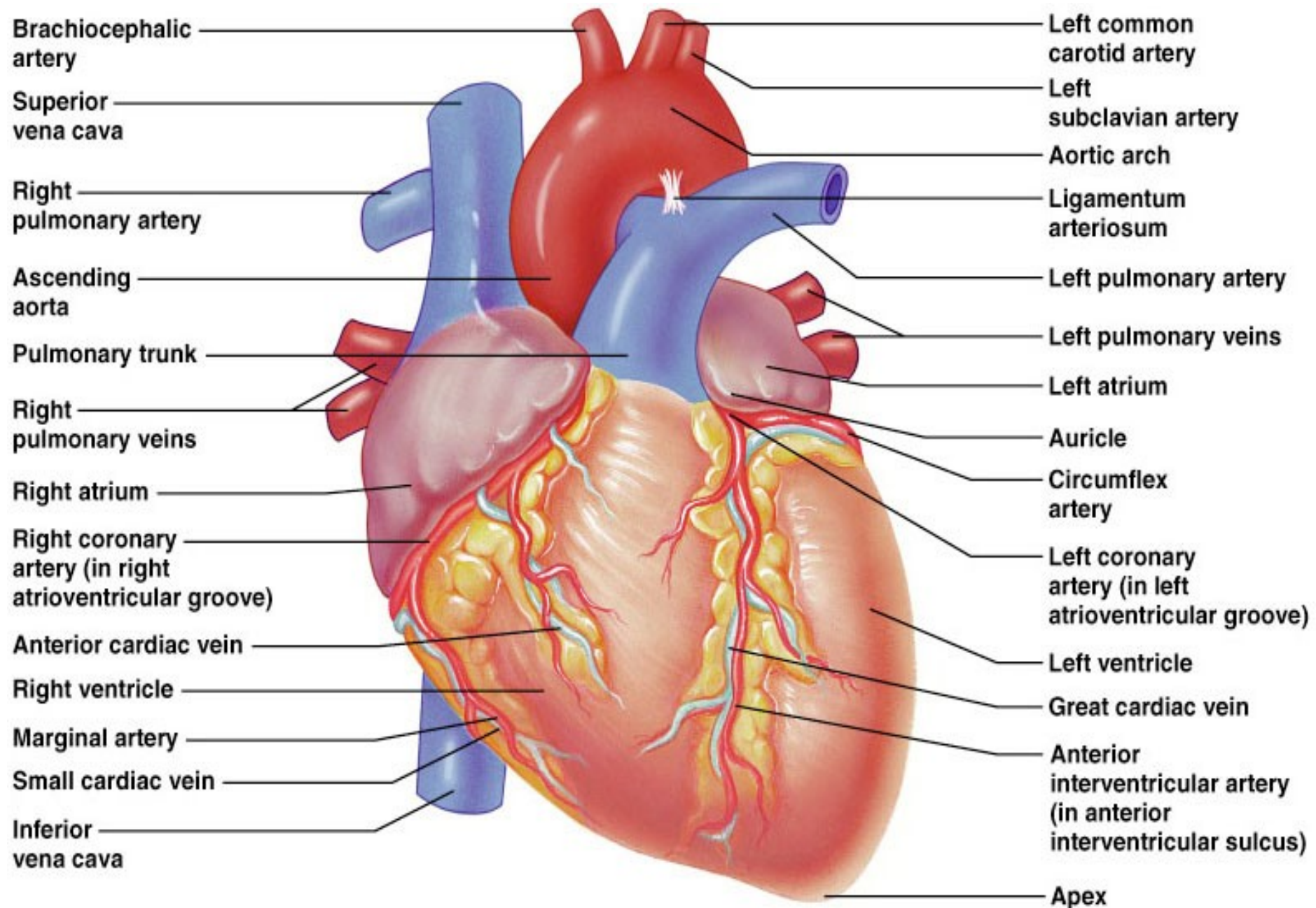
Vital Statistics

- **Constant Activity:** It never stops working, beating roughly 115,000 times per day
- **Blood Volume:** The heart pumps about 2,000 gallons (47 Barrel or 7,570 litter) of blood daily, traveling 12,000 miles (19,312 Km).
- **Lifespan Beats:** Over a long lifetime, the heart may beat more than 3.5 billion times.
- **Blood Vessel Network:** The body contains roughly 60,000 miles (96, 560 Km) of blood vessels.
- **Resting Rate:** A healthy adult heart typically beats between 60 and 100 times per minute
- **Independent Electrical System:** The heart has its own electrical supply, meaning it can continue to beat even when separated from the body, provided it has an adequate oxygen supply
- The average heartbeat of a woman **is about 8 beats a minute faster** than a man's heartbeat.
- Other than **the cornea**, every cell in the human body (75 trillion cells) gets blood from the heart

The Heart

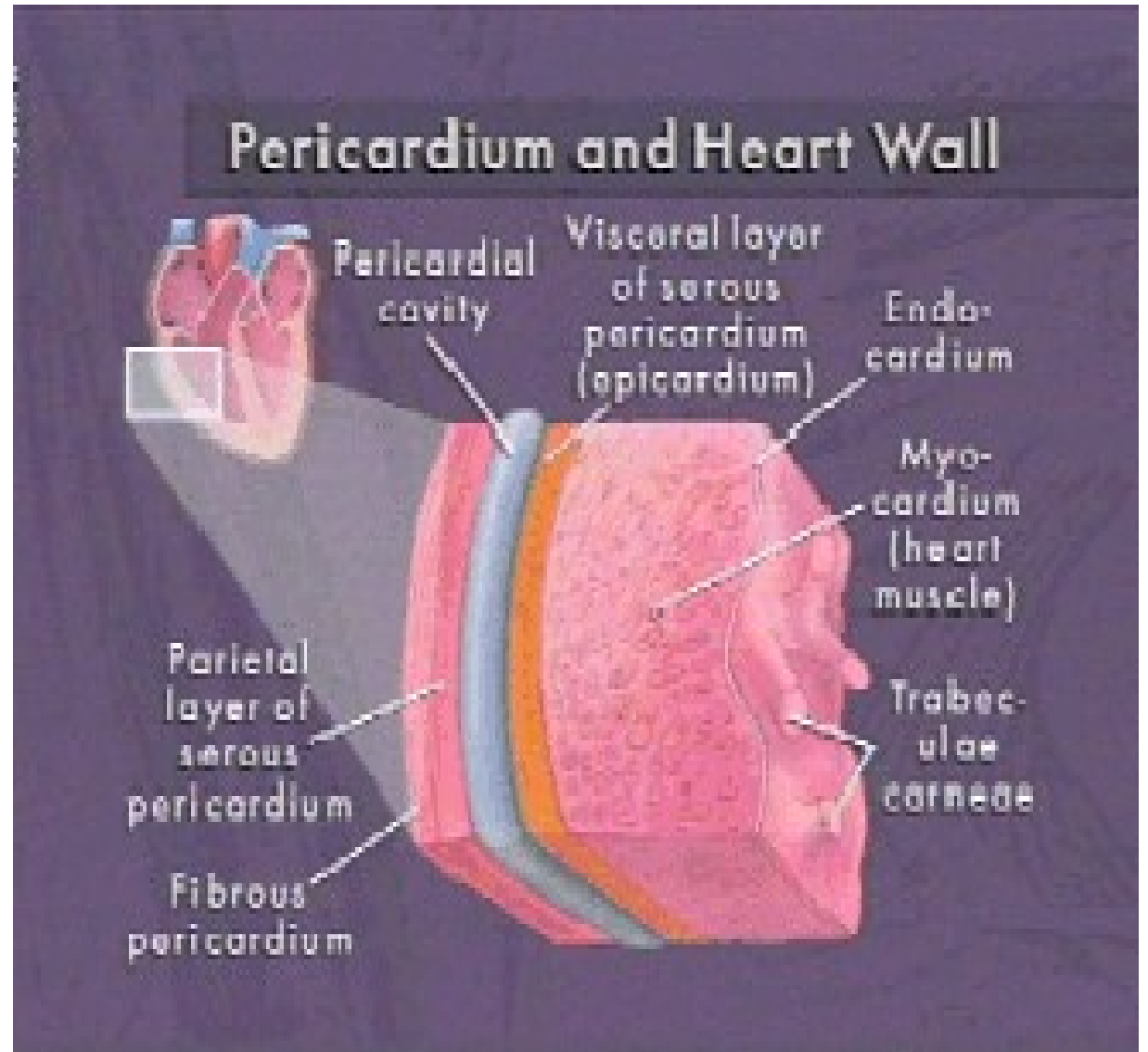
- Heart is the hollow, muscular organ that plays a central pumping role
- The heart is composed of three major types of cardiac muscle:
 - *Atrial muscle, Ventricular muscle, and*
 - Specialized *excitatory and conductive muscle* fibers
- Vertically divided into left & right sides by a structure called **septum**
- Composed of 4-chambers: 2 atria and 2 ventricles
- Has 4 valves:
 - 2 AV valves (TCV &BCV) and
 - 2 SLV (PV &AV)
- Size: Approximately equivalent to clenched fist
- Weight: 280 to 320 grams in average adults
- Located in the mediastinum





Histology of the heart

- Surfaces/Layers: The heart is composed of three layers:
 1. **Endocardium**: - innermost layer; epithelial tissue that lines the entire circulatory system
 - 2 - **Myocardium** - thickest layer; consists of cardiac muscle
 - 3 - **Epicardium** - thin, external membrane around the heart



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The Pericardium

- Is a fibrous closed sac investing the entire heart and cardiac portion of great vessels.
 - contains a **small amount of fluid** for lubrication - facilitating the continuous movement of the enclosed heart.
 - Contains:
 - a. **Visceral layer (epicardium)** immediately lining the outer surface of the heart and
 - b. **Parietal layer** forming protective outer lining of the pericardial sac.
- ☐ The sac contains a small amount of **fluid** for lubrication - facilitating the continuous movement of the enclosed heart.

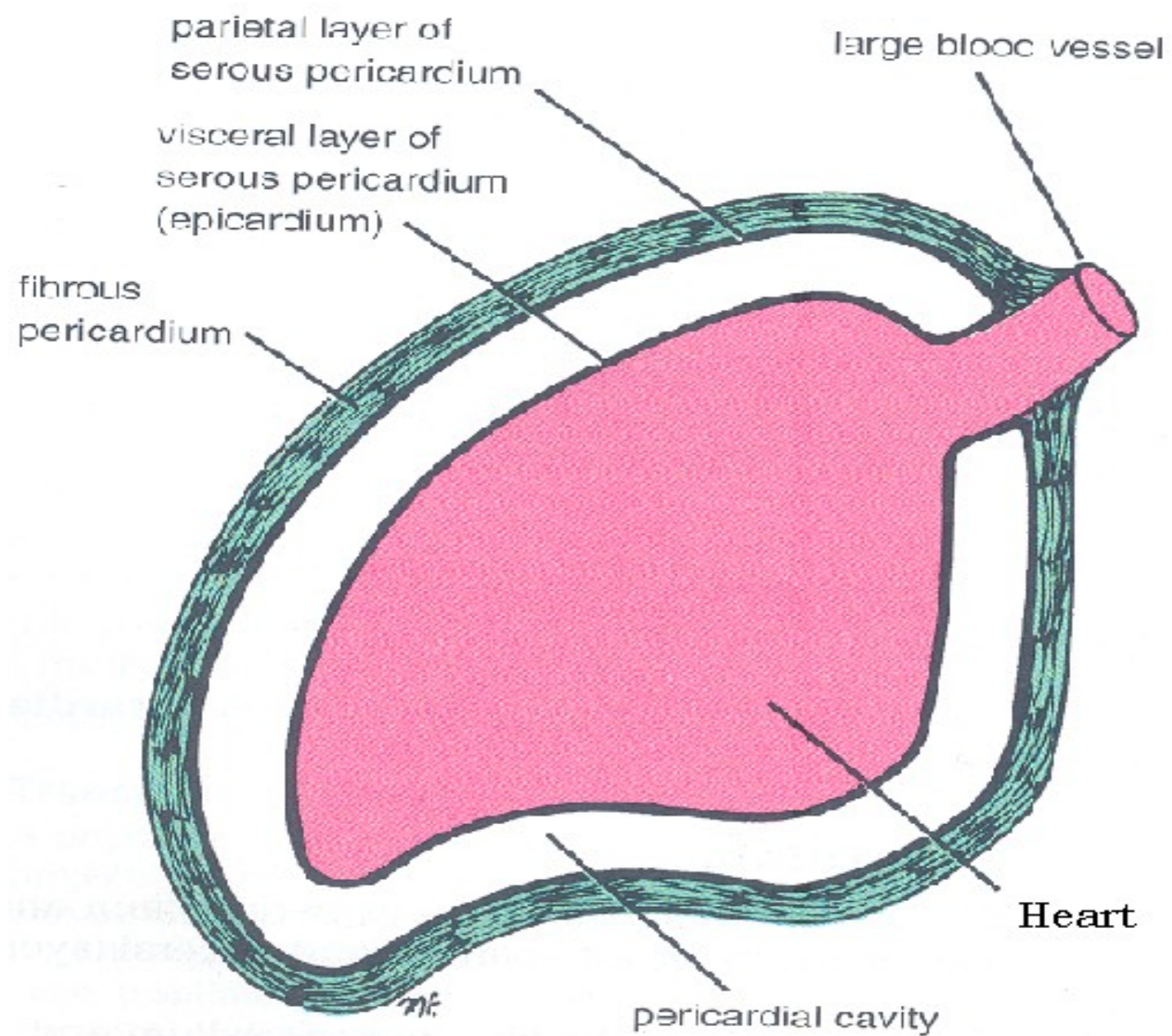


Fig..Different layers of the pericardium

Chambers of the Heart

- **The Atria:** Two thin-walled overlying muscular sheaths, serving as **reservoirs and pumps**
- **The Ventricles:** **Thicker-walled** portion of the heart that pumps blood from the low-pressure venous system into the higher pressure arterial system.

Cardiac Valves

- Thin flaps of flexible, endothelium-covered fibrous tissue firmly attached to fibrous rings at the base of the heart
- Responsible for the **unidirectional flow of blood** through the heart.

Types of Cardiac valves

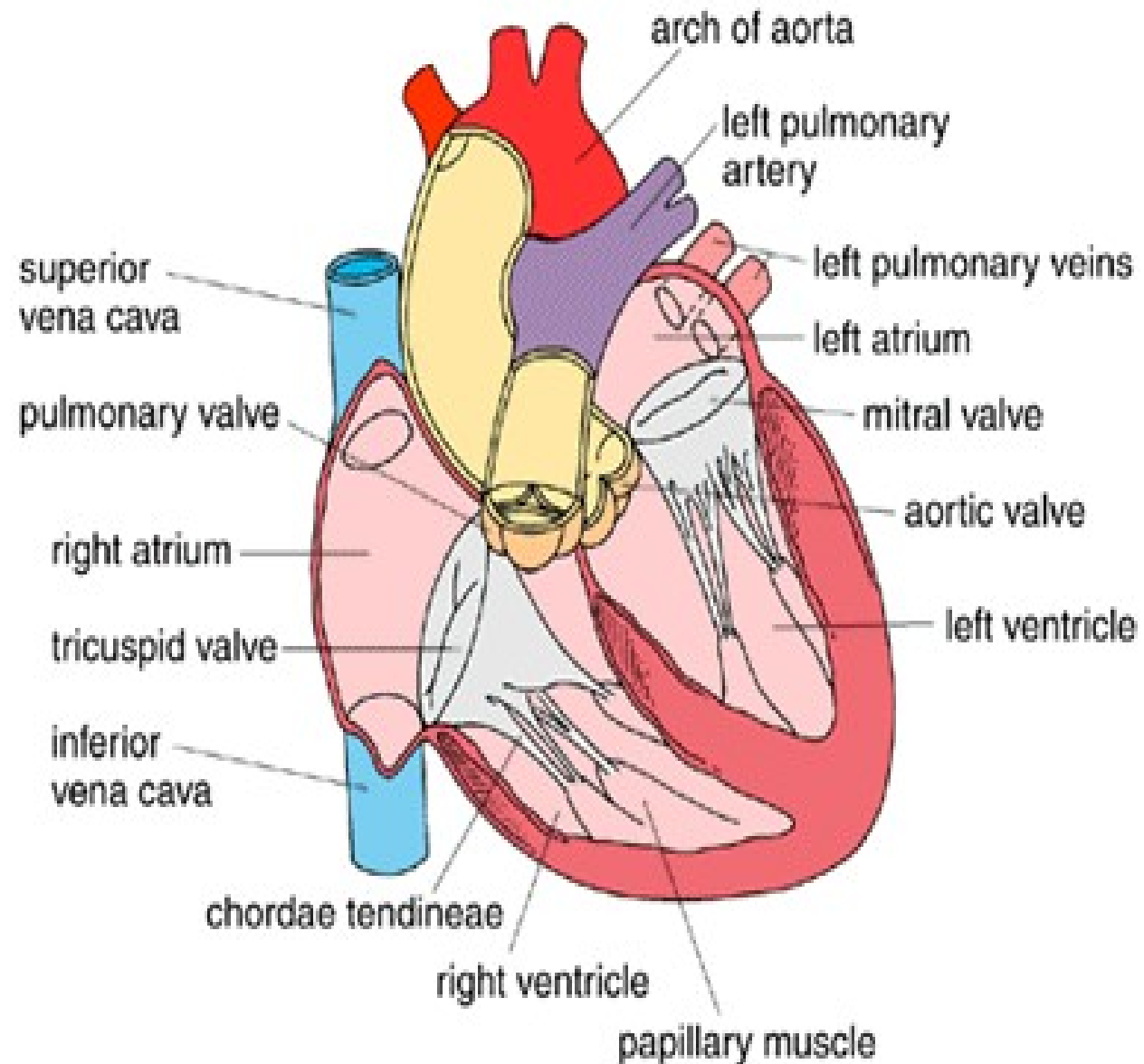
■ The Mitral and Tricuspid (atrio-ventricular-AV) valves

- Thin-walled and located b/n the atria and the ventricles.
- Mitral (two cusps) valve lies b/n left atrium and left ventricle
- Tricuspid (three cusps) valve lies b/n Rt atrium and Rt ventricle
- Chordae tendineae (fine strong filaments arising from the strong **papillary muscles**) and extending to valve cusps are attached to the free edges of AV valves.
- Papillary muscles contract when ventricles contract
- AV valves open when pressure in the atria is greater than pressure in the ventricles (i.e., during ventricular diastole) & closed when pressure in the ventricles is greater than pressure in the atria (i.e., during ventricular systole)

■ The semilunar valves

- Constitute the aortic and pulmonary valves located at the exits of the right and left ventricles.
- All valves consist of connective tissue (not cardiac muscle tissue) and, therefore, open & close passively.
 - a. **Aortic valve:** is three-cusped and allows blood to flow into the aortic tree and through their cusps to the left and right main coronary arteries
 - b. **Pulmonary valve:** allows blood to flow into the pulmonary artery.
- SL valves open when pressure in the ventricles is greater than pressure in the arteries (i.e., during ventricular systole) and close when pressure in the pulmonary trunk and aorta is greater than pressure in the ventricles (i.e., during ventricular diastole).

Types of Cardiac valves...cont'd



Pathway of Blood Through the Heart and Lungs

- Right atrium → tricuspid valve → right ventricle
- Right ventricle → pulmonary semilunar valve
pulmonary arteries → lungs
- Lungs → pulmonary veins → left atrium
- Left atrium → bicuspid valve → left ventricle
- Left ventricle → aortic semilunar valve → aorta
- Aorta → systemic circulation

Cardiac muscle (myocardium)

It is composed of 3 types of cardiac muscles

1. The **atrial muscle**
2. The **ventricular muscle**
3. The specialized excitatory and conductive muscle fibres
(**Authorhythmic Cells**)
 - Atrial and ventricular muscles are contractile components
(Contain too many actin & myosin)
 - The atrial and the ventricular muscles are separated by the
fibrous skeleton of the heart

Function: -form cardiac valves

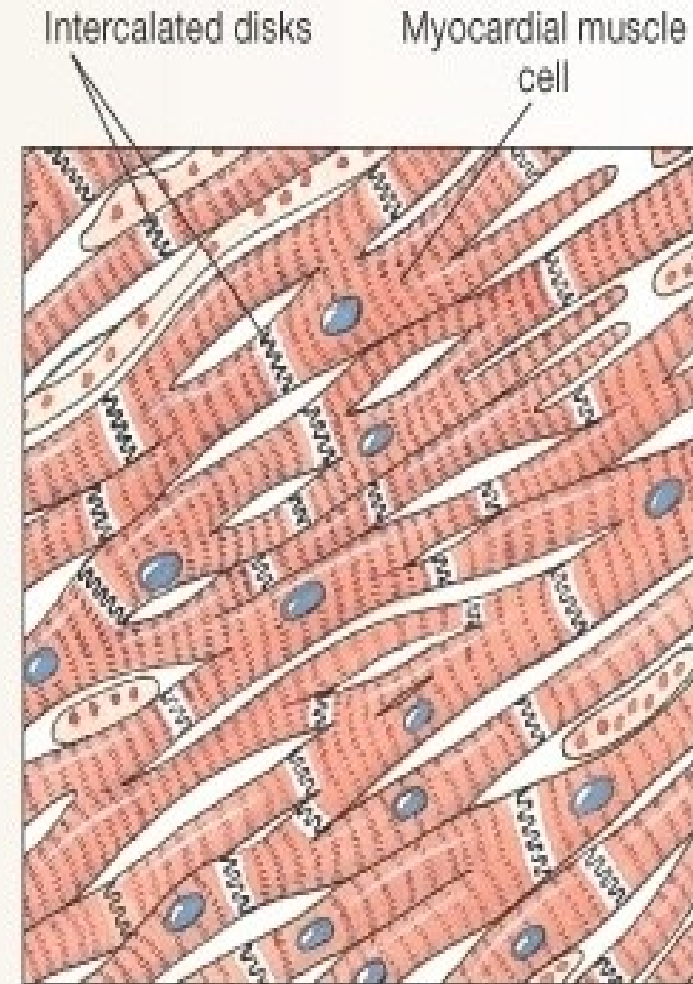
-serve as a means of attachment and insertion
of
cardiac muscles

Microscopic Structure of the Myocardium

- The myocardium consists of the **contractile machinery** with a series of overlapping sheets of **muscle bundles**
- Cardiac muscle tissue forms two **functional syncytia** or units that contract in a **synchronous or wavelike** fashion important for the pumping action of the heart.
 - a. The **atrial syncytium**: constitutes walls of atria.
 - b. The **ventricular syncytium**: constitutes walls of ventricles.
- APs are conducted from **atrial syncytium** to **ventricular syncytium** through the AV bundle.

...cont'd

- ❑ Adjacent cells join end-to-end at dark structures (end-to-end junctions) called **intercalated discs** which are actually cell membranes that separate individual cardiac muscle cells.
- ❑ Cardiac cells contain two types of specialized junctions at the intercalated discs.
 - a. **Desmosomes**: which act like rivets and **hold the cells tightly together**
 - b. **Gap junctions**: which permit action potentials to easily spread from one cardiac muscle cell to adjacent cells.
- ❑ Cardiac muscle has many **mitochondria** and utilizes most of the O_2

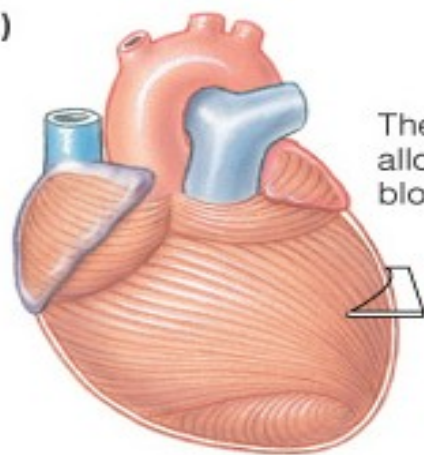


Myocardial muscle cells are branched, have a single nucleus, and are attached to each other by specialized junctions known as intercalated discs.

Structure of Cardiac Myocyte

- **Myocyte** is 10-20 μm in diameter and 50-70 μm in length with a single central nucleus.
- It is packed with long contractile bundles (1 μm) called **myofibrils** each of which is composed of smaller units called **sarcomeres (fig.below)** that give the myocyte its characteristic striated appearance and serves as the contractile unit of the muscle.
- **Myocytes** do not contract unless stimulated electrically by pacemaker tissue.

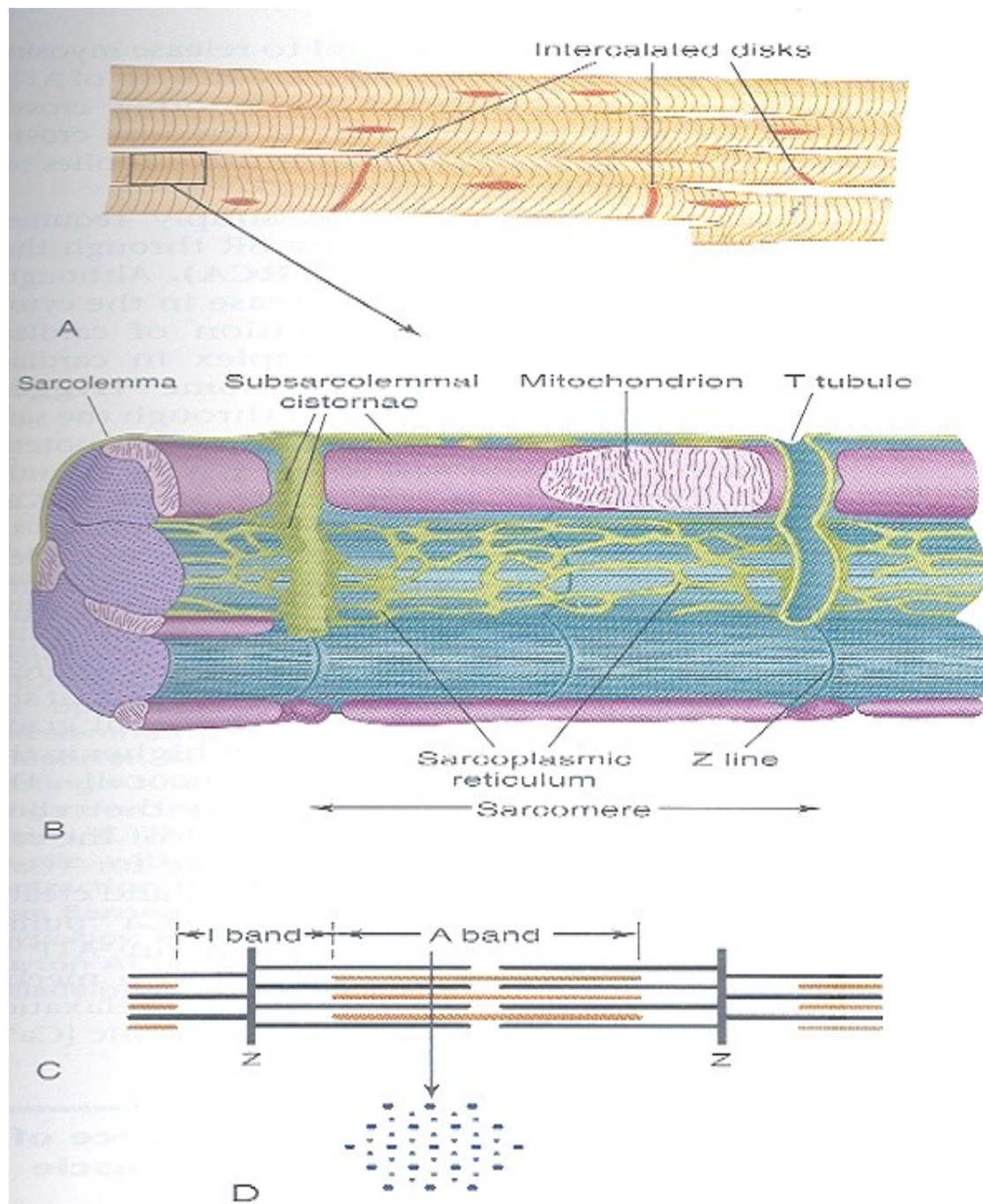
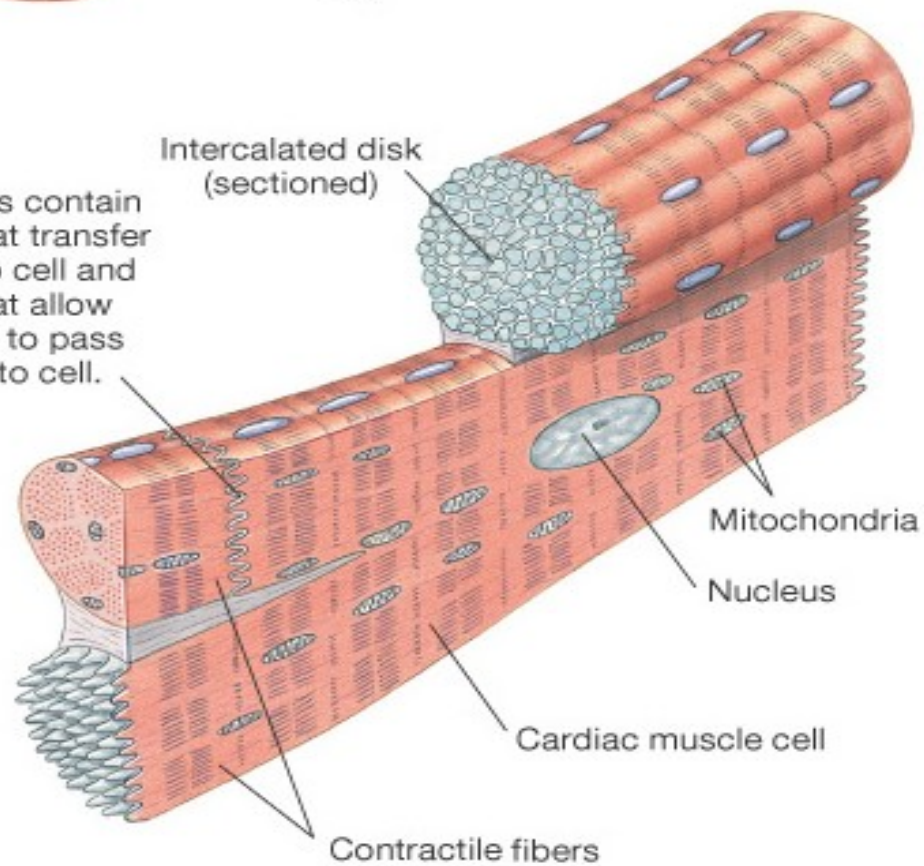
(a)



The spiral arrangement of ventricular muscle allows ventricular contraction to squeeze the blood upward from the apex of the heart.

(b)

Intercalated disks contain desmosomes that transfer force from cell to cell and gap junctions that allow electrical signals to pass rapidly from cell to cell.



Properties of cardiac muscles

Cardiac muscles have 4 physiological properties, which are essential for the function of the heart as the central pumping organ.

These are

- ✓ Autorhythmicity
- ✓ Excitability
- ✓ Conductivity
- ✓ Contractility

Authorhythmicity

- Autorhythmicity is the ability of the heart to generate cardiac impulse and to beat regularly independent of any external stimulus.
- The heart is provided with specialized excitatory and conductive muscle systems having two functions:
 1. They generate rhythmic cardiac impulse
 2. They conduct cardiac impulse

Pacemaker Tissue of the Heart

- The SA node in the right atrium is the primary pacemaker of the heart.
- The AV node located b/n rt atrium and rt ventricle is the secondary pacemaker.
- Both the SA and AV nodes contain two cell types (round and elongated); the round cells are **more sparse** and the elongated cells **more preponderate**.

Specialized Conductive Tissue of the Heart

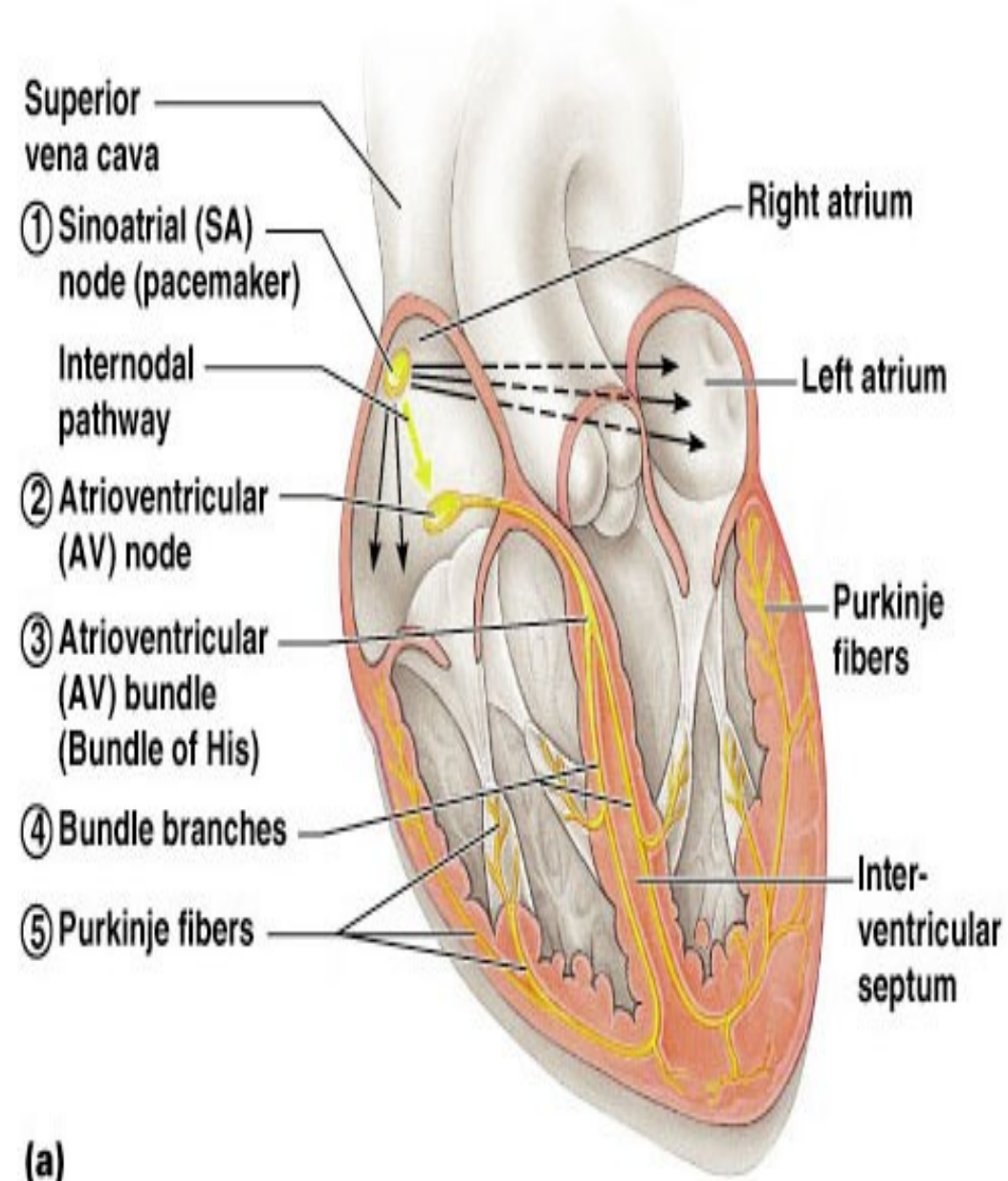
- Consists of **Purkinje fibers** ramifying over the sub-endocardial surfaces of both ventricles.
- **Purkinje cells** are broad cells (70-80 μm in diameter) compared with ventricular myocardial cells (10-15 μm in diameter).

Specialized excitatory and conductive system of the heart

It is comprised of the following components:

1. **Sino-atrial node (SA-node):** in which the normal (80-120 x/min) rhythmical self-excitatory impulse is generated
2. **Internodal pathways:** conduct impulse from the SA-node to the **Atrioventricular node (AV-node)**
3. **AV-node:** in which impulse from the atria delayed to be conducted to the ventricle. Site of nodal delay. Rhythm=40-60 x/min
4. **Atrioventricular bundle (bundle of His):** which conducts impulse from the atria to the ventricle.
Rhythm=20-40 x/min
5. **Purkinje fibers:** conduct cardiac impulse to the ventricles:
Rhythm=20-40 x/min

Specialized excitatory and conductive system of the heart



Velocity of conduction of AP in cardiac muscles

Structures	Conduction velocity (m/s)
SA-node	0.05
Internodal fibers	1.0
Atrial muscle	0.3
AV-node	0.05
AV-bundle	1.0
Purkinje fibres	4.0
Ventricular muscles	1.0

Sinoatrial node (pace maker)

- ❖ Located on the posterior wall of the R-atrium inferior to the opening of the SVC
- ❖ SA-nodal fibres have a special capability to discharge repetitive and rhythmic cardiac impulse.
- ❖ It generates impulse at a frequency of 80-120/min
- ❖ Determines the frequency of heart beat
- ❖ RMP of SA-nodal fibers is -55 to -60 mv
- ❖ SA-nodal discharge decreases with age. It can be calculated using a formula:

$$\text{Intrinsic SA-nodal rhythm} = 118.1 - (0.57 \times \text{Age})$$

Factors affecting autorhythmicity

1. ANS:

SyNS: Innervates SA-node, AV-node, conductive and contractile muscles

↑rhythmicity, +ve chronotropic effect

Mechanism: increases the slope of pacemaker potential of SA-node by decreasing K⁺ permeability

PaNS: Innervates SA-node, AV-node, atrial muscles

Inhibits autorhythmicity, -ve chronotropic effect

Strong vagal stimulation leads to **cardiac arrest**

Mechanism: decreases the slope of pacemaker potential by increasing K⁺ conductance in the SA-node

2. Temperature: ↑BT by 1°C, ↑HR by 20 beats/min
↑BT increases SA- nodal discharge

Factors affecting autorhythmicity cont'd...

3. Drugs: AD, NAD, T3/T4 increase cardiac rhythmicity
they have +ve chronotropic effect

Cholinergic drugs: Ach, pilocarpin → -ve
chronotropic effect

4. Electrolytes:

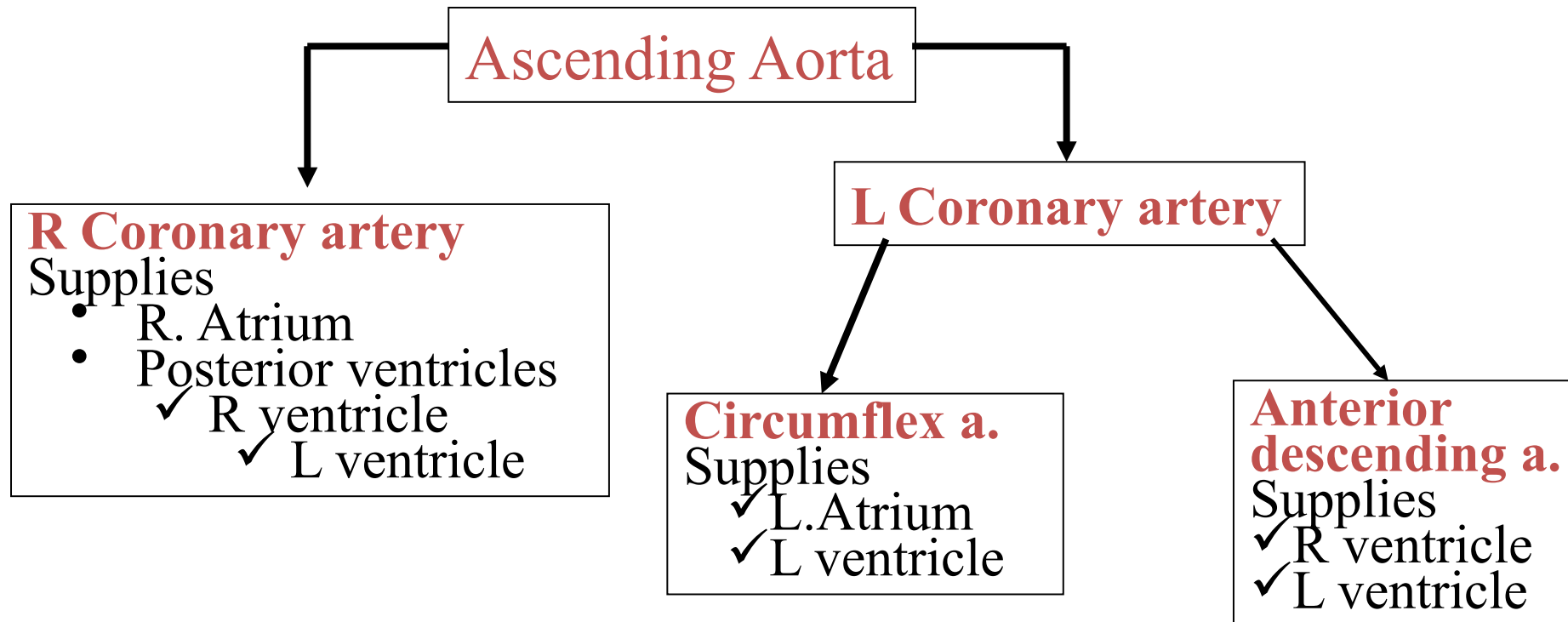
Hyperkalemia ($[K^+]$ 5-9 meq/L) leads to bradycardia by hyperpolarizing the membrane.

5. Mechanical factors: Atrial wall stretch by ↑ blood volume, ↑ HR

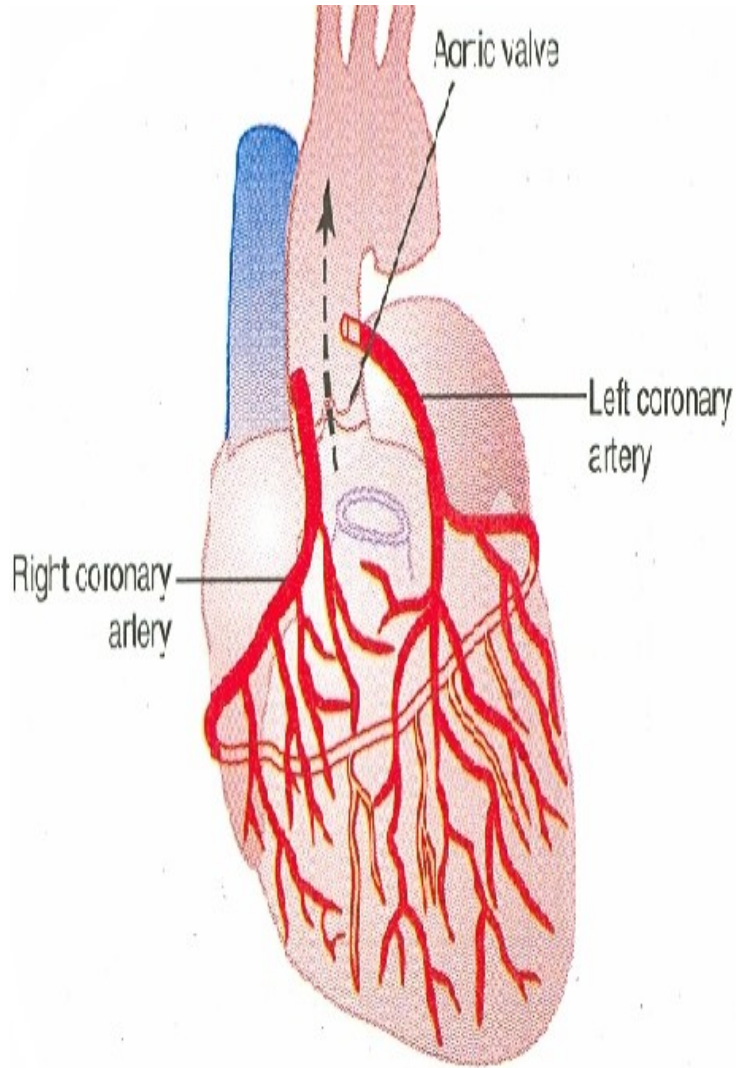
AD=adrenalin; NAD=nor adrenalin

Blood supply to the heart

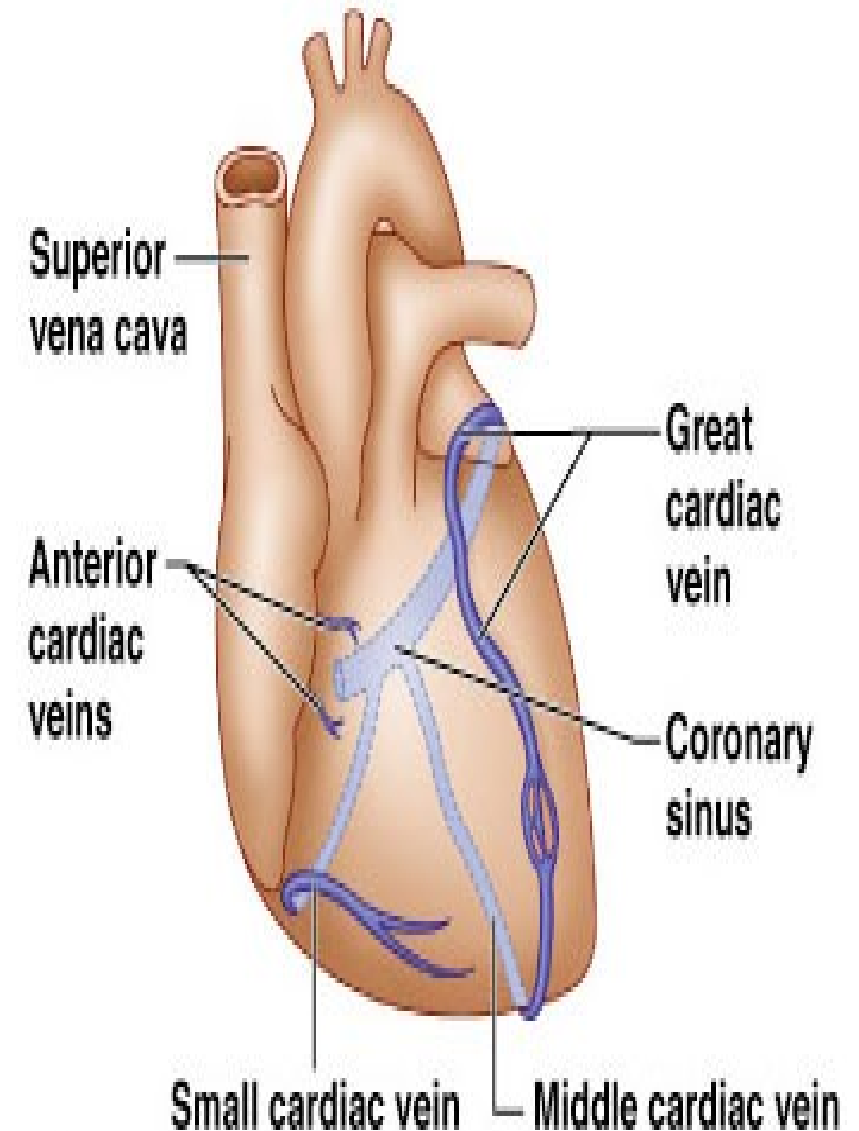
- The heart receives arterial blood from the **coronary artery**, which is the branch of **ascending aorta**.
- Resting coronary blood flow = **250 ml/min, 5% CO**
- **Coronary arterial diseases (Angina pectoris, MI)**



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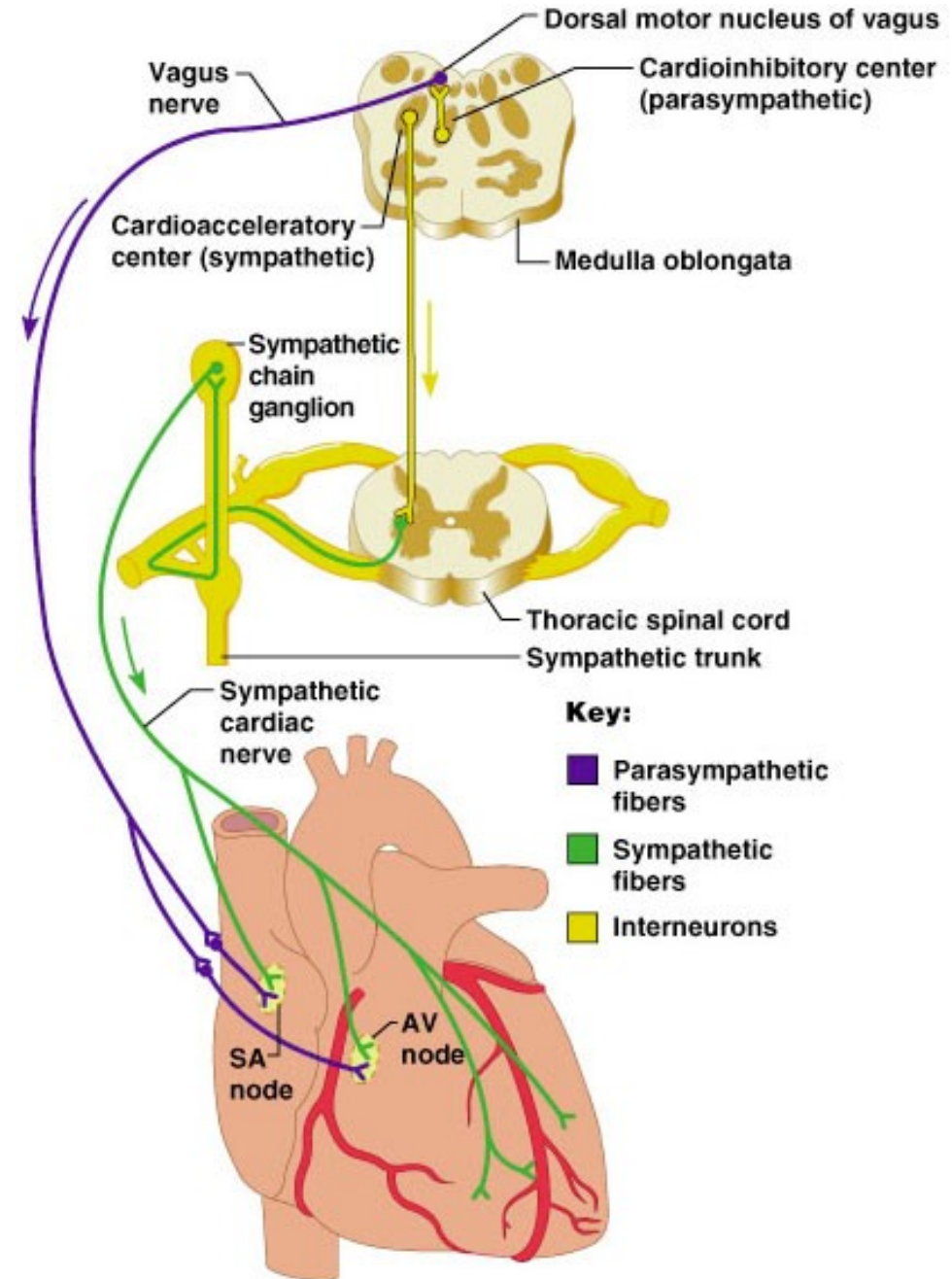


- The major venous drainage system of the human heart consists of four intercommunicating parts which generally open into the right atrium.
- The major cardiac veins include
 - the great vein,
 - the left marginal vein,
 - the posterior vein of the left ventricle and
 - the middle and small cardiac veins.



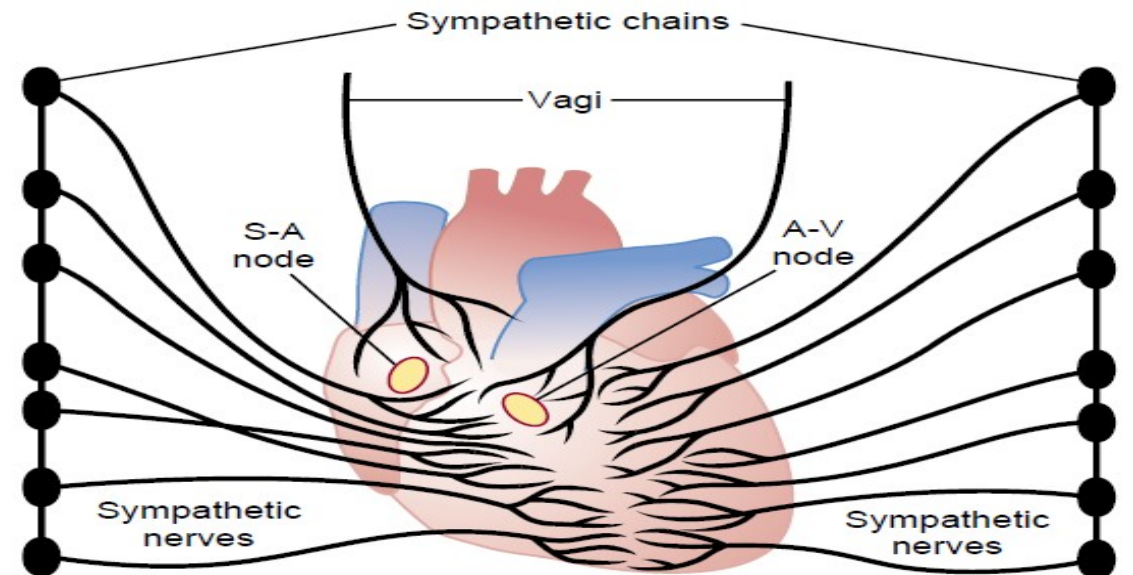
Innervation of the Heart

- Heart has dual autonomic innervation from both SNS and PNS with afferent and efferent components.
- The **sympathetic nerve supply** to the heart is controlled by the **medullary vasoconstrictor/ cardio accelerator center**
- Preganglionic sympathetic fibers arise from the lateral horn of the upper 5-thoracic spinal segments
- Postganglionic sympathetic fibers arise from the cervical and thoracic ganglia and proceed to supply atria, ventricles and nodal areas



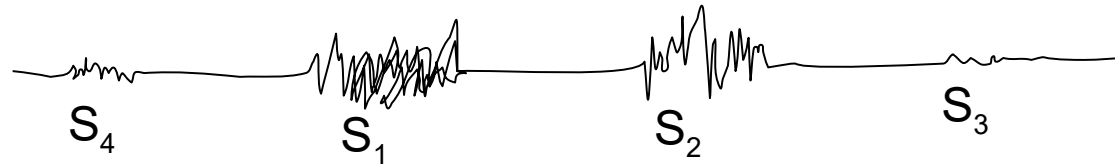
Innervation of the heart cont'd...

- The parasympathetic nerve supply to the heart is controlled by **the vasodilator/ cardio inhibitor center**.
- Preganglionic parasympathetic fibers arise from cardio inhibitory center in the medulla and proceed as vagal fibers to relay in terminal ganglia in the wall of the atria
- Short postganglionic fibers arise from terminal ganglia and supply the atria, SA-node and the AV-node
- The right vagus has a strong influence on SA node, while the left vagus has dominant effect on AV node
- Ventricles are not supplied by vagus nerve



Heart Sounds

- Heart sounds (lub-dub) are associated with **closing** of heart valves.
- 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)
- Means of identification
 1. Auscultation: - direct/immediate auscultation
- Stethoscope mediated auscultation
 2. Phonocardiography based recording



S_1 : is always audible, has a LUB-sound

S_2 : is always audible, has DUB-sound

Continuous heart sound: Lub-Dub, Lub-Dub

Duration of Lub is shorter than that of Dub

S_3 : is audible in children & in adults during exercise

S_4 : is audible very rarely.



Heart Sounds (cont'd)

S₁: First heart sound

- **Cause:** sudden closure of AV-valves.
- **Timing:** occurs at the beginning of ventricular systole
- **Best site of auscultation**
 - ❖ 5th LIS (left intercostal space), at mid-clavicular line
 - ❖ 4th LIS, near the sternum

Clinical Correlate: Loud S1 suggests tachycardia or mitral stenosis; soft S1 suggests first-degree heart block or mitral regurgitation

S₂: Second heart sound

- **Cause:** Sudden closure of semilunar valves
- **Timing:** occurs at the beginning of ventricular diastole
- **Best site of auscultation**
 - ❖ 2nd RIS, near the sternum
 - ❖ 2nd LIS, near the sternum

Heart Sounds (cont'd)

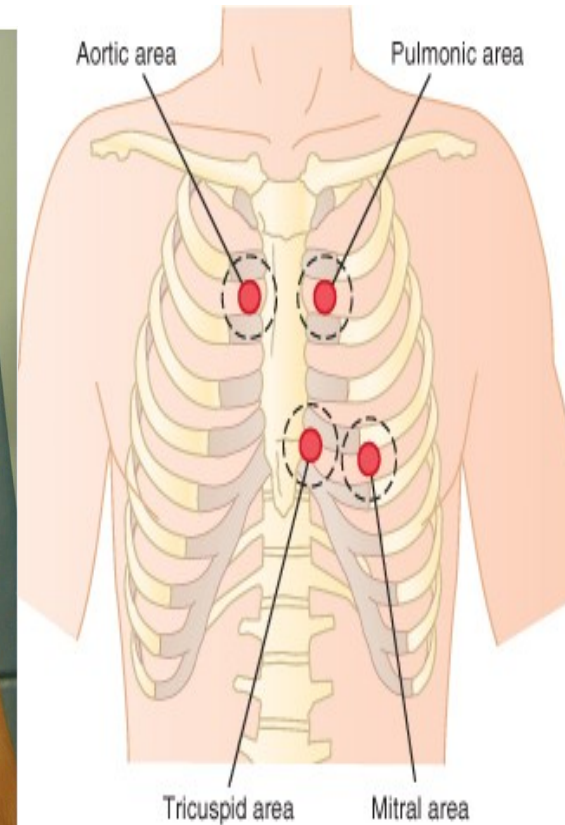
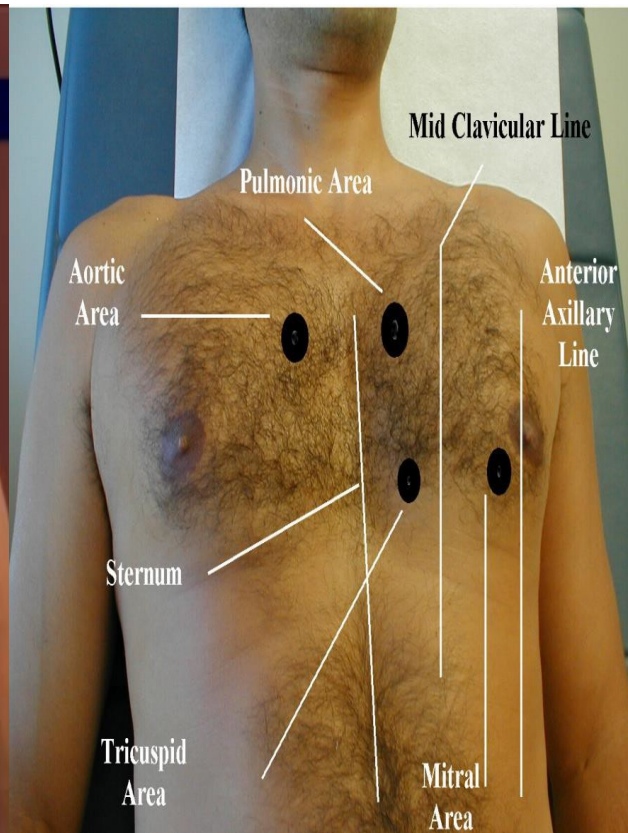
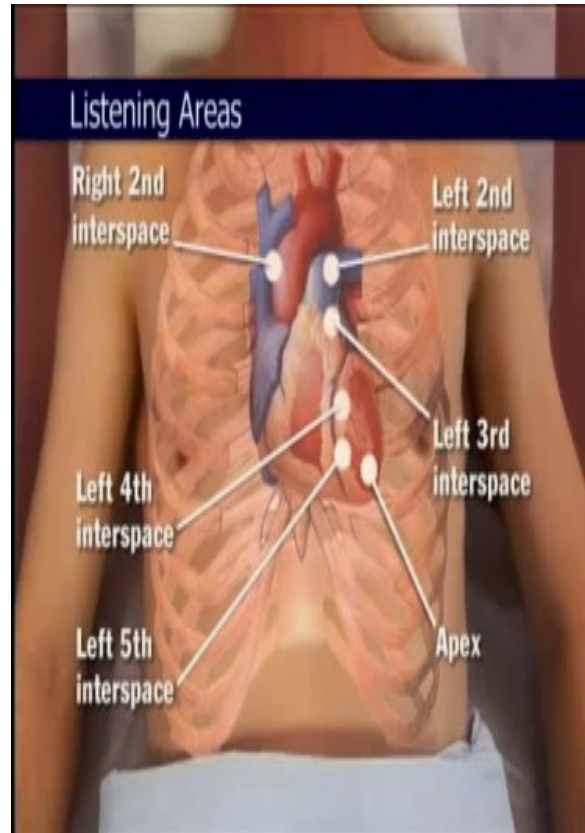
S₃: Third heart sound

- **Cause:** rapid filling of the ventricles with blood during ventricular diastole.
 - It is audible in children and in adults during exercise.
- **Best site of auscultation:** at the apex of the heart beneath the xiphoid process.

S₄: Fourth heart sound

- **Cause:** rapid ventricular filling during atrial systole
- **Timing:** during atrial systole
- **Best site of auscultation:** at the apex of the heart

Areas of Auscultation [areas of Listening]



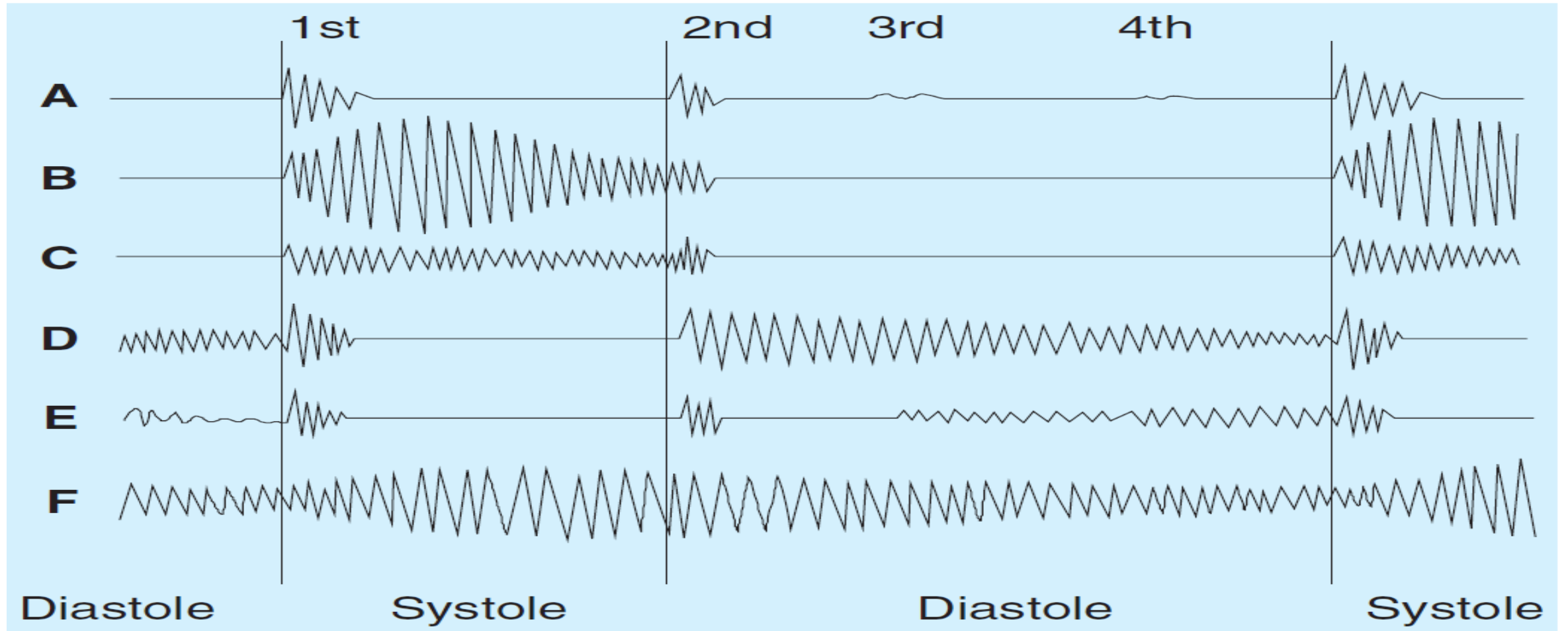
Abnormal Heart Sounds

- **Heart Murmurs (Turbulent Flow)**
- **Systolic Murmurs:** Occur between S1 and S2
 - **Ejection Murmurs:** Caused by aortic/pulmonary stenosis (e.g., aortic stenosis).
 - **Regurgitant Murmurs:** Caused by mitral/tricuspid insufficiency (e.g., mitral regurgitation).
- **Diastolic Murmurs:** Occur between S2 and the next S1
 - **Clinical Correlate:** Usually indicate valvular pathology, such as aortic regurgitation or mitral stenosis
- **Continuous Murmurs:** Heard throughout both systole and diastole.
 - **Clinical Correlate:** Often indicate patent ductus arteriosus (PDA)

Heart murmurs cont'd

Site of abnormality	Type of abnormality	Type of murmur
Aortic or pulmonary valve	Stenosis Insufficiency	Systolic Diastolic
Mitral or tricuspid valve	Stenosis Insufficiency	Diastolic Systolic
Interventricular septum	Congenital hole	Systolic
Aorta	Coarctation	Systolic
Ductus arteriosus	Patent	Continuous
Blood	Anaemia	Systolic

Heart murmurs cont'd



Cardiac murmurs on phonocardiography: A, normal phonocardiogram; B, systolic murmur (in aortic stenosis); C, systolic murmur (in mitral regurgitation); D, diastolic murmur (in aortic regurgitation); E, diastolic murmur (in mitral stenosis) and F, continuous murmur (in patent ductus arteriosus)

Pathophysiology-Clinical Correlates

- Cardiovascular diseases (CVDs) are the leading cause of death and disability-adjusted life years (DALYs) worldwide, with an estimated **19.8 million** deaths in 2022, comprising approximately **32% of all global deaths**.
- As of 2023, cases reached 626 million, **with nearly 80% of CVD deaths occurring in low- and middle-income countries**
- The burden is rising due to population aging and growth, **with 35.6 million annual deaths projected by 2050**

Pathophysiology.....

- Cardiac muscle dysfunction results from structural, electrical, or metabolic damage to the myocardium, causing impaired contraction (systolic) or filling (diastolic), often leading to heart failure
- **Systolic Dysfunction (Reduced EF):** Primarily characterized by impaired contraction (e.g., in dilated cardiomyopathy) where the ventricles cannot pump effectively, resulting in reduced stroke volume.
- **Diastolic Dysfunction (Preserved EF):** Primarily characterized by stiffening or thickening of the heart muscle (e.g., in hypertrophic or restrictive cardiomyopathy), which impairs relaxation and filling, leading to increased filling pressures.

Major Causes of Dysfunction

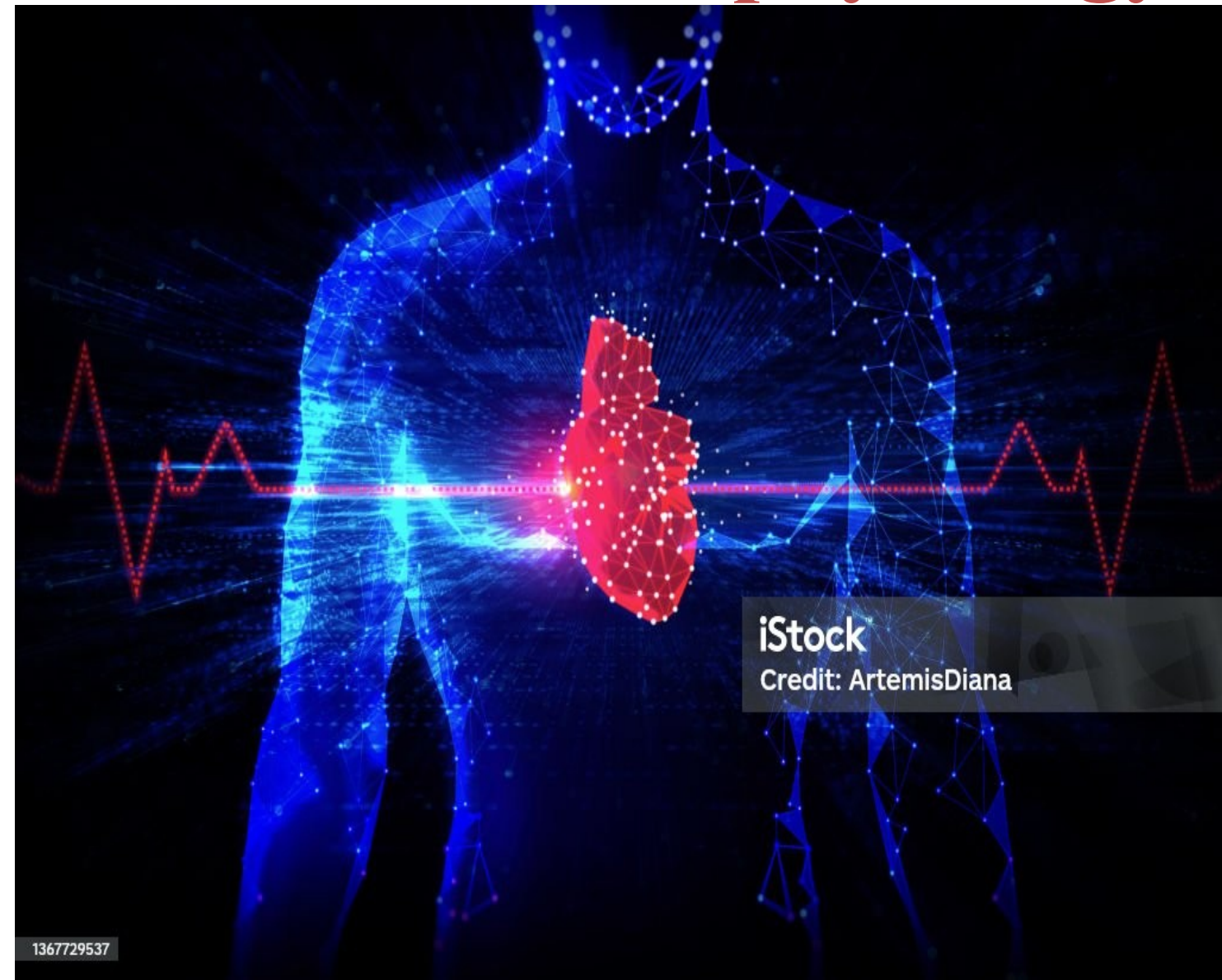
- **Cardiomyopathies:** Genetic or acquired diseases of the myocardium (dilated, hypertrophic, restrictive, or arrhythmogenic).
- **Ischemic Injury:** Coronary artery disease reducing blood flow.
- **Inflammation:** Myocarditis, often due to viral infections.
- **Metabolic/Endocrine Factors:** Chronic hypertension, obesity, or diabetes.

Pathophysiology.....

Types Cardiomyopathy:

- **Dilated cardiomyopathy (DCM):** This condition is dilation and impaired contraction of one or both ventricles that cannot be explained by coronary artery disease, valvular disease, or hypertension
- **Hypertrophic cardiomyopathy (HCM):** This condition is an autosomal dominant disease leading to a thickened (hypertrophied) left ventricle
- **Restrictive cardiomyopathy (RCM):** This condition results in impaired ventricular filling in the setting of non-dilated ventricles.
- **Ischemic cardiomyopathy (ICM)** is a form of dilated cardiomyopathy related to coronary artery disease (CAD)
- **Heart Failure:** impairment of ventricular filling and systolic ejection of blood due to structural and functional defects of the myocardium
- **Myocarditis:** an inflammatory disease of the myocardium most commonly caused by acute rheumatic fever or viral infections

Electrophysiology of the heart



iStock
Credit: ArtemisDiana



ACTION POTENTIALS IN CARDIAC MUSCLE:

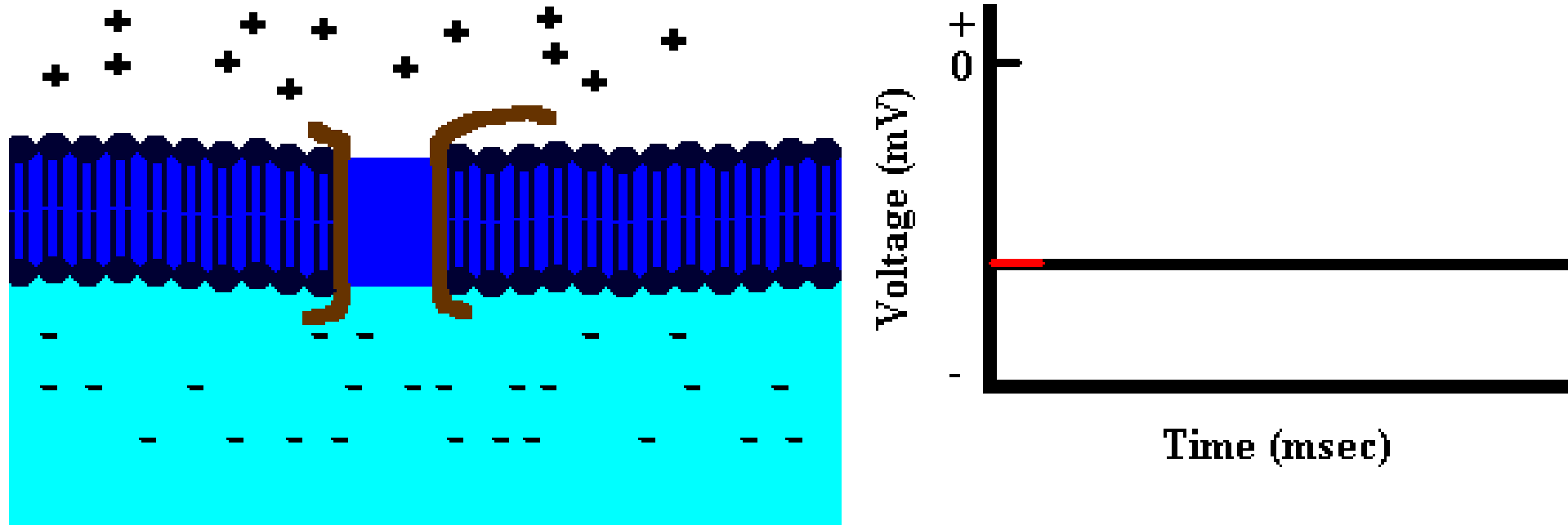
Membrane Potential

- Definition:
 - All cells have a voltage difference across their plasma membrane. This is called **membrane potential** (V_m).
- Resting membrane potential (RMP):
 - It is the membrane potential (V_m) at rest
 - At rest, there are electropositivity out and electronegativity inside the cell membrane of the neuron.
 - The RMP of a typical neuron is -90 mv
- Causes of the RMP:
 1. An outward diffusion of K^+ through K^+ leak channels. The ECF is very high in Na^+ while the ICF is very high in K^+ . As a result, K^+ is constantly leaking out of the cell.
 2. The Na^+/K^+ pump is constantly pumping 3 Na^+ ions out and 2 K^+ ions in for every ATP used. Thus more positive charge is leaving than entering.
 2. There are protein anions (i.e. negatively charged proteins) within the ICF that cannot travel through the PM.

Action Potential

- Definition:

- An immediate change of the RMP into depolarization that is followed by reestablishment of the RMP (repolarization) is called **action potential** or **nerve impulse**.
- If RMP changes from -90 mV to threshold level (-75 mV) → voltage gated Na^+ channels open → Na^+ influx → **depolarization** → → → Na^+ channels become inactivated soon.
- Opening of voltage gated K^+ channels → K^+ efflux → **repolarization**.



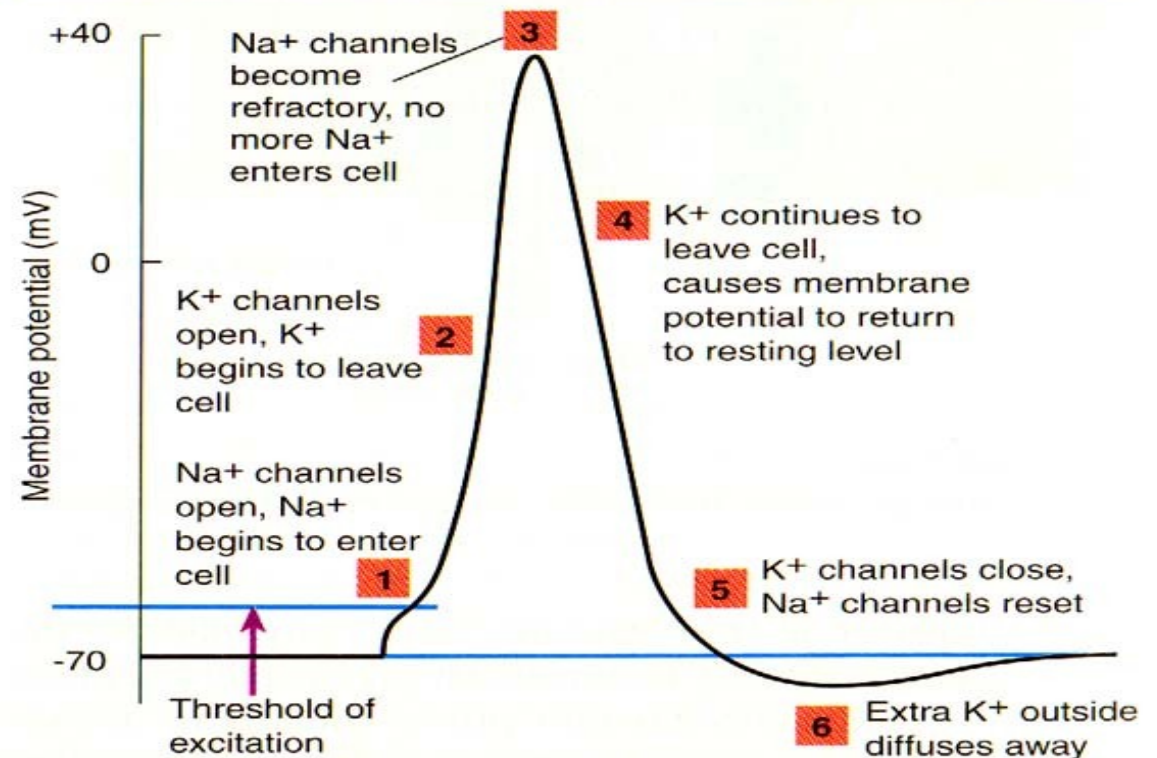
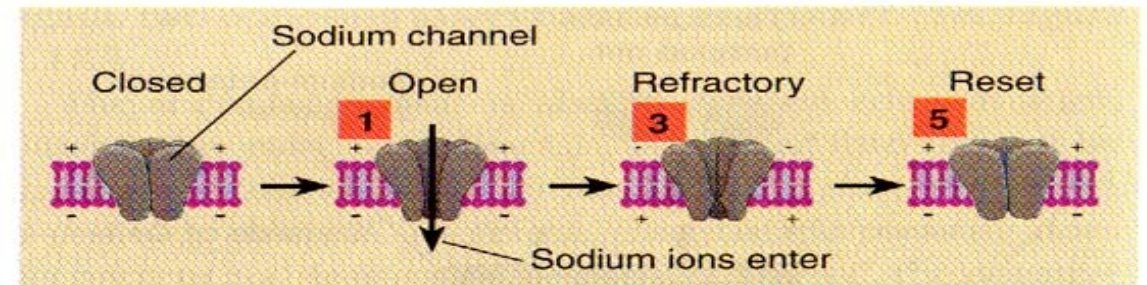
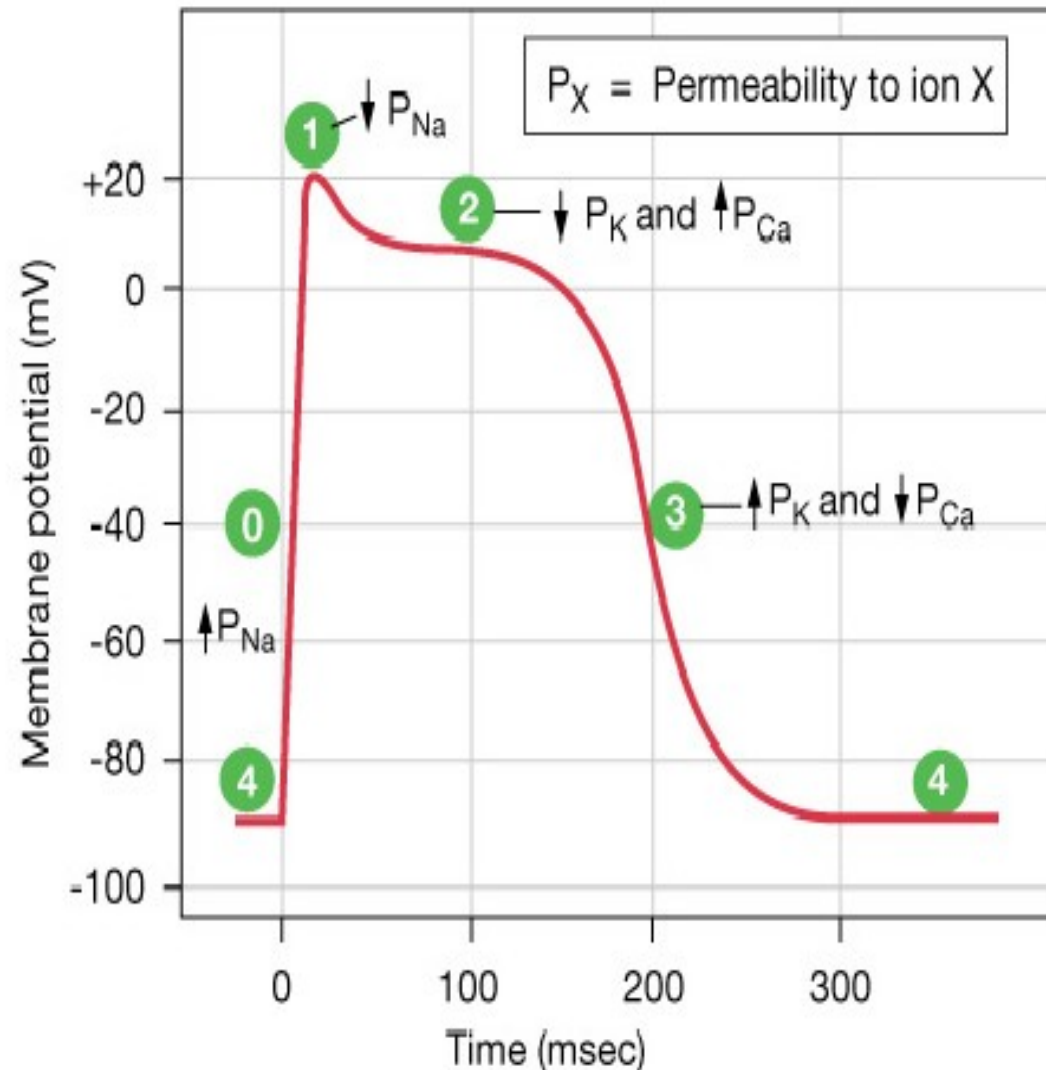
Action Potential....

- The heart beats in the absence of any nervous connections because the electrical (pacemaker) activity that generates the heartbeat resides within the cardiac muscle
- The electrical activity of cardiac cells depends on the ionic gradients across their plasma membranes and changes in permeability to selected ions brought about by the opening and closing of cation channels
- The ion concentration gradients that determine transmembrane potentials are created and maintained by active transport
- The transport of Na and K is accomplished by the plasma membrane Na/K-ATPase
- Calcium is partially transported by means of a Ca²⁺-ATPase and partially by an antiporter that uses energy derived from the Na electrochemical gradient to remove Ca²⁺ from the cell

Action Potential....

- Once the ventricular cell membrane is brought to threshold, voltage-gated Na channels open, causing the initial rapid upswing of the action potential(phase 0).
- The opening of Na channels causes Na permeability to increase (??)
- As permeability to Na exceeds permeability to K, the membrane potential approaches the Na equilibrium potential, and the inside of the cell becomes positively charged relative to the outside.
- Averages about 105 millivolts, which means that the intracellular potential rises from a very negative value, about -85 millivolts, between beats to a slightly positive value, about +20 millivolts, during each beat
- Phase 1 of the ventricular action potential is caused by a decrease in the number of open Na channels and the opening of a particular type of K channel. These changes tend to repolarize the membrane slightly.
- The plateau of phase 2 results from a combination of the closing of K channels and the opening of voltage-gated Ca²⁺ channels
- The return of the membrane potential (phase 3, or repolarization) to the resting state is caused by the closing of Ca²⁺ channels and the opening of particular classes of K channels

Action Potential- Cardiac vs Skeletal muscle??



Refractory Period of Cardiac Muscle

Cardiac cell is unresponsive to further stimuli. During the phase of contraction, the heart is unexcitable (refractory) to stimulation.

There are two degrees of refractoriness:

- **Relative refractoriness:** Only APs of smaller amplitudes and rates of rise can be generated.
- **Absolute refractoriness:** No other AP can be triggered, even by extremely strong stimuli. Because of this refractoriness, the heart muscle cannot be tetanized
- The normal refractory period of the ventricle is **0.25 to 0.30 second**, which is about the duration of the prolonged plateau action potential
- There is an additional *relative refractory period* of about 0.05 second during which the muscle is more difficult to excite than normal but nevertheless can be excited by a very strong excitatory signal
- The refractory period of atrial muscle is much shorter than that for the ventricles (**about 0.15 second for the atria compared** with 0.25 to 0.30 second for the ventricles)
- Cardiac muscle has a long refractory period than the skeletal muscle.

What is the importance??

Electrical Activity of the Pacemaker

- When the electrical activity of a cell from the SA or AV node is compared with that of a ventricular muscle cell, three important differences are observed
 - (1) the presence of a pacemaker potential,
 - (2) the slow rise of the action potential, and
 - (3) the lack of a well-defined plateau
- The pacemaker potential results from changes in the permeability of the nodal cell membrane to the major cations First, K channels, primarily responsible for repolarization, begin to close

Electrical Activity ...cont'd

- Resting membrane potential” of the sinus nodal fiber between discharges has a negativity of about **-55 to -60** millivolts, in comparison with -85 to **-90 millivolts** for the ventricular muscle fiber.
- The cause of this lesser negativity is that the cell membranes of the sinus fibers are naturally leaky to **sodium and calcium ions**, and positive charges of the entering sodium and calcium ions neutralize much of the intracellular negativity
- Because of the high sodium ion concentration in the extracellular fluid outside the nodal fiber, as well as a moderate number of already open sodium channels, positive sodium ions from outside the fibers normally tend to leak to the inside.
- Therefore, between heartbeats, influx of positively charged sodium ions causes a slow rise in the resting membrane potential in the positive direction

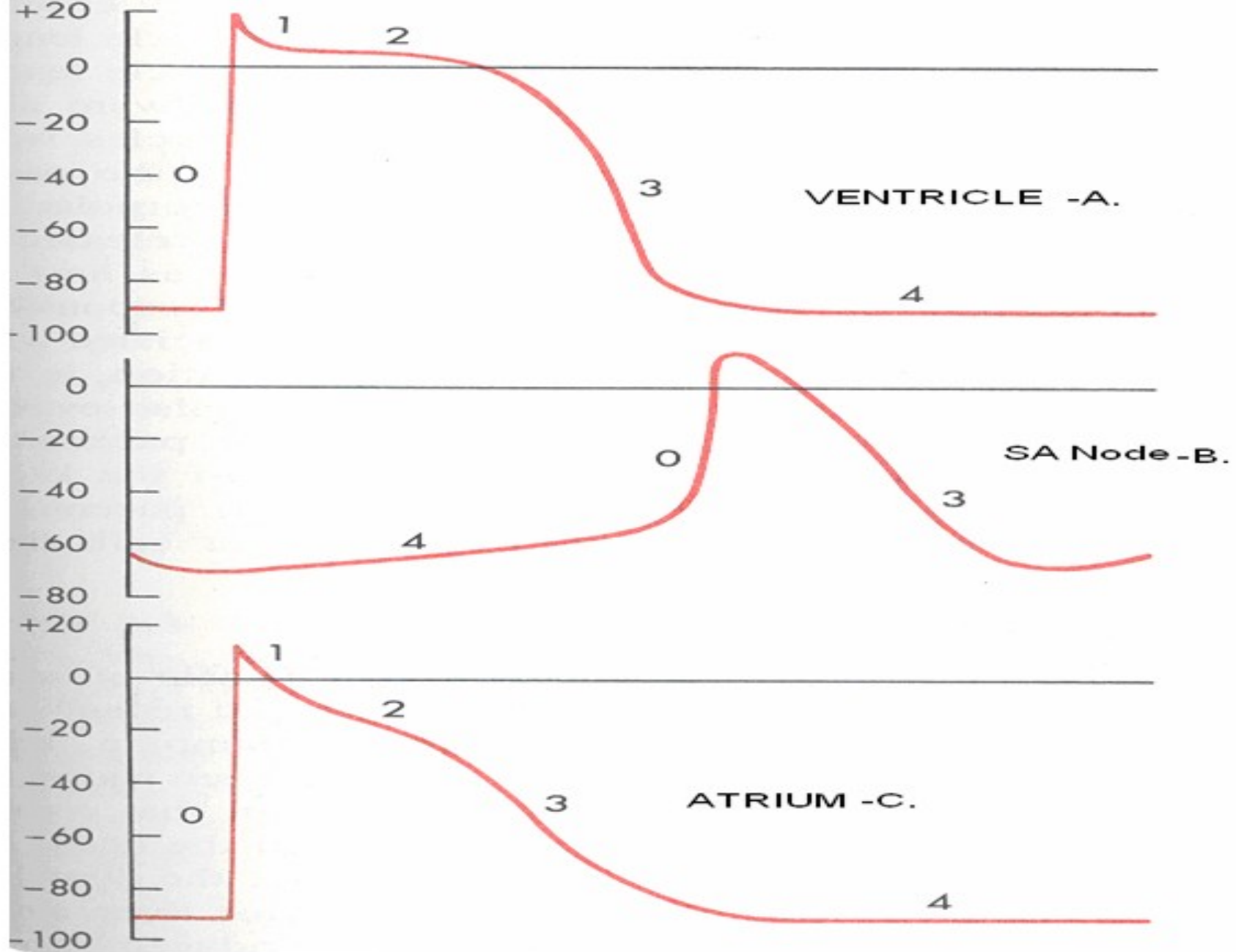


Fig..Action potentials (mv)

Electromechanical Properties of The Heart

- Cardiac function is **not fully dependent on intact nervous pathways**.
- It continues to beat after full denervation because of the intrinsic properties it has.

A. Automaticity (Myogeny): Ability to start pace making is most highly developed in nodal tissues but can be observed in any piece of cardiac tissue.

- Inherent rates:
 - SA node =70-80 bpm, AV node =60-65 bpm
 - Atria = 60-65 bpm, Ventricles =30-40 bpm

...cont'd

SA node is understood to be primary pacemaker by the following findings:

- i. First region to display electrical activity is SA node.
- ii. Crushing and localized cooling of SA node leads to bradycardia
- iii. Topical application of drugs and humoral agents to SA node leads to alteration of heart rate

B. Conductivity:

Most highly developed in PNW which is feebly contractile. Conduction velocity is some 6 times that of the rest of myocardium.

...cont'd

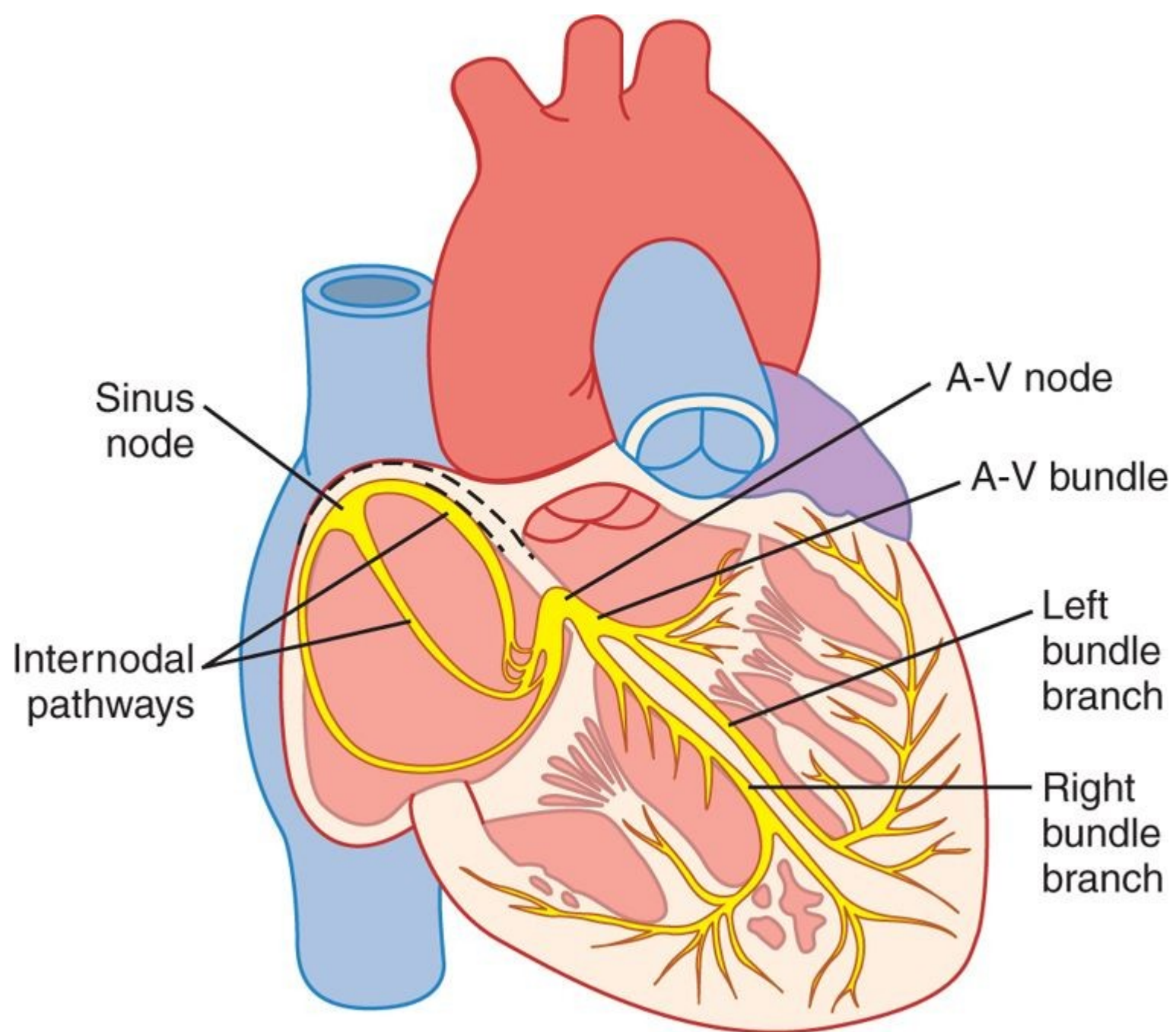
C. Contractility: Atrial and ventricular muscles are:

1. Highly specialized for contraction.
2. Follow the ALL-or-NONE response pattern

D. Refractoriness: Cardiac cell is unresponsive to further stimuli. During the phase of contraction, the heart is unexcitable (refractory) to stimulation. There are two degrees of refractoriness:

Relative refractoriness: Only APs of smaller amplitudes and rates of rise can be generated.

Absolute refractoriness: No other AP can be triggered, even by extremely strong stimuli. Because of this refractoriness, the heart muscle cannot be tetanized.



Effect of ANS on the Electromechanical activities of the Heart

- NS controls various aspects of the characteristics of the heart including the **frequency and vigor** (chronotropic (**heart rate**) and inotropic (**pump strength and stroke**) effects) at which it beats.
- The frequency of pacemaker (SA node) firing is influenced by underlying changes in ionic conductance and ionic flow through the plasma membrane.
- It is controlled by the activity of both divisions of the ANS
 - Stimulation of symp. NS $\rightarrow \uparrow$ HR
 - Stimulation of parasymp. NS $\rightarrow \downarrow$ HR

Molecular mechanism of ANS??

...cont'd

- The parasymp. transmitter Ach acting on cell membrane of SA node makes the membrane more permeable to K^+ .
- This produces two effects:
 1. repolarization or RMP
 2. reduction in rate of formation of pre-potential
- These conditions lead to longer time for the Pre-potential to reach threshold thus making HR

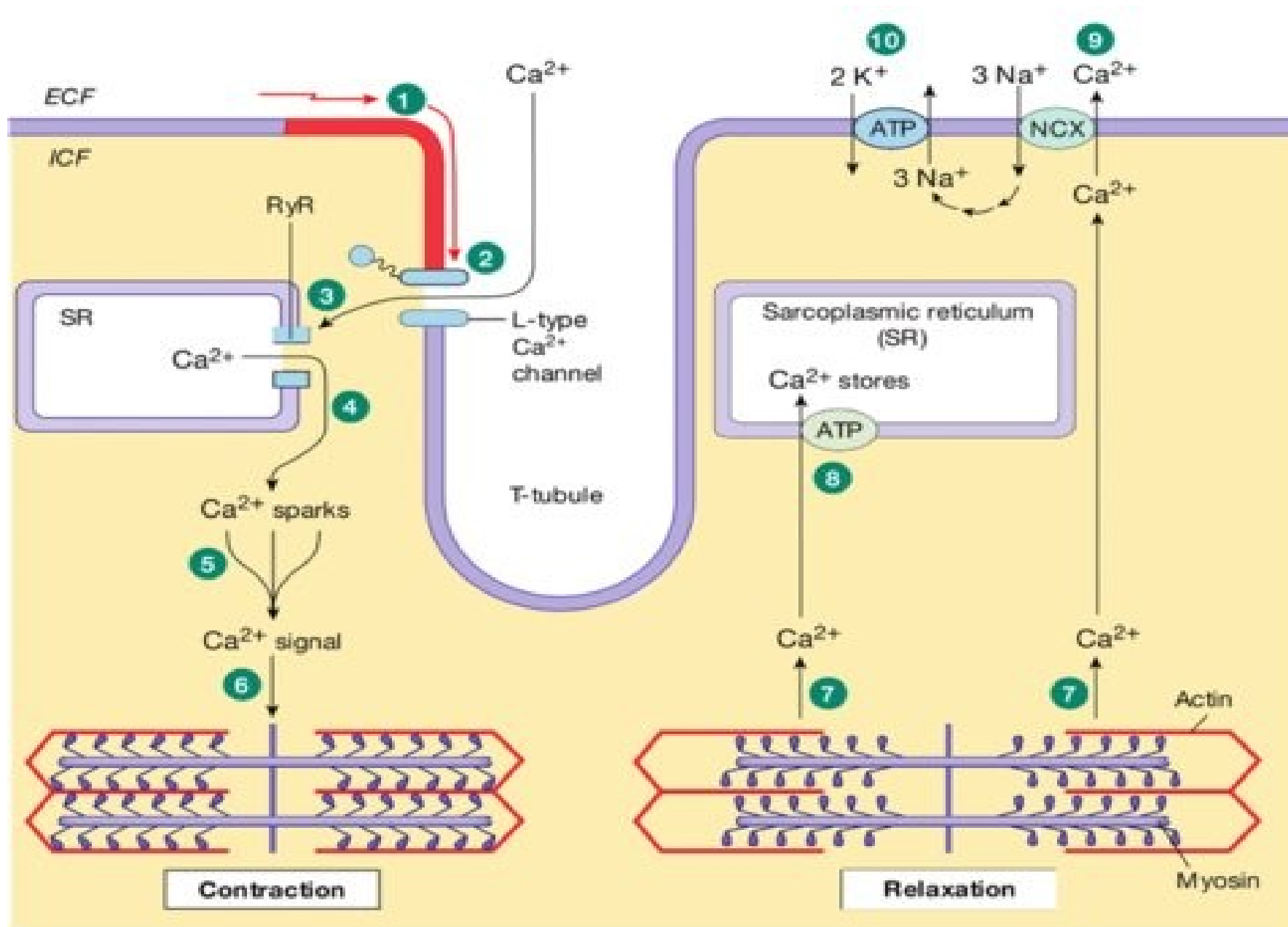
Excitation-contraction coupling in cardiac muscle

- It is a mechanism by which an action potential causes contraction of myocardium.
- Calcium ions provide the link b/n **electrical excitation** and **muscle contraction**
- The influx of Ca^{++} from the ECF during excitation (plateau) triggers the release of Ca^{++} from the SR.
- The free cytosolic Ca^{++} activates contraction of the myofilament (systole).
- Relaxation (diastole) occurs as a result of uptake of Ca^{++} by the SR and extrusion of intracellular Ca^{++} by $\text{Na}^{+} - \text{Ca}^{++}$ exchange

...cont'd

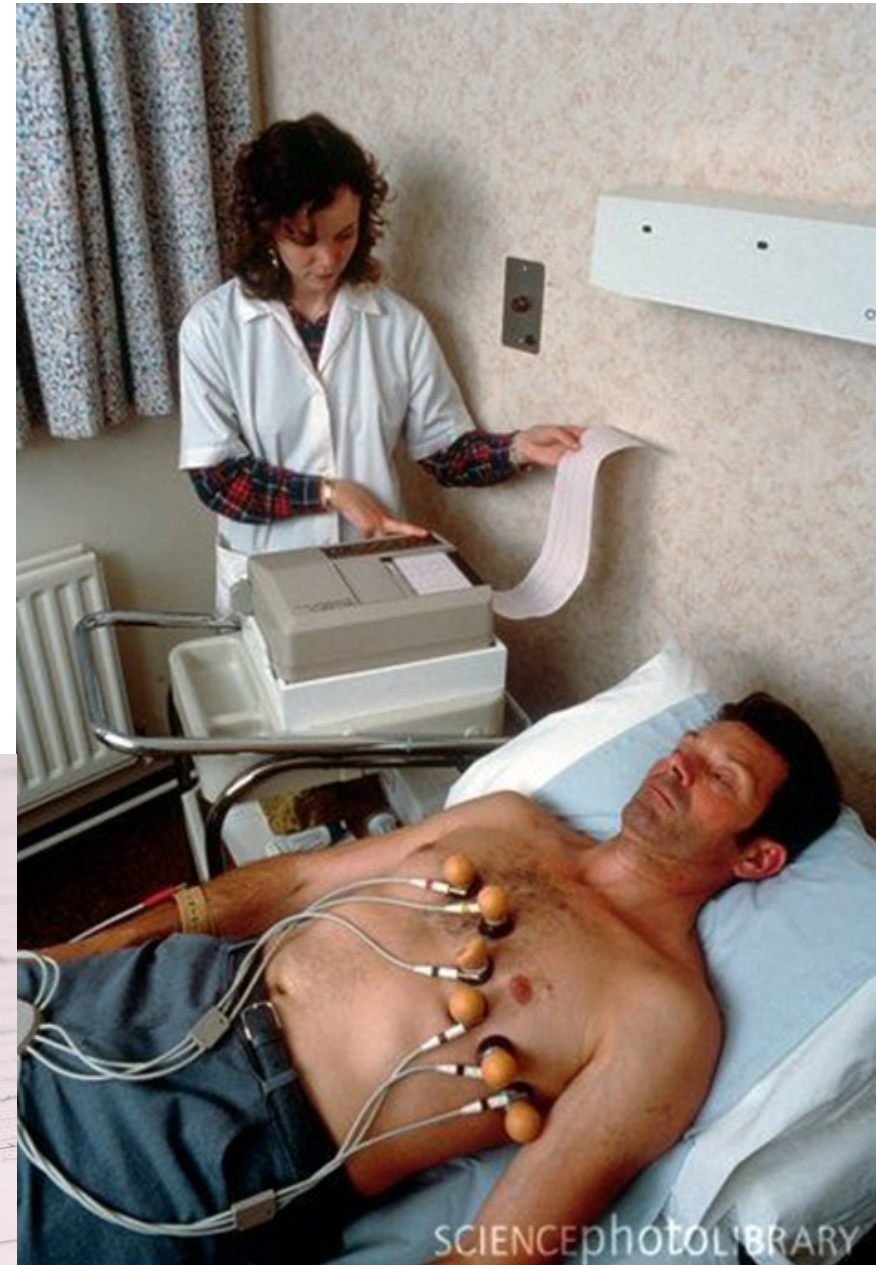
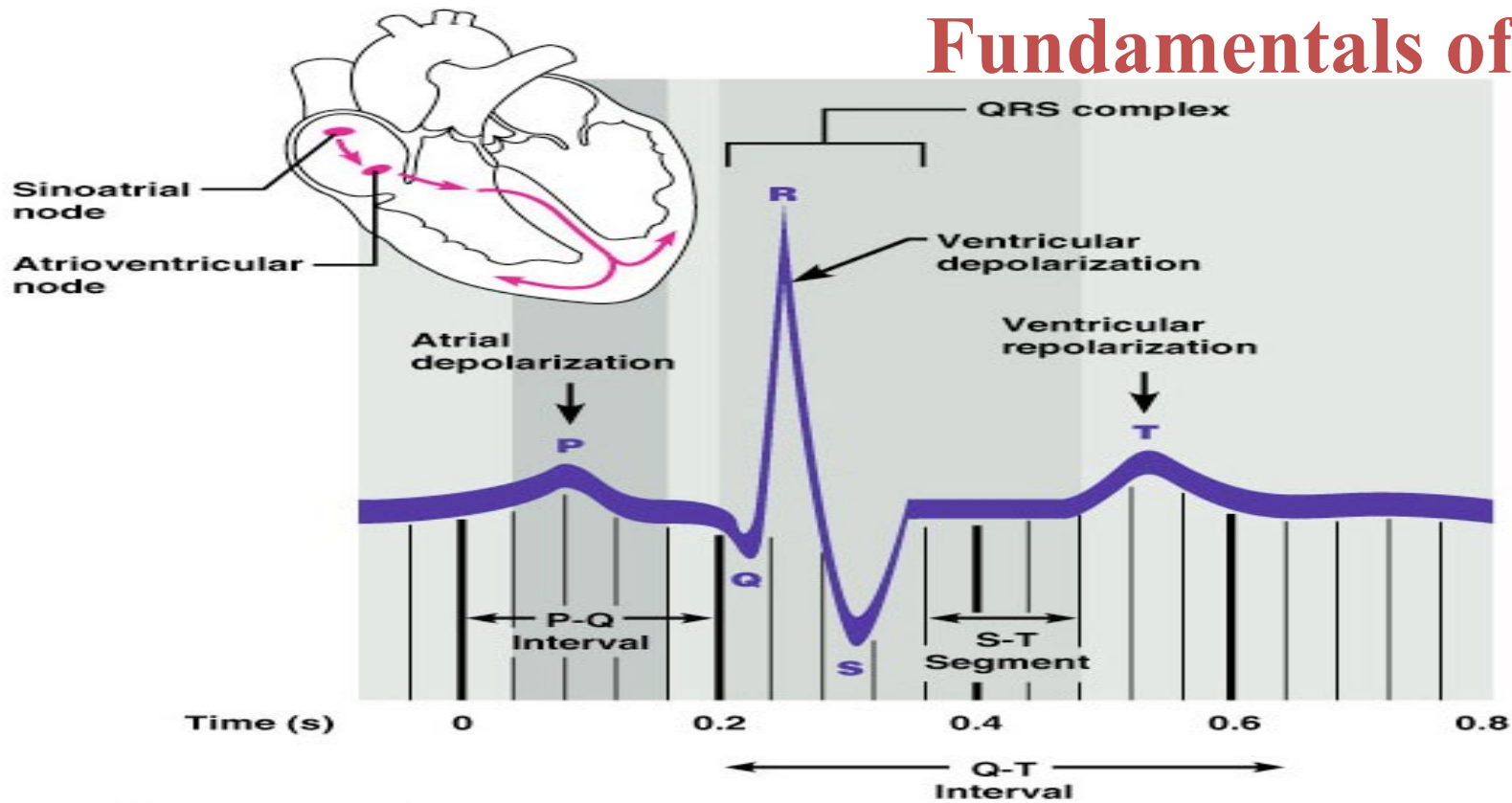
- It has the following sequences
Auto-rhythmicity → Membrane depolarization → T-tubule depolarization → Release of Ca^{2+} from SR + from T-tubule (ECF) → Ca^{2+} activates contractile molecules → Sliding of filaments → Contraction
- The action potential of cardiac muscle has a **prolonged refractory period**.
- Advantage: prevents tetanic contraction of the heart

What major differences did you observe between excitation-contraction coupling process between cardiac and skeletal muscle cells??



- 1 Action potential enters from adjacent cell.
- 2 Voltage-gated Ca^{2+} channels open. Ca^{2+} enters cell.
- 3 Ca^{2+} induces Ca^{2+} release through ryanodine receptor-channels (RyR).
- 4 Local release causes Ca^{2+} spark.
- 5 Summed Ca^{2+} sparks create a Ca^{2+} signal.
- 6 Ca^{2+} ions bind to troponin to initiate contraction.
- 7 Relaxation occurs when Ca^{2+} unbinds from troponin.
- 8 Ca^{2+} is pumped back into the sarcoplasmic reticulum for storage.
- 9 Ca^{2+} is exchanged with Na^{+} by the NCX antiporter.
- 10 Na^{+} gradient is maintained by the Na^{+} - K^{+} -ATPase.

Fundamentals of Electrocardiography



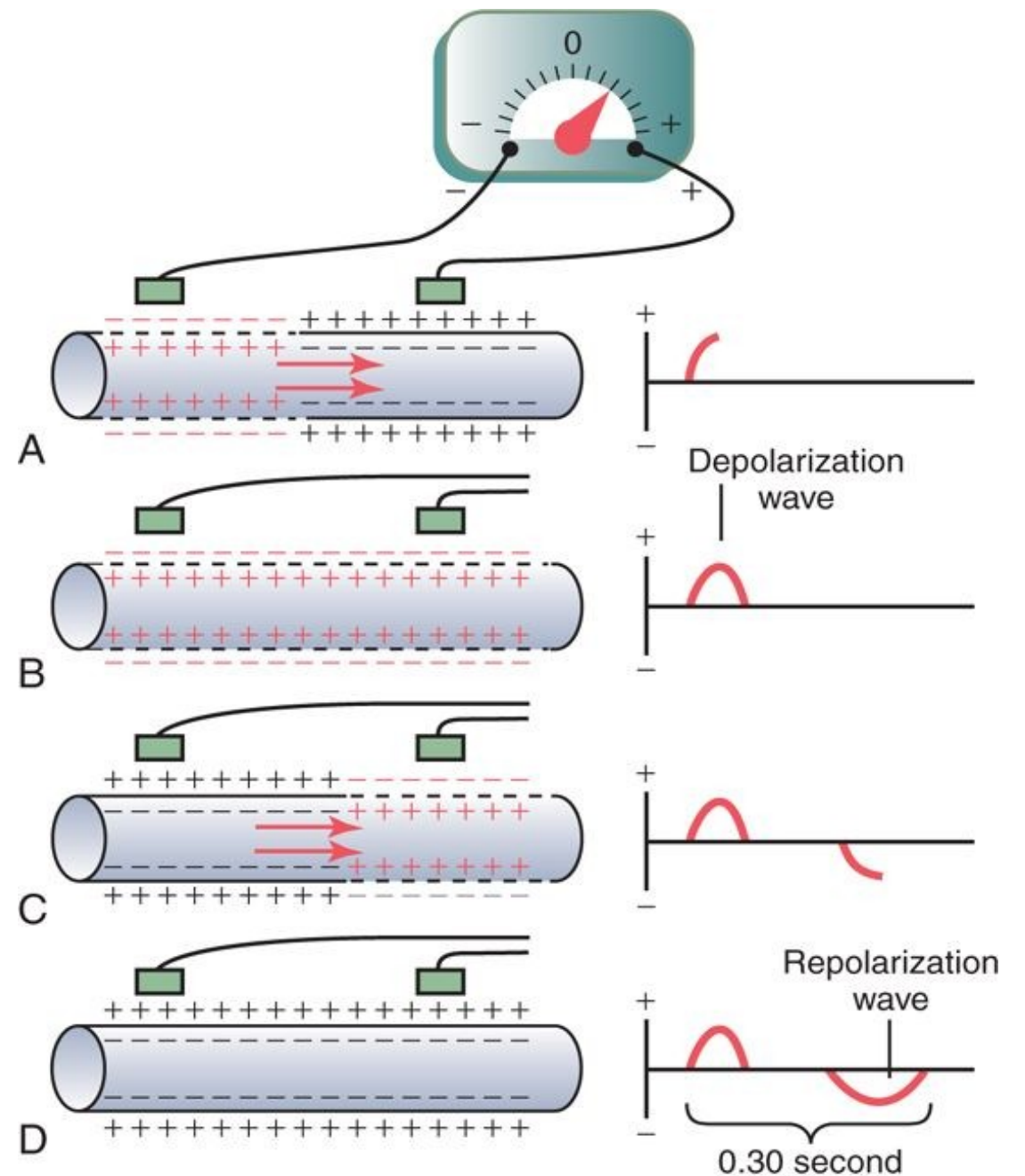
Fundamentals of Electrocardiography

- ECG is the electrical activity of the heart that is conducted through the body fluid and recorded from the surface of the body using highly sensitive electrodes.
- Recorded from specific sites of the body in graphic form (Fig. below) relating voltage (vertical axis) with time (horizontal axis).
- It is the sum of all the depolarization occurring in the cardiac muscle fibers during each cardiac cycle
- Developed by W. Einthoven in Leiden and A. Waller in London in 1909.

When the cardiac impulse passes through the heart, **electrical current** also spreads from the heart into the adjacent tissues surrounding the heart and to all the way to **the surface of the body**

The body is a good conductor of electricity because tissue fluids have a high concentration of ions that move (creating a current) in response to potential differences

If electrodes are placed on the skin on opposite sides of the heart, electrical potentials generated by the current can be recorded; the recording is known as an ***Electrocardiogram (ECG)***

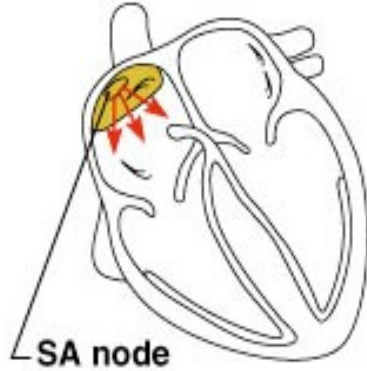


CHARACTERISTICS OF THE NORMAL ELECTROCARDIOGRAM

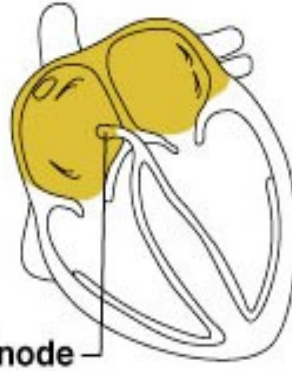
- The normal ECG is composed of a **P wave**, a **QRS complex**, and a **T wave**.
- The QRS complex is often, three separate waves: the Q wave, the R wave, and the S wave.
- The *P wave* is caused by electrical potentials generated when the atria depolarize before atrial contraction begins.
- The *QRS complex* is caused by potentials generated when the ventricles depolarize before contraction, that is, as the depolarization wave spreads through the ventricles.
- Therefore, both the P wave and the components of the QRS complex are *depolarization waves*.
- The *T wave* is caused by potentials generated as the ventricles recover from the state of depolarization.
- The T wave is known as a *repolarization wave*.
- Thus, the ECG is composed of both depolarization and repolarization waves

Normal ECG...

SA node generates impulse; atrial excitation begins



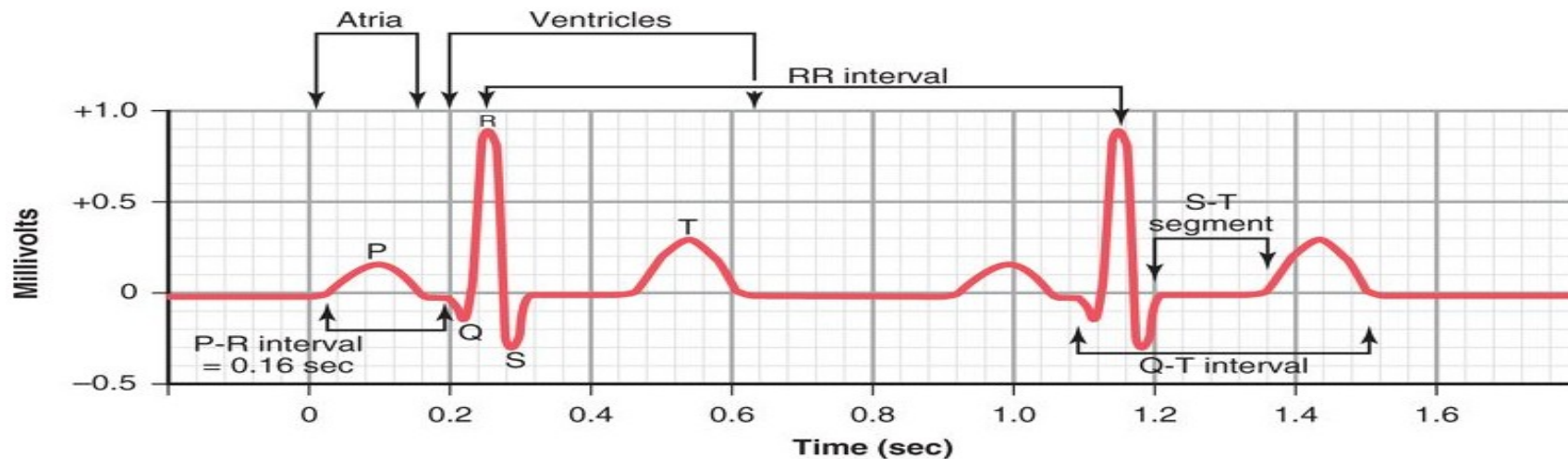
Impulse delayed at AV node



Impulse passes to heart apex; ventricular excitation begins



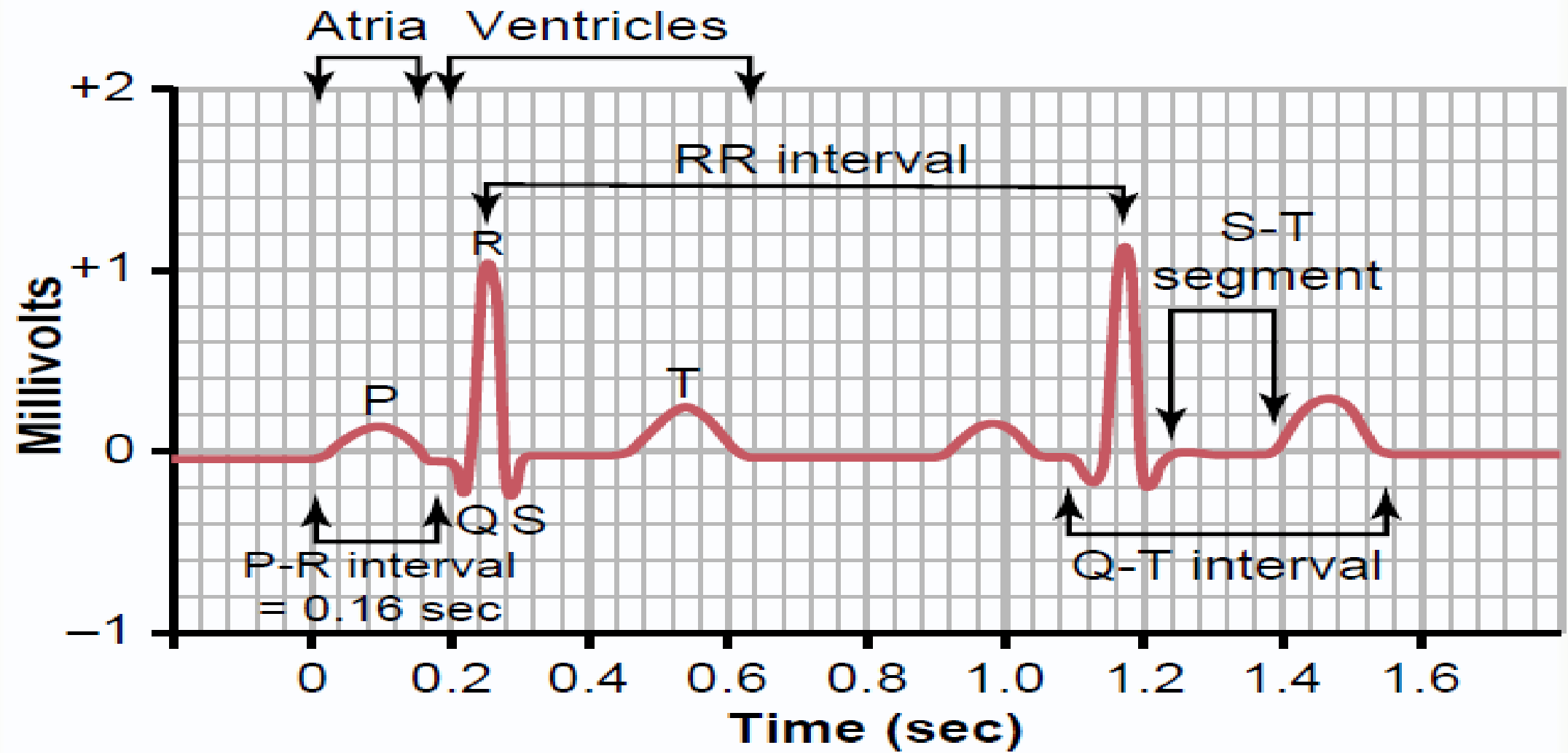
Ventricular excitation complete



Electrocardiogram cont'd

- *PR interval* represents the time between the onset of atrial depolarization and the onset of ventricular depolarization (normally ranges from 0.12 to 0.20 seconds)
- *ST segment* is the period at which the entire ventricle is depolarized and roughly corresponds to the plateau phase of the ventricular action potential
 - ✓ The ST segment is important in the diagnosis of *ventricular ischemia*, in which the ST segment can become *either depressed or elevated*
- *QT interval* is duration of both ventricular depolarization and repolarization, roughly estimates the duration of ventricular action potentials (range from 0.2 to 0.4 seconds depending on heart rate)

Electrocardiogram cont'd



Electrocardiogram cont'd

TABLE 2-4 SUMMARY OF ECG WAVES, INTERVALS, AND SEGMENTS

ECG COMPONENT	REPRESENTS	NORMAL DURATION (S)
P wave	Atrial depolarization	0.08–0.10
QRS complex	Ventricular depolarization	0.06–0.10
T wave	Ventricular repolarization	¹
PR interval	Atrial depolarization plus AV nodal delay	0.12–0.20
ST segment	Isoelectric period of depolarized ventricles	¹
QT interval	Length of depolarization plus repolarization— corresponds to action potential duration	0.20–0.40 ²

¹Duration not normally measured.

²High heart rates reduce the action potential duration and therefore the QT interval.

Electrocardiogram cont'd

- ✓ a wave of depolarization (instantaneous mean electrical vector) traveling toward a positive electrode results in a positive deflection in the ECG trace
- ✓ a wave of depolarization traveling away from a positive electrode results in a negative deflection
- ✓ a wave of repolarization traveling toward a positive electrode results in a negative deflection
- ✓ a wave of repolarization traveling away from a positive electrode results in a positive deflection
- ✓ a wave of depolarization or repolarization oriented perpendicular to an electrode axis produces no *net deflection*
- ✓ the instantaneous amplitude of the measured potentials depends upon the orientation of the positive electrode relative to the mean electrical vector
- ✓ voltage amplitude (positive or negative) is directly related to the mass of tissue undergoing depolarization or repolarization

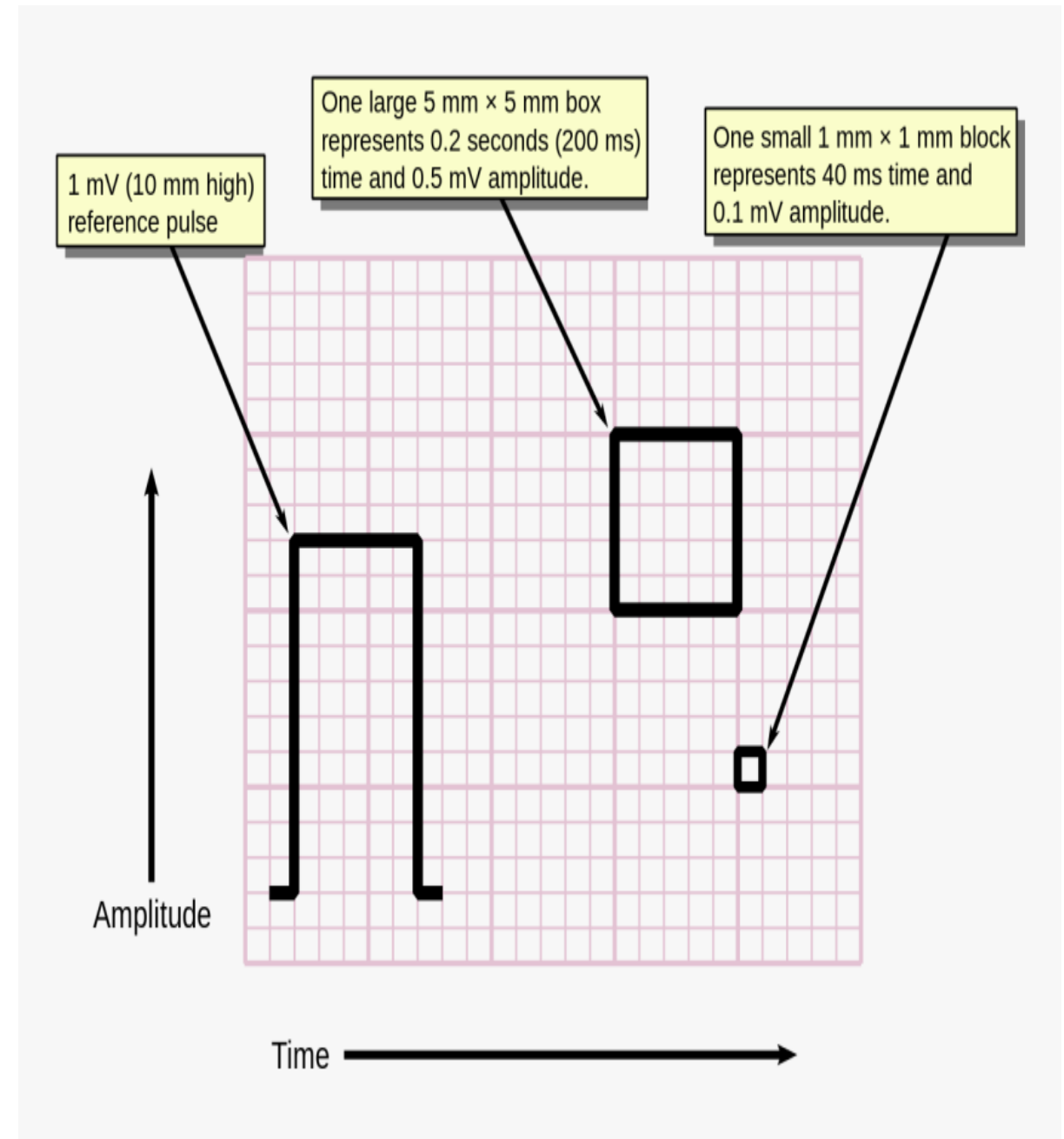
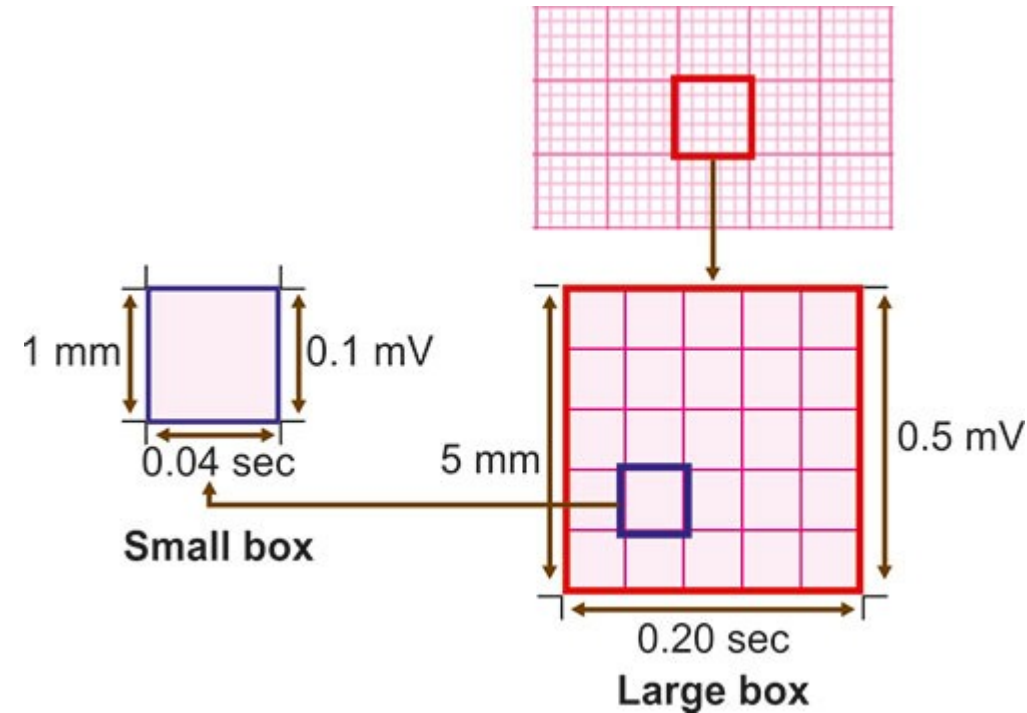
ECG Paper Conventions

Horizontal Axis: Time and Duration

- The standard paper speed is **25 mm/second**.
- **Small Square (1 mm)**: Represents 0.04 seconds (40 ms).
- **Large Square (5 mm)**: Represents 0.20 seconds (200 ms).
- **1 Second**: Equal to 5 large squares (25mm).

Vertical Axis: Voltage and Amplitude

- The paper is calibrated so that a 1-millivolt (mV) electrical signal creates a 10 mm vertical deflection.
- **Small Square (1 mm)**: Represents 0.1 mV
- **Large Square (5 mm)**: Represents 0.5 mV
- **1.0 mV**: Equal to 10 mm (2 large squares).



ECG Recordings

- There are 3 types of recording:
 1. Bipolar Limb Leads
 2. Augmented (Unipolar) Limb Leads
 3. Precordial (Chest) Leads

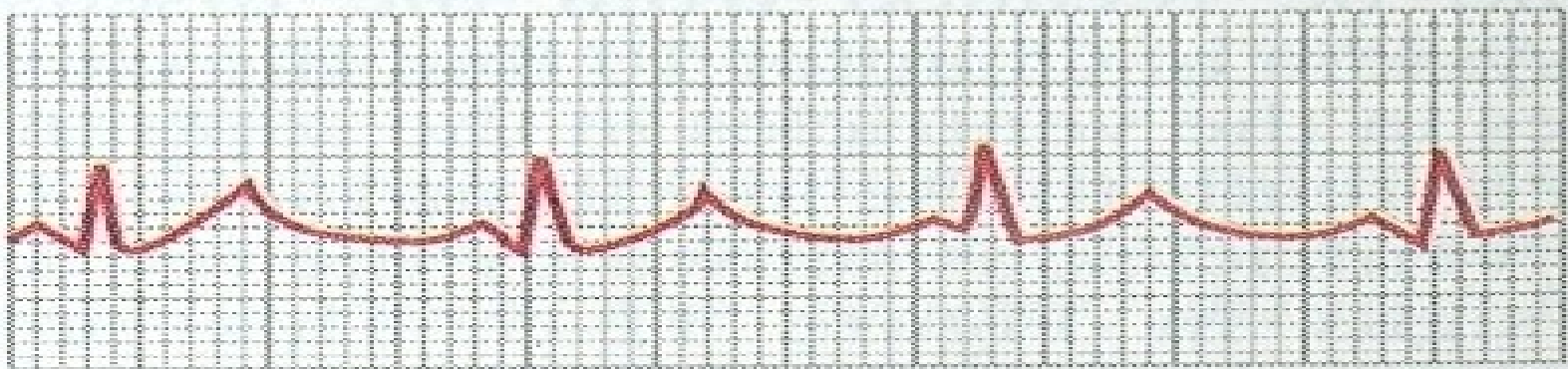
A. Bipolar Limb Leads

- Record voltage b/n two electrodes (leads) placed on the wrists and legs.
- These leads include:
 1. **Lead I= LA-RA**: electric potential d/ce b/n Lft arm & Rt arm
 2. **Lead II=LL-RA**: electric potential d/ce b/n Lft leg & Rt arm
 3. **Lead III= LL-LA**: electric potential d/ce b/n Lft arm & Lft leg.

Table 13.9 Electrocardiograph (ECG) Leads

Name of Lead	Placement of Electrodes
<i>Bipolar limb leads</i>	
I	Right arm and left arm
II	Right arm and left leg
III	Left arm and left leg
<i>Unipolar limb leads</i>	
AVR	Right arm
AVL	Left arm
AVF	Left leg
<i>Unipolar chest leads</i>	
V ₁	4 th intercostal space to the right of the sternum
V ₂	4 th intercostal space to the left of the sternum
V ₃	5 th intercostal space to the left of the sternum
V ₄	5 th intercostal space in line with the middle of the clavicle (collarbone)
V ₅	5 th intercostal space to the left of V ₄
V ₆	5 th intercostal space in line with the middle of the axilla (underarm)

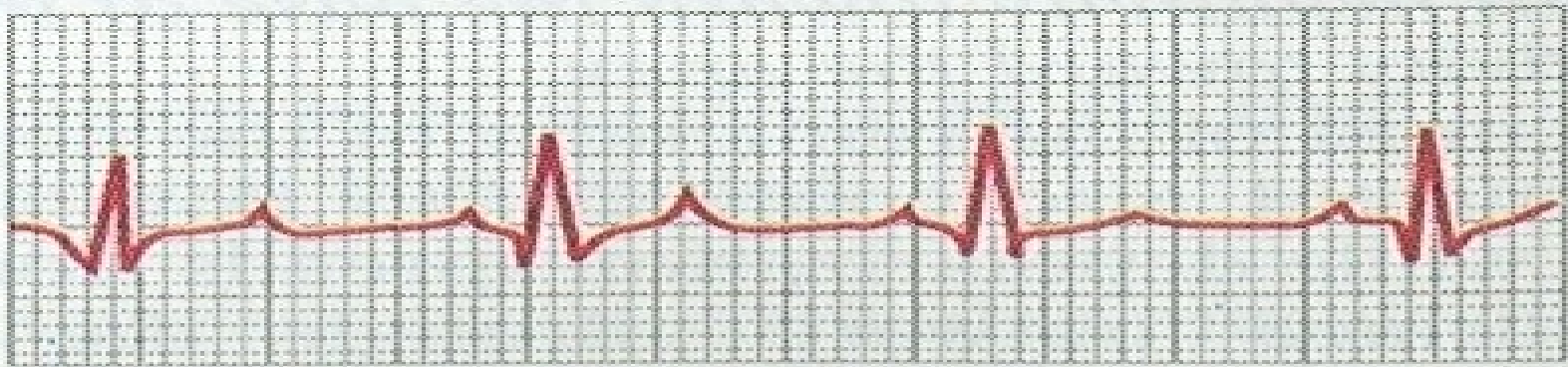
I



II



III



ECG Recordings...cont'd

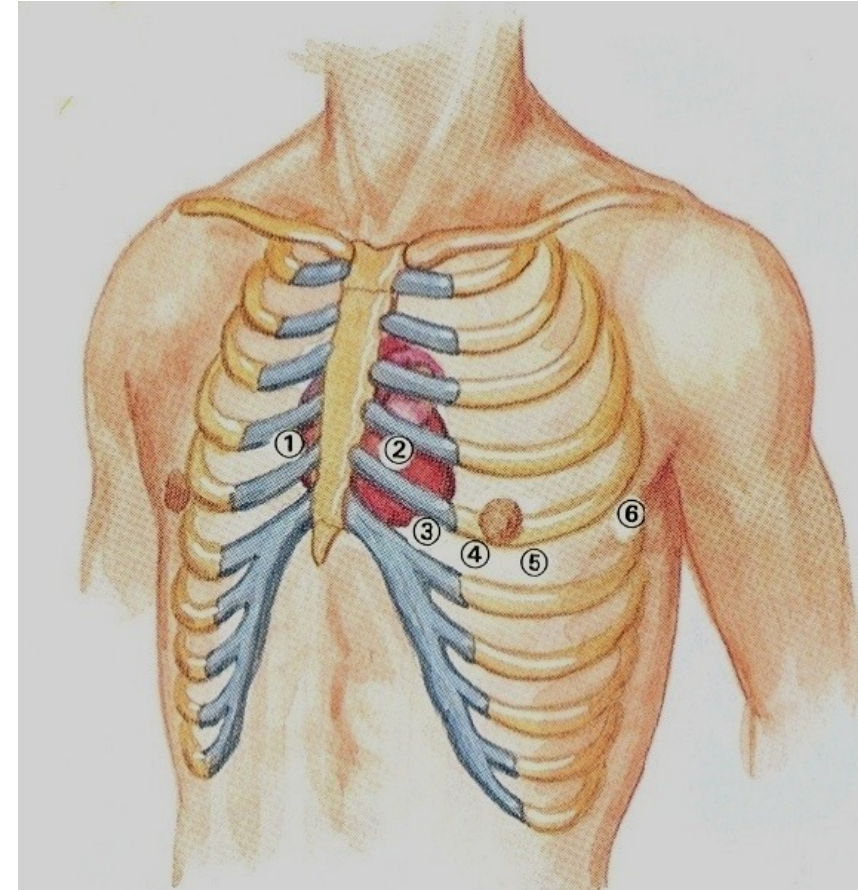
B. Unipolar (Augmented) Limb Leads

- Voltage is recorded b/n a single “exploratory electrode” placed on the body and an electrode that is built into the electrocardiograph and maintained at zero potential (ground).
- In this system, two of the limbs are connected through electrical resistances to the **negative (indifferent) terminal** of the electrocardiograph and the 3rd limb is connected to the **positive terminal**.
- When positive terminal is placed on right arm, the lead is designated as **aVR**; when on the left arm **aVL**, and when on left leg, **aVF** (Fig. below).

ECG Recordings...cont'd

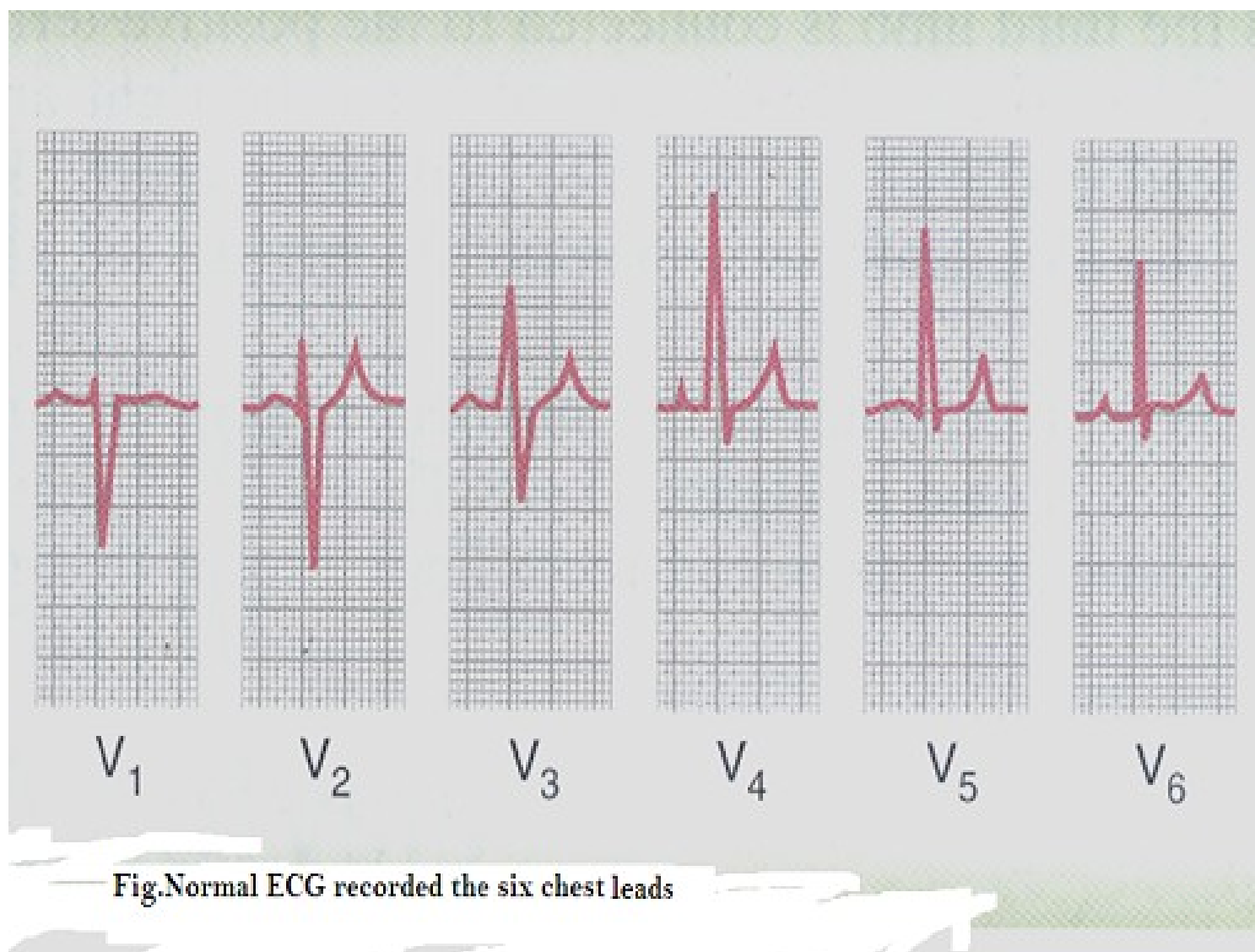
C. Precordial (Chest) Limb Leads

- Unipolar leads labeled V1 through V6.
- One electrode is connected to the **positive terminal** of the Electrocardiograph and the **negative electrode (indifferent electrode)** is connected through equal electrical resistances to the right arm, left arm, and left leg.
- Electrodes are placed on the chest.



...cont'd

- Heart surfaces are close to the chest wall and each chest lead records mainly the electrical potential of the cardiac muscle immediately beneath the electrode,
 - i.e., relatively minute abnormalities in the ventricles can cause marked changes in the ECG.
- QRS in
 - V1, V2, are negative because the chest electrodes are nearer the base of the heart (direction of electronegativity).
 - V3 is in b/n.
 - QRS of leads v4-v6 are positive because they are nearer the apex (direction of electro positivity).



Application of ECG

- By analyzing electric potential fluctuations, the physician can get some insight into:
 - anatomical orientation of the heart,
 - heart rate determination
 - relative size of heart chambers,
 - a variety of disturbances of rhythm and conduction
arrhythmia and conduction block
 - extent, location and progress of ischaemic damage
(myocardial infarction)
 - hypertrophy
 - pericarditis and myocarditis

Some Examples of ECG Abnormalities

A. Axis Deviation

- Occurs when the mean electrical axis is deviated either to the left ($<-30^\circ$) or to the right (>90 to 110°).

Causes:

- i. Change in the position of the heart
 - Left axis deviation: expiration, supine position, obesity
 - Right axis deviation: inspiration, upright position, tall and lanky stature, overload and hypertrophy of right ventricle.
- ii. Pathologic conditions: congenital heart disease, right ventricular hypertrophy, acquired disease of myocardium: ischaemic heart disease, degeneration of the conducting system, e.g. BBB.

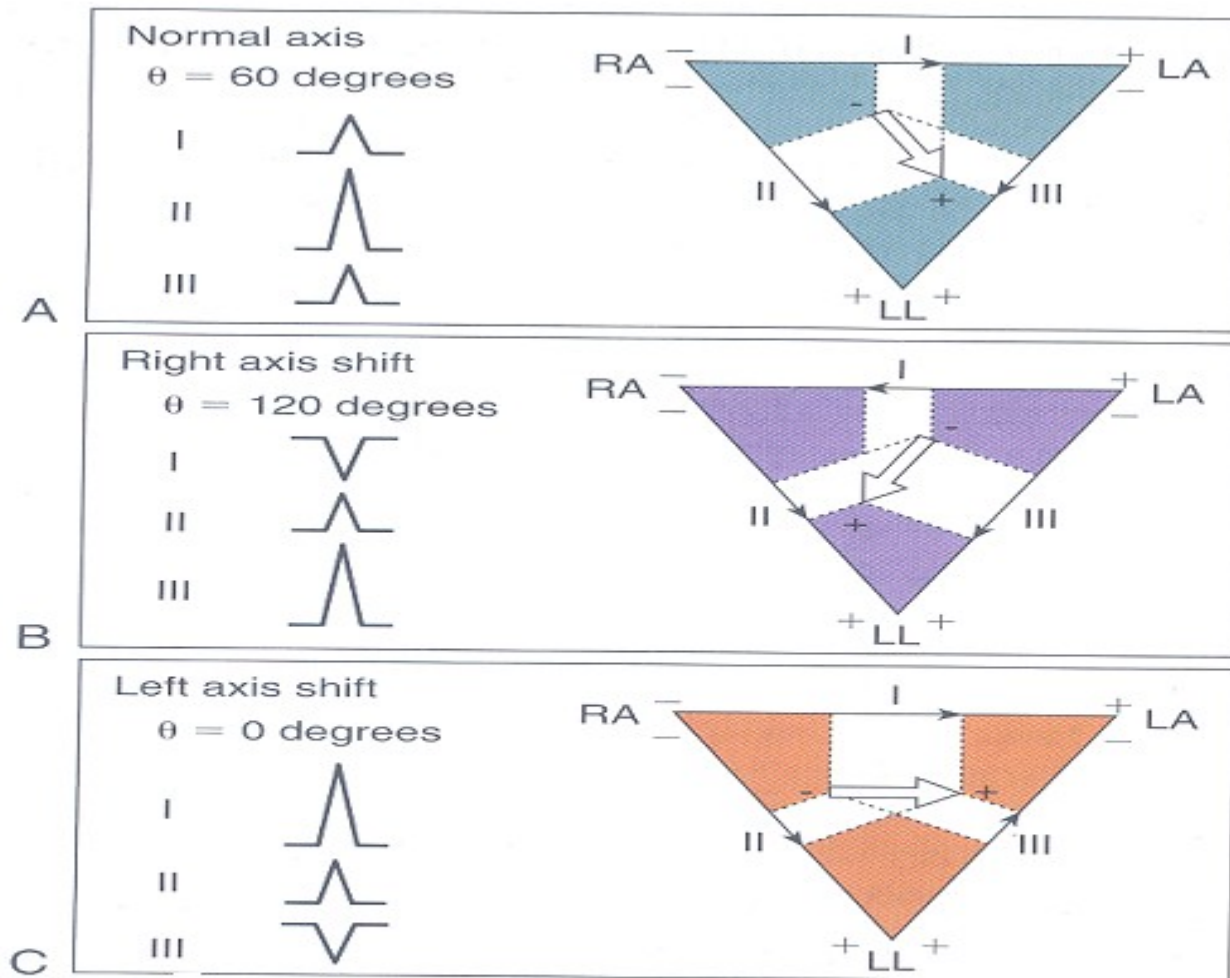


Fig.. Magnitude and direction of the QRS complexes in limb leads I, II, and III when the mean electrical axis (Q) is 60 degrees (A), 120 degrees (B), and 0 degrees (C).

Some Examples of ECG ...cont'd

B. Conduction Block

- This may occur anywhere between the S.A node and the ventricles.
- A block between the SA node and the AV node occurs very rarely. One of the more common sites of block is A-V node.
- Three degrees of heart block are usually distinguished.

a. First degree heart block

- PR interval prolonged, >0.21 sec, usually due to A-V nodal delay.
- May be caused by increased vagal tone and systemic diseases
- Has no effect on the pumping ability of the ventricles

Some Examples of ECG ...cont'd

b. Second Degree Heart Block

- Associated with organic heart disease
- Not all P waves are followed by QRS complexes.

1. Partial progressive AV block, normal waves.

- PR interval increases progressively leading to dropped beat (Wenckebach phenomenon, Mobitz type I, 2:1 block)

2. Constant block : PR interval relatively fixed; the atrio-ventricular rate is a constant small number ratio, e.g. 2:1, 3:1,

3. Periodic block, Mobitz type II, PR waves not fixed. Ratio of transmission: 6: 5, 8:7, etc.

Some Examples of ECG ...cont'd

c. Third Degree (complete) Heart Block

- Impulses completely interrupted between atria and ventricles

1. QRS complexes dissociated from P waves
2. Ventricular rate much slowed (35-40 beats/min)
3. Fainting often occurs due to cerebral ischaemia

Cause: organic heart disease is often the cause.

Significance:

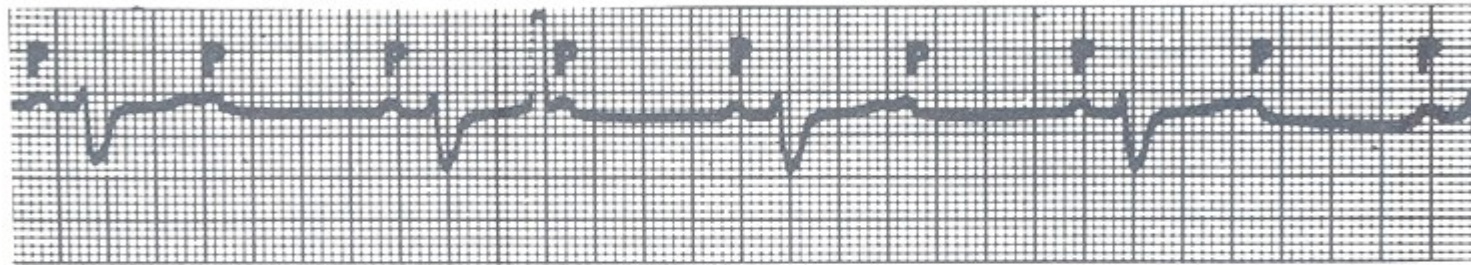
- ✓ May be associated with prolonged ventricular stand still.
- ✓ Cardiac arrest, if prolonged, can lead to cerebral ischaemia, syncope, or death.
- ✓ This condition is termed **Stokes-Adams Syndrome**.

A



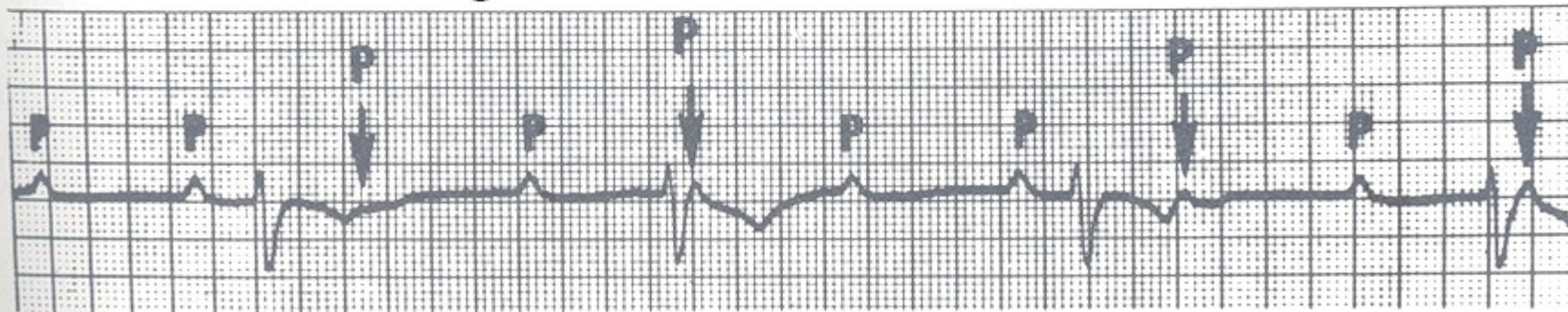
1st degree heart block

B



2nd degree heart block, 2:1

C



3rd degree heart block

Some Examples of ECG ...cont'd

- d. Wolf-Parkinson-White syndrome** (Ventricular pre-excitation or accelerated conduction)
 - results from aberrant conduction pathway b/n atria and the ventricles.
 - It eliminates the normal delay of impulse transmission.
- d. Bundle branch block:**
 - blockage of the right or left common bundles or the left anterior or posterior fascicles.

Some Examples of ECG ...cont'd

C. Arrhythmias

- Impulse initiation or impulse propagation are disturbed.
- Disturbances of impulse initiation include those that arise from the SA node and those that originate from various ectopic foci.
- Conduction blocks and reentrant rhythms are the principal disturbances of impulse propagation.

Some Examples of ECG ...cont'd

C. Arrhythmias

Type	Arrhythmias of the Heart
1. Extra systole (premature beat)	• Abnormal contraction produced during diastole • Caused by an ectopic foci • It could be nodal, atrial, or ventricular extra systole
2. Paroxysmal atrial tachycardia (PAT)	• Atrial beat >160 beats/min • A complication of MI; broad and bizarre QRS complexes • Caused by ectopic focus
3. Paroxysmal ventricular tachycardia (PVT)	• Ventricular beat =200 beats/min, • Caused by ectopic focus

4. Atrial flutter	• Fast and regular atrial beats (200-400 b/min). • Different parts contracting • Caused by re-entry, a wave of excitation propagating continuously within a closed circuit- a ring-like pathway around ends of vena cavae; 2:3, 3:1 block
5. Atrial fibrillation	• Fast and irregular uncoordinated atrial beats (400-600 b/min), dissociated from QRS • Caused by multiple re-entry. • Small and numerous circuits, “F” waves in ECG. • No p- wave. • May be caused by: mitral valve dx, coronary atherosclerosis, thyrotoxicosis, etc.
6. Ventricular fibrillation	• Fast and irregular uncoordinated QRS and ventricular beats (400-600b/min) • Caused by multiple re-entry • May lead to cardiac stand still • Circulation stops • May be caused by electrocution • May lead to death if cardiac massage is not performed

Some Examples of ECG ...cont'd

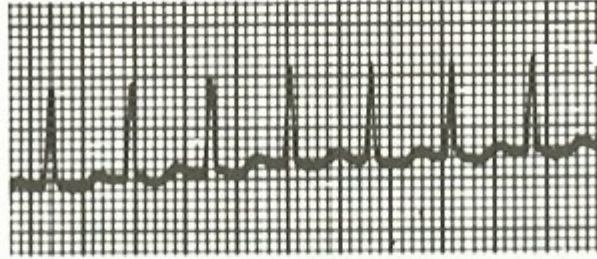
D. Voltage Abnormalities

- May be determined by making use of the convention for voltage calibration
 - i. Increased voltage
 - If **QRS** of 3 standard leads is greater than 4mV, then the heart is said to be hypertrophic.
 - ii. Prolonged duration of QRS is caused by:
 - Delayed conduction of impulses
 - Block of Purkinje system
 - iii. Spiked T-wave caused by hyperkalaemia
 - iv. Decreased voltages are caused by:
 - Muscle abnormalities-like myocardial infarction
 - Conduction problem like 1st to 3rd degree heart block
 - Ant. rotation of cardiac apex.

Normal		
Digoxin therapy		• ST "scooped" depression • Mild PR prolongation
Hyperkalemia		• Tall "peaked" T-wave
Severe hyperkalemia		• Flattened P • Widened QRS
Hypokalemia		• ST depression, flattened T • Prominent U wave
Hypercalcemia		• Shortened QT interval
Hypocalcemia		• Prolonged QT interval

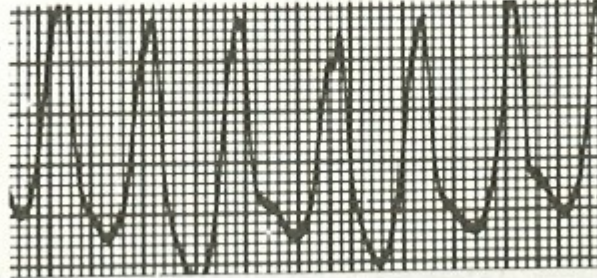
ST segment and T wave abnormalities caused by different conditions

A.

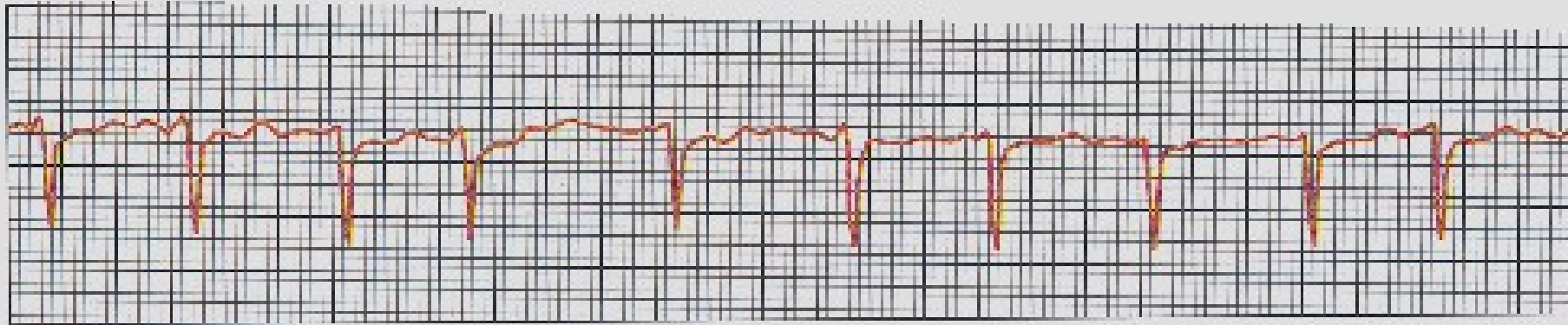


Paroxysmal atrial tachycardia

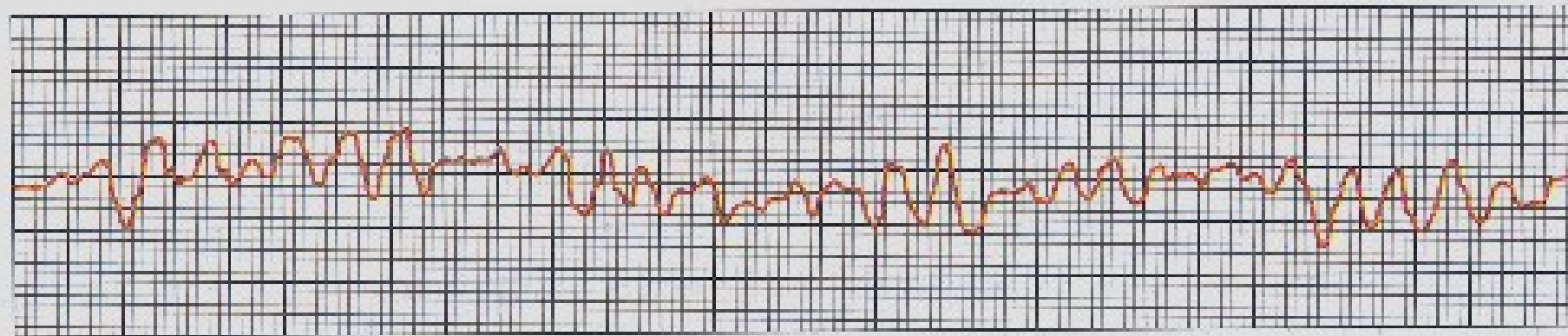
B.



Ventricular tachycardia



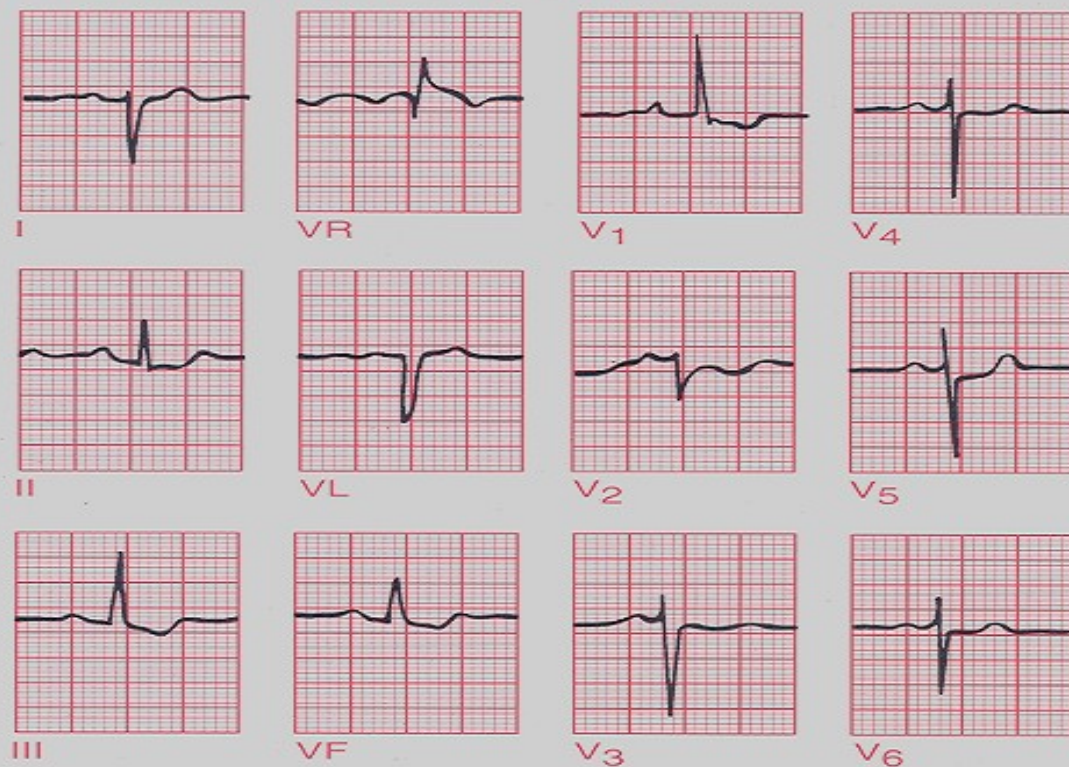
Atrial fibrillation



Ventricular fibrillation

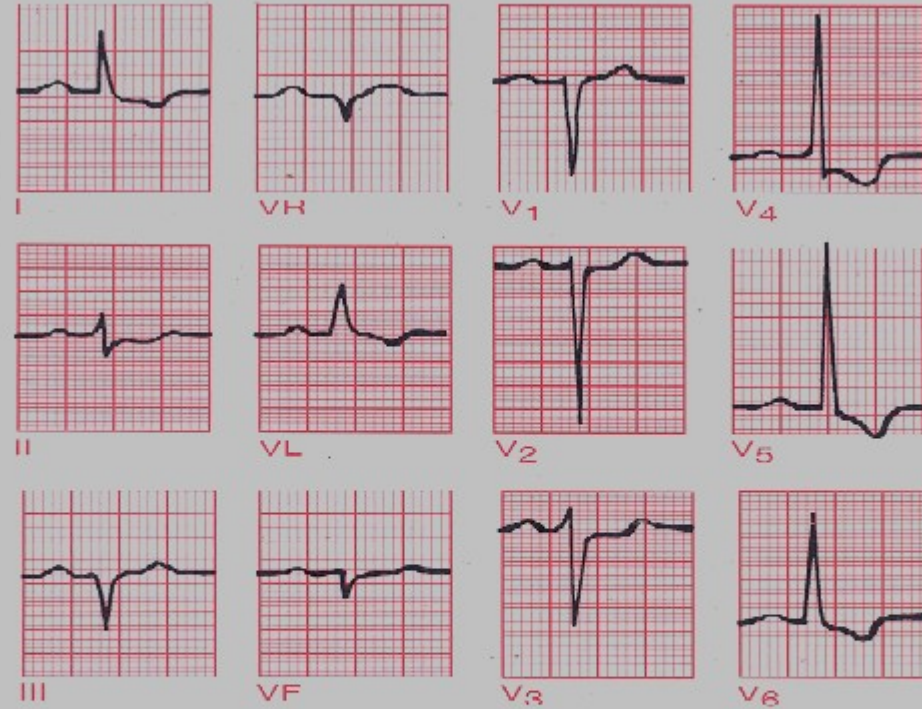
Fig..Fibrillation ECG

RIGHT VENTRICULAR HYPERTROPHY



Note: Sinus rhythm
Peaked P wave in V₁
Right axis deviation (deep S wave in lead I)
Tall R wave in V₁
Deep S wave in V₆
Clockwise rotation (no septal complex seen)
Inverted T waves in V₁ and V₂.

LEFT VENTRICULAR HYPERTROPHY



Note: Sinus rhythm

Normal axis (the R and S waves are equal in lead II, so the axis is at the limit of normality)

Height of R wave in V_5 is 35 mm

Deep S waves in V_1 and V_2

Inverted T waves in leads I, VL, V_1 – V_6 .

Some Examples of ECG ...cont'd

D. Myocardial Infarction

- A sudden, irreversible, ischaemic injury due to coronary arterial narrowing or occlusion with sustained damage to a portion of the myocardium.
- Associated with electrocardiographic, metabolic, and haemodynamic changes

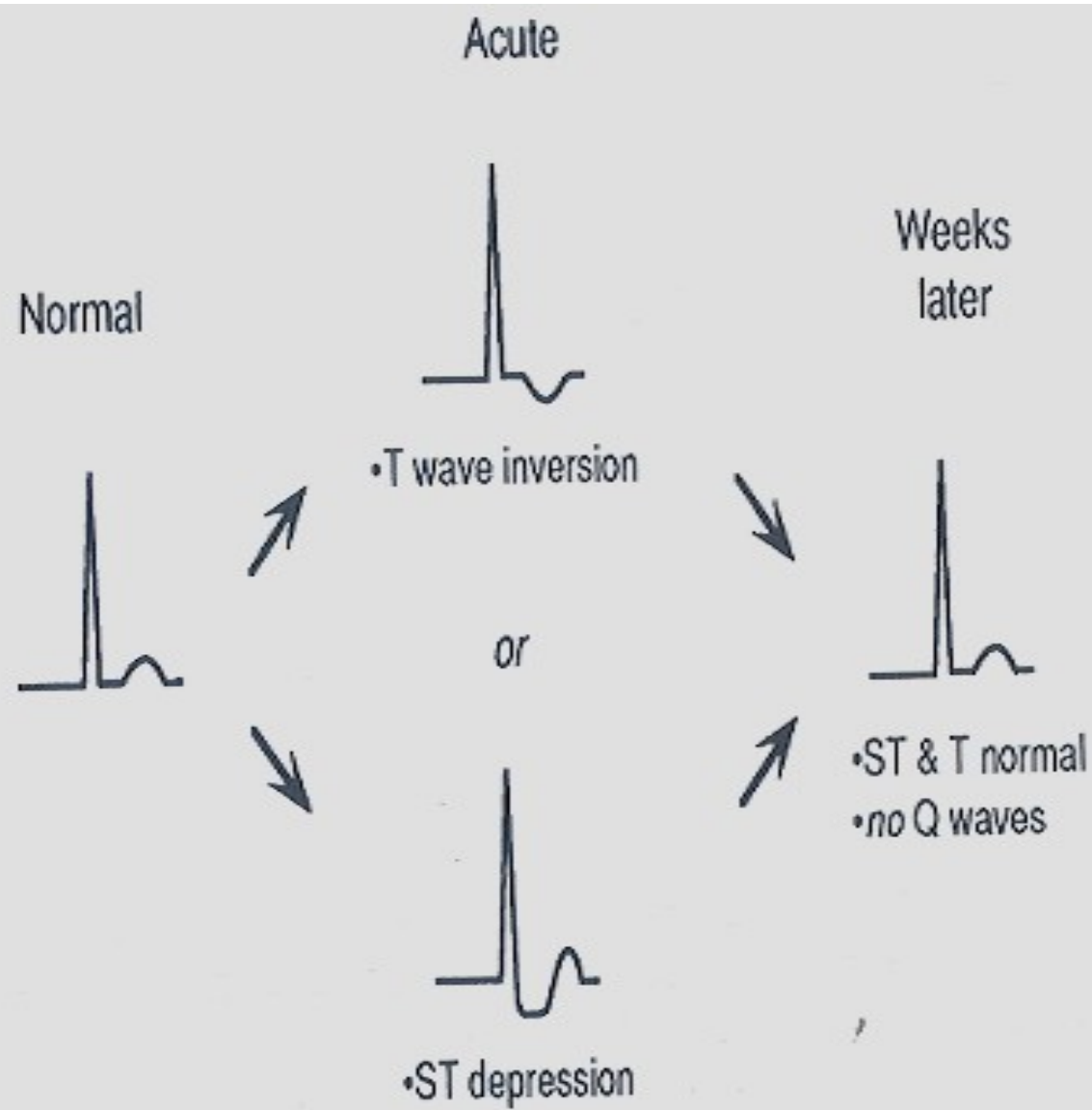


Fig.. ECG abnormalities in unstable angina and non-ST-elevation myocardial infarction.

ST-Elevation Myocardial Infarction

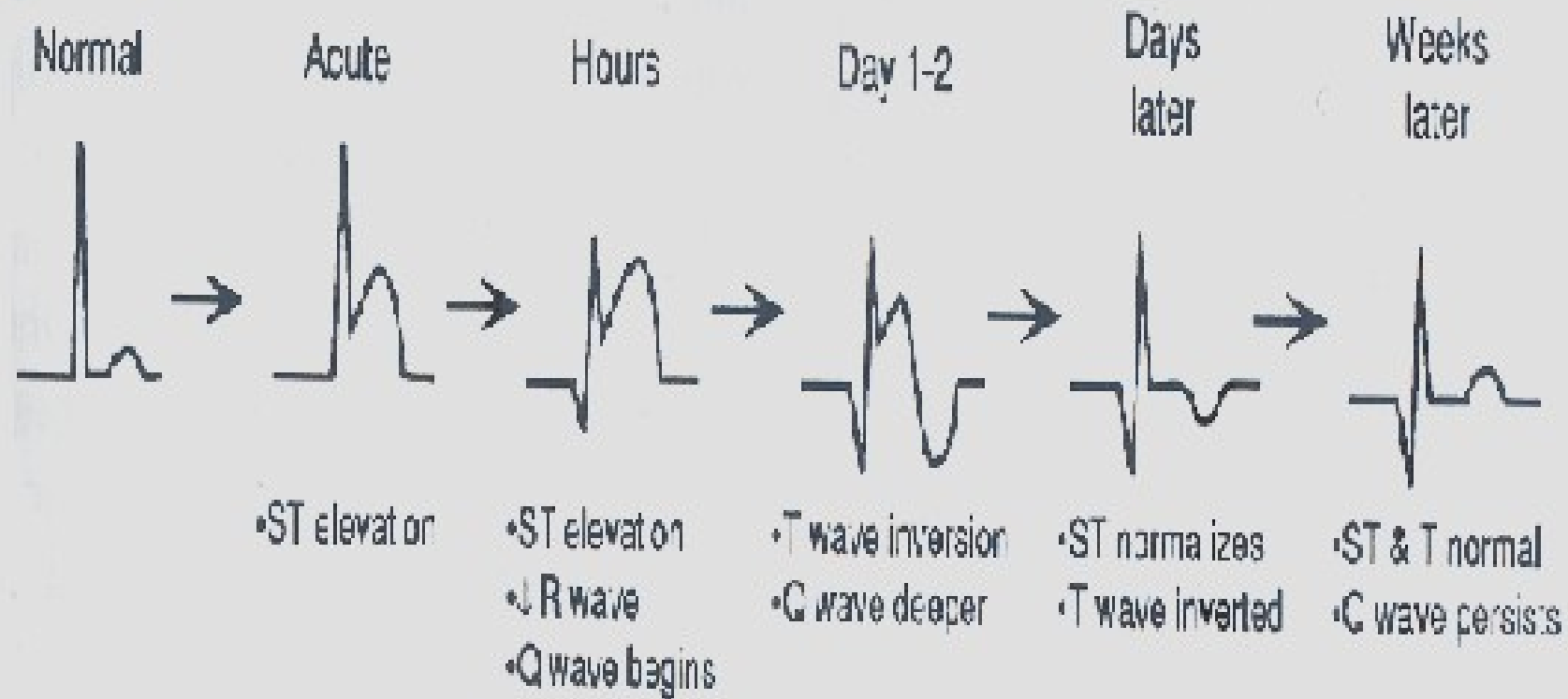
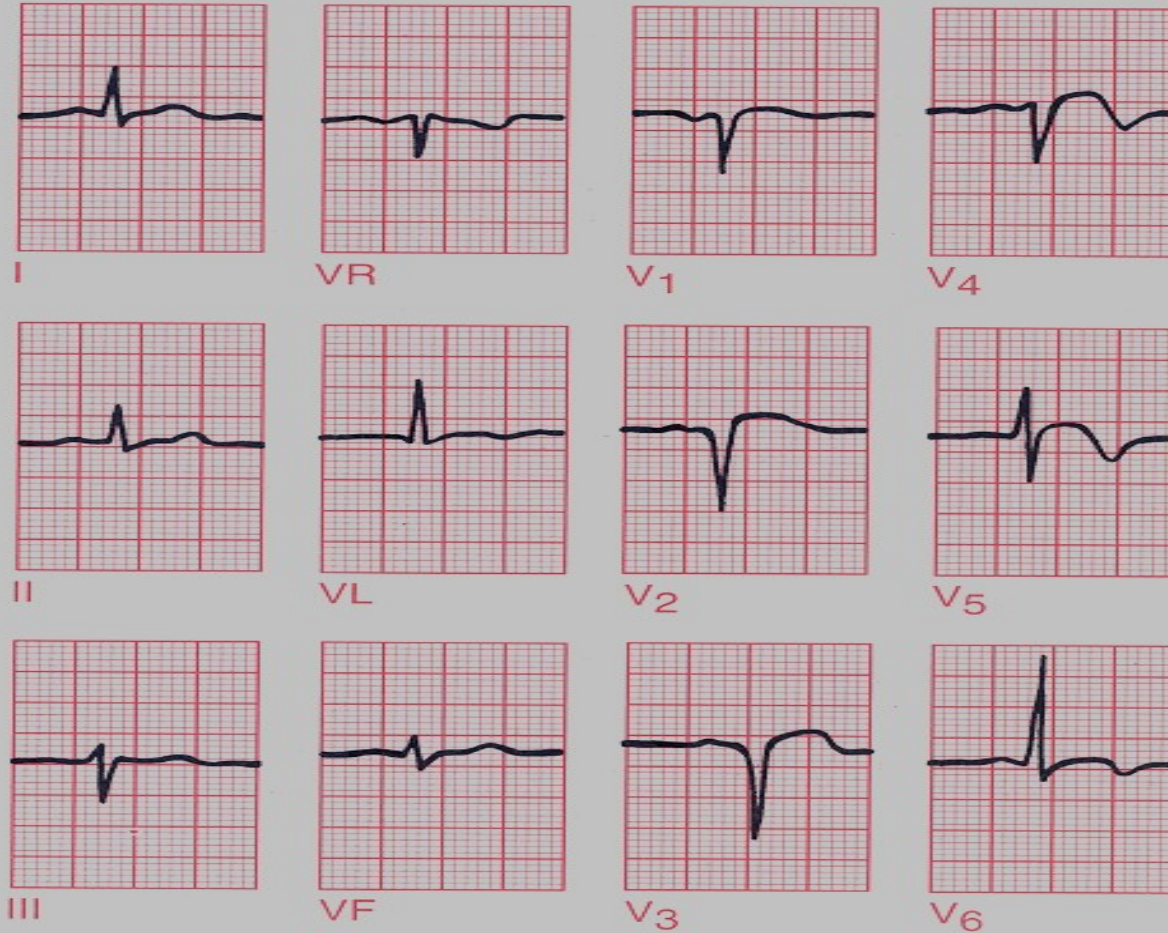


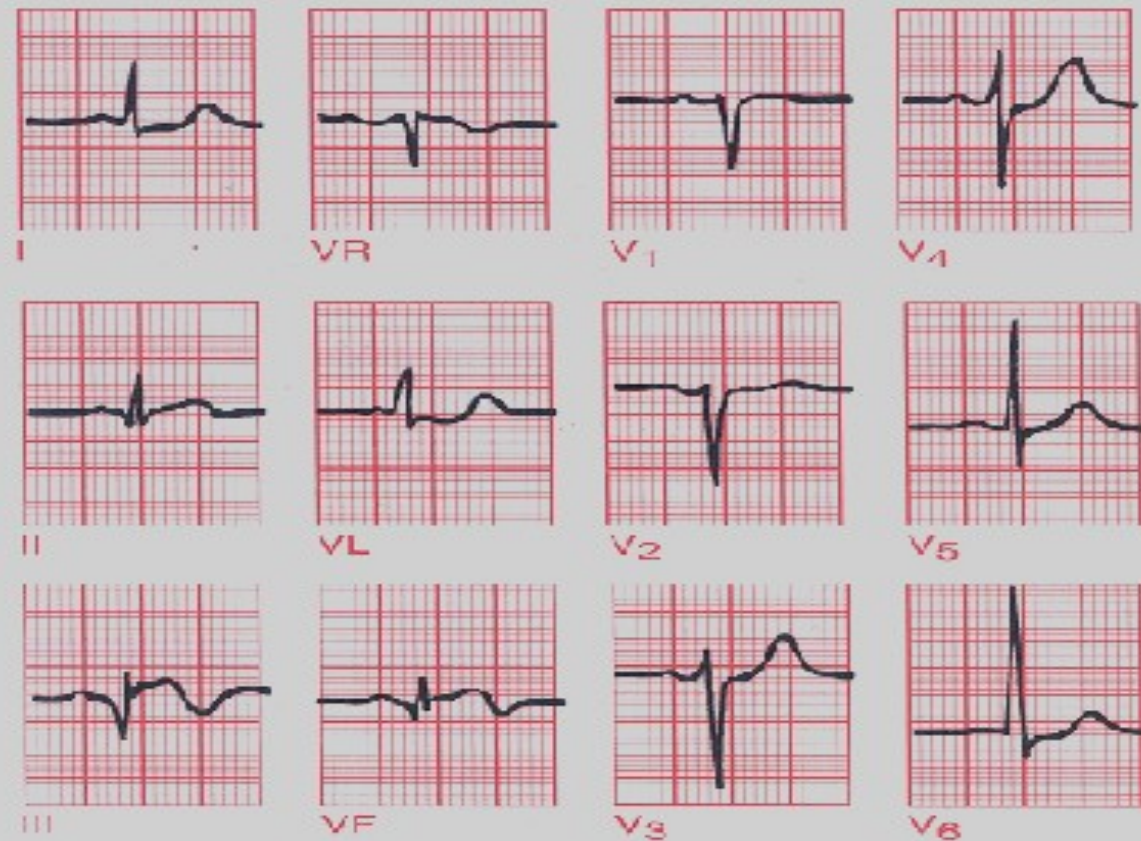
Fig... ECG evolution during ST-elevation myocardial infarction.

ANTERIOR INFARCTION

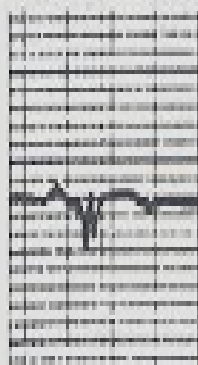


Note: Sinus rhythm
Normal axis
Q waves in leads V_2 and V_3
Inverted T waves in leads V_4 – V_6 .

INFERIOR INFARCTION



Note. Sinus rhythm
Normal axis
Q waves in leads III and VF
Raised ST segments in leads III and VF
Depressed (ischaemic) ST segments in leads VL and V4
Normal QRS complexes and T waves in the chest leads.



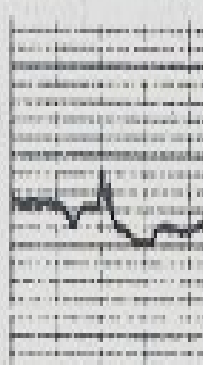
1



2



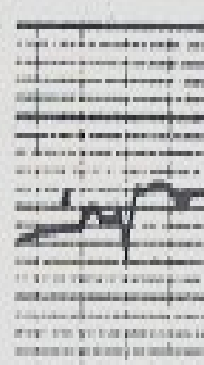
3



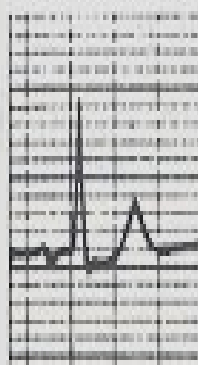
aVr



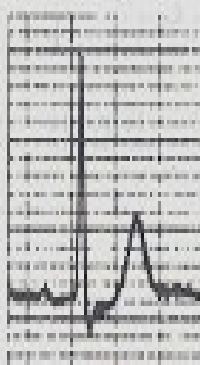
aVl



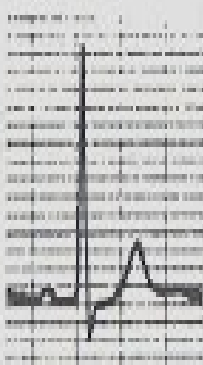
aVf



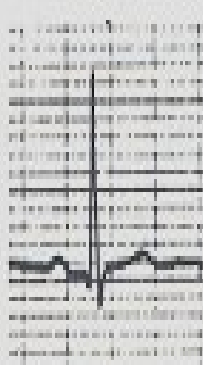
V1



V2



V3



V4



V5



V6

Established posterior myocardial infarction.

Some Examples of ECG ...cont'd

D. Congestive Heart Failure

- A Heart is unable to pump blood at a rate commensurate with systemic metabolic requirements.
- Causes may be **cardiac or extra cardiac**.
 - i. **Cardiac** causes may be:
 - Myocardial contractile failure due to **ischaemic heart dx, myocarditis or cardiomyopathy**
 - A disorder which prevents proper filling or emptying of heart such as **valvular stenosis or regurgitation or pericardial dx**.
 - ii. **Extra cardiac** causes
 - High pressure overloads such as **hypertension or high output overloads such as renal failure, anaemia, AV fistula or thyrotoxicosis**

Heart Rate

- ⊕ HR is the *number of cardiac cycles per minute*.
- ⊕ **Normal HR:** 60 - 100 beats/minute
 - If < 60 beats/minute = *bradycardia*, normal in the athletes
 - If > 100 beats/minute = *tachycardia*
- ⊕ **HR varies with the following factors**
 1. **Age:** - higher in new-born infants (120 beats/min)
 2. **Sex:** - higher in females (85 beats/min)
 3. **Time of the day:** - ↓ in morning, ↑ in evening.
 4. **Resting & sleep:** - ↓ the HR.
 5. **Physical training:** low in athletes (45-60 beats/min).
 6. **Body position:** - ↑ in standing & ↓ in supine positions.
- ⊕ ***How to count the HR?***
 - Counting arterial **pulsation, heart sound & ECG cycles**

Regulation of Heart Rate (HR)

1. Nervous factors
2. Physical factors, e.g. temperature
3. Mechanical factors, e.g. right atrial distension
4. Chemical factors, e.g. catecholamines

Regulation of HR, cont'd

HR is determined by *rate of discharge of impulse from SA-node*.

1. Factors that directly stimulate SA-node:

- ✦ Body temperature = $\uparrow$ HR
- ✦ Right-atrial distension $\rightarrow \uparrow$ Blood volume = $\uparrow$ HR
 - ✓ Right-atrial distension $\rightarrow$ Stretch receptors $\rightarrow$ Medullary cardiovascular centre $\rightarrow$ Sympathetic stimulation $\rightarrow \uparrow$ HR
- ✦ Catecholamines: AD, NAD = $\uparrow$ HR

2. Factors influencing SA-nodal discharge by stimulating the medullary cardiovascular centre.

- ✓ Stimulation of higher centre (cerebral cortex, HT)
- ✓ Chemical content of the blood ($\downarrow$ PO₂, $\uparrow$ PCO₂, $\uparrow$ H⁺)
- ✓ Cardiovascular reflexes (baroreceptor, chemoreceptor)
- ✓ Stress (pain, fear, excitement)

Cardiac cycle

- Activities in the heart in a single beat
- Contraction (systole) and relaxation (diastole) of cardiac chambers
- A single cardiac cycle comprised of
 - Atrial systole and atrial diastole
 - Ventricular diastole + ventricular systole
- 75 cycles completed per minute
- Duration of each cycle = 0.8 second
 - Ventricular diastole = 0.5 second
 - Ventricular systole = 0.3 second

Means of exploring activities accomplished in a cardiac cycle

1. Auscultation of heart sounds, phonograph
2. ECG tracing
3. Measuring aortic pressure
4. Measuring atrial pressure
5. Measuring ventricular pressure and volumes

Phases of the Cardiac Cycle

- Includes the following events:
 - **Electrical events** of summated ECG voltage changes-P wave, QRS complex,T wave and U wave in some cases
 - **Mechanical events** of myocardial diastole and systole with opening and closing of cardiac valves, pressure and volume changes, heart sounds, atrial (jugular) a,c,v, waves.
- Phases:
 - I. Atrial and Ventricular diastole
 - II. Phase of systole

Phases of ...cont'd

A. Phase of Diastole (Atrial and ventricular diastole):

- Atrial pressure rises before the tricuspid valve opens during diastole. This is mirrored by **v-wave**
- Atria and ventricles are relaxed.
- As pressure falls below that in the atrium, it leads to opening of AV valves and closing of semi-lunar valves
- Blood flows into the heart filling both atrial and ventricular chambers throughout diastole.
- Rapid filling, **proto-diastole** occurs for **0.04 sec.**
- When pulse rate is slow and diastole is long, inflow is slowed in the latter part of diastole→ a period of **diastasis.**

Phases of ...cont'd

- The 3rd heart sound begins with end of rapid inflow and ends with beginning of diastasis.
- Proto-diastole + diastasis = 70% of inflow
- About 70% of the ventricular filling occurs passively.
- The remaining 30% of filling occurs by active atrial systole
- P-wave formation begins with diastasis and ends with atrial systole.
- The beginning of the 1st heart sound and the end of QRS complex coincide with isovolumetric or isometric contraction (0.02-0.06 sec).
- 2nd heart sound occurs with closure of semilunar valves and during isometric relaxation (0.09-0.12 sec)

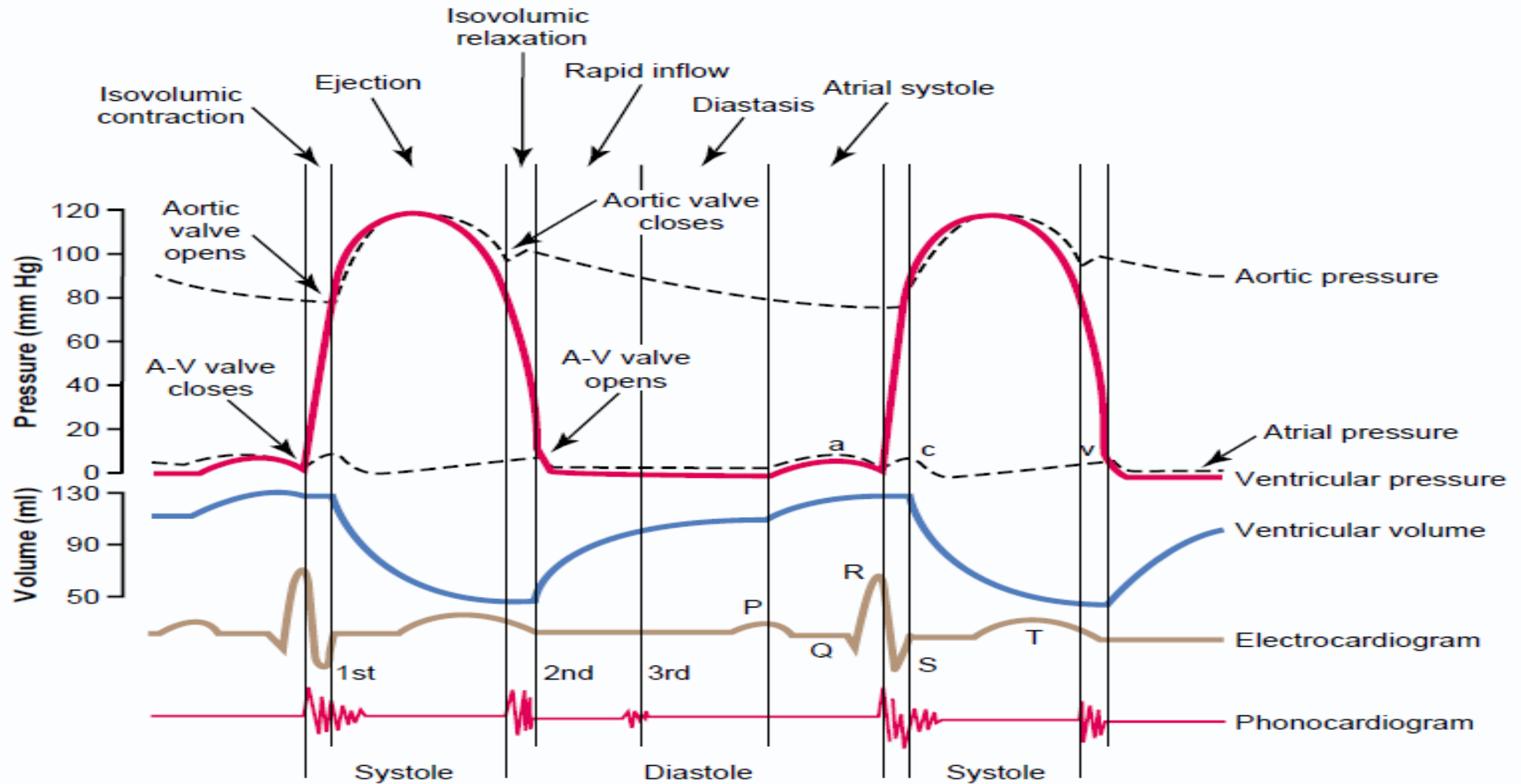


Figure Events of the cardiac cycle for left ventricular function (left atrial pressure, left ventricular pressure, aortic pressure, ventricular volume, the electrocardiogram, and the phonocardiogram)

Phases of ...cont'd

B. Phase of Systole

a. Atrial systole (contraction):

- Propels an additional blood of about 30% of EDV into ventricles
 - Preceded by P-wave
 - $EDV = 120 \text{ mL}$
 - Forms a-wave, formed by atrial systole. This is one of the three atrial pressure waves (a,c,v) transmitted to thoracic and neck veins (Fig.below)
- $\uparrow$ ventricular volume
 - $\uparrow$ ventricular pressure (atrial kick)

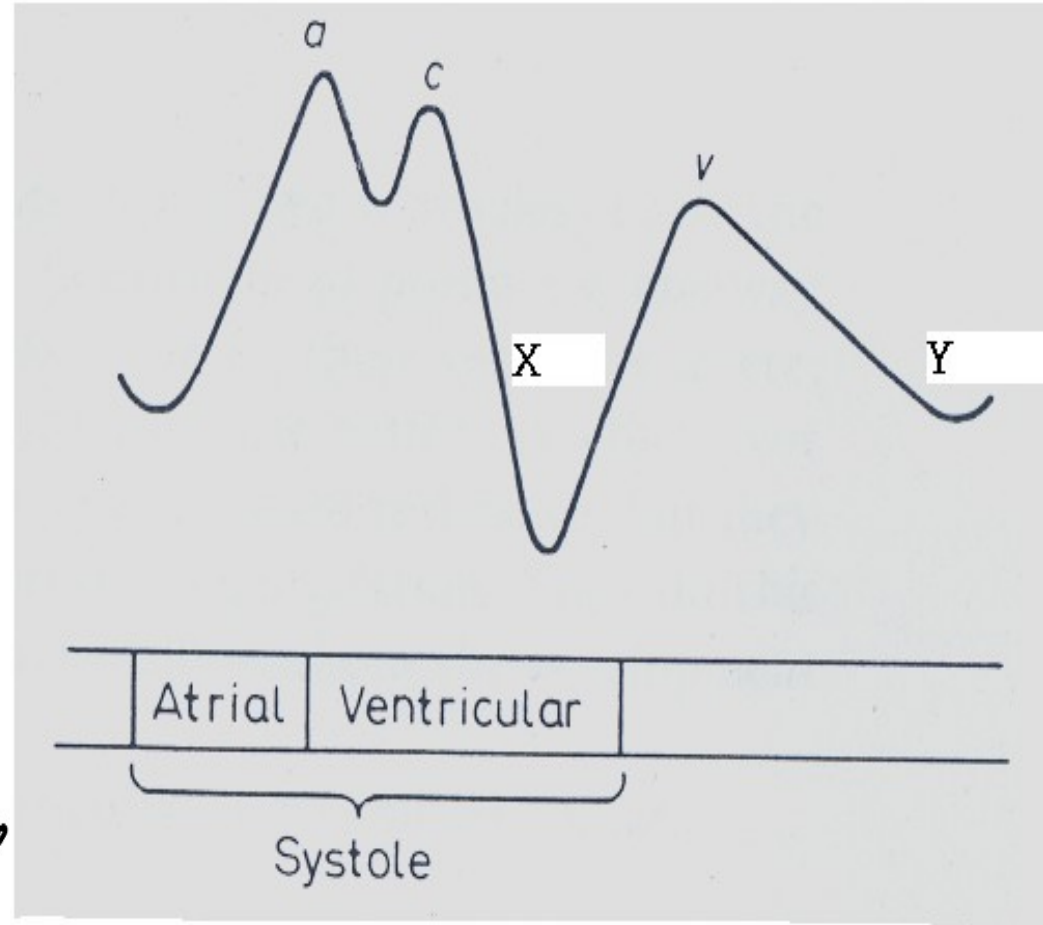
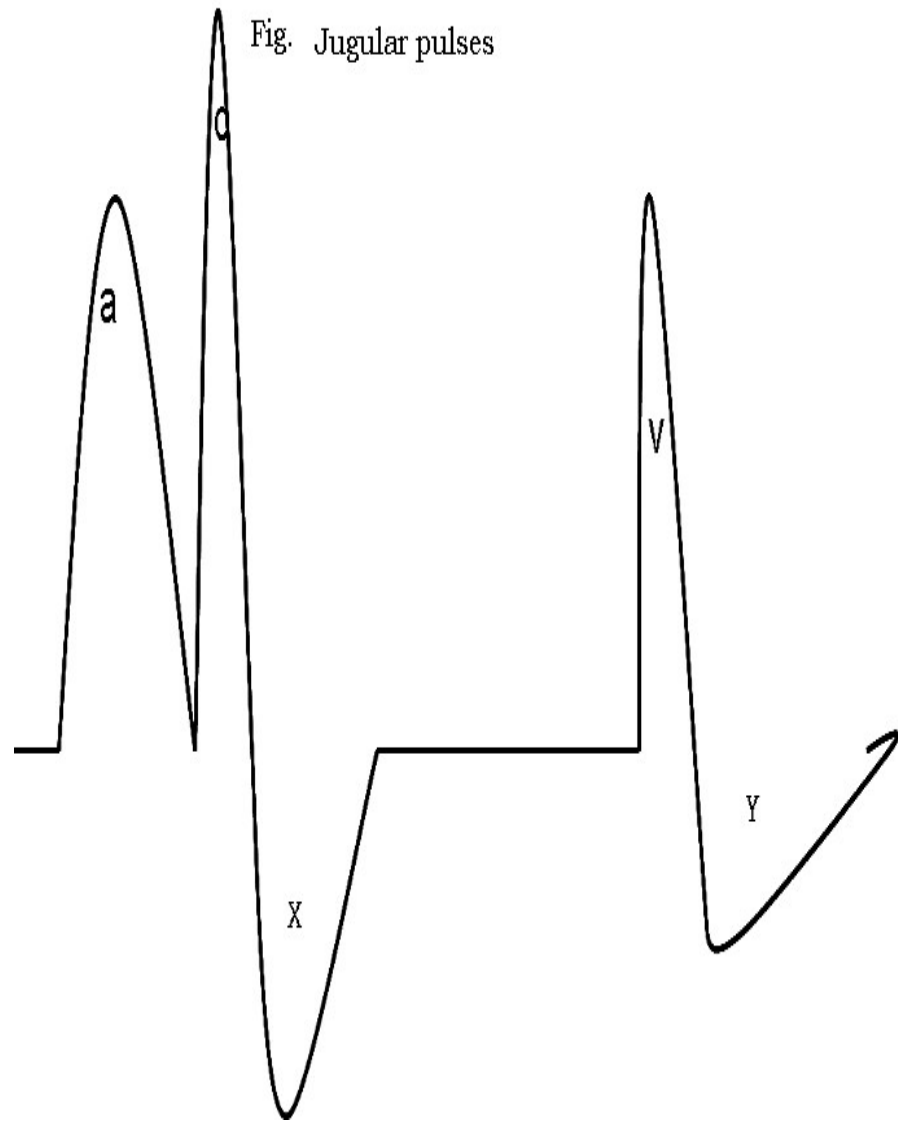


Fig..The Venous Pressure Pulse waves.

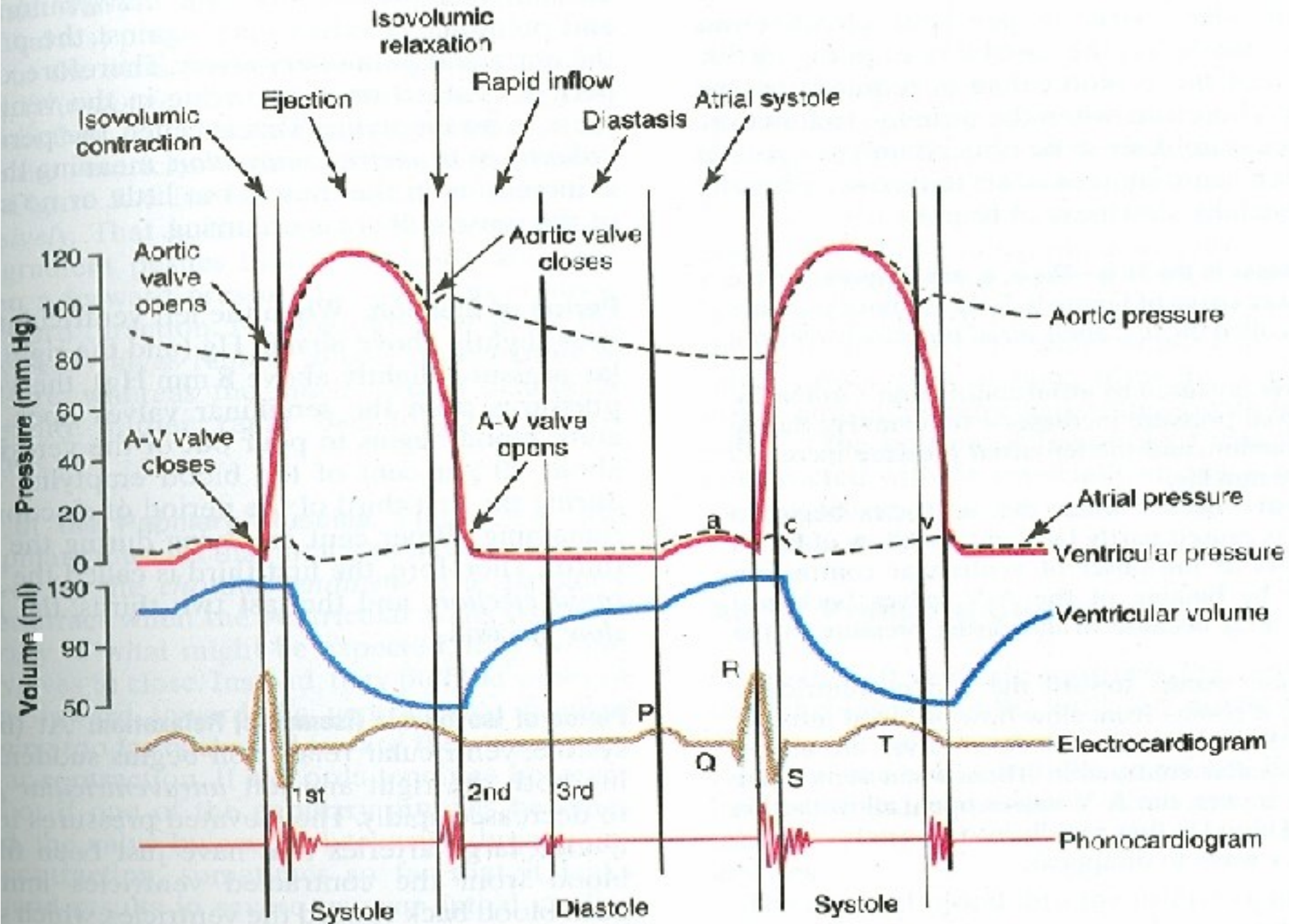
Phases of ...cont'd

b. Ventricular Systole

- During this period:
- Isovolumetric contraction (0.02-0.06 sec) occurs
- i.e, there will be:
 - No change in ventricular volume or length of ventricular muscle fibers
 - Rise of BP in left ventricle from 4 to 80 mmHg
 - Closed AV valves
 - Closed S-L valves
 - Aortic and pulmonary pressures are at their lowest levels

Phases of ...cont'd

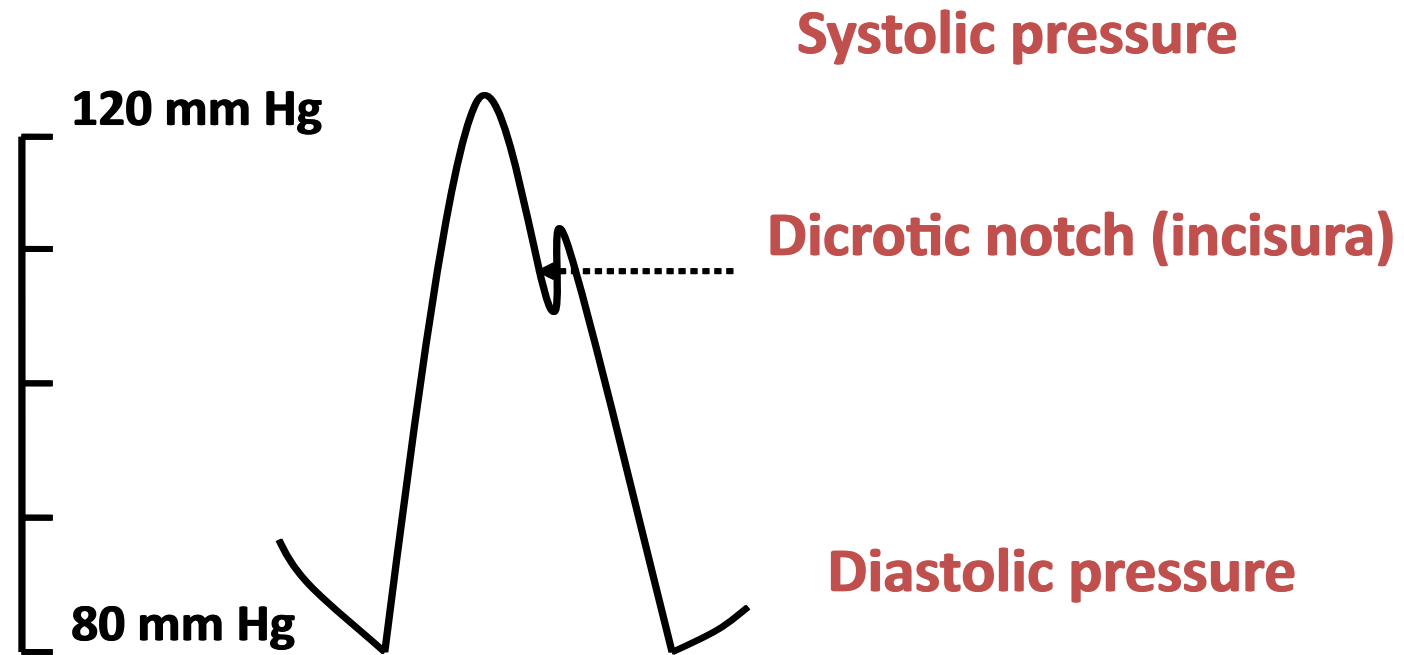
- **c-wave** as transmitted manifestation of the rise in atrial pressure is produced by the bulging of tricuspid valve into atria during iso-volumetric contraction (**ventricular systole**).
- **c-wave** is synchronous with pulse wave in carotid artery, and hence the use of the letter “c”,
- During isovolumetric contraction part of the first heart sound develops, QRS complex develops fully and Ventricular pressure rises from 4mmHg to 80 mmHg
- At the end of isovolumetric contraction:
 - S-L valves open
 - Ventricular pressure rises from 80 mmHg to 120 mmHg
 - **Rapid ejection period (0.10 sec) occurs...70% of SV**
 - **Slow (reduced) ejection (0.08 sec) follows,,,30% of SV**
 - Isovolumetric relaxation follows



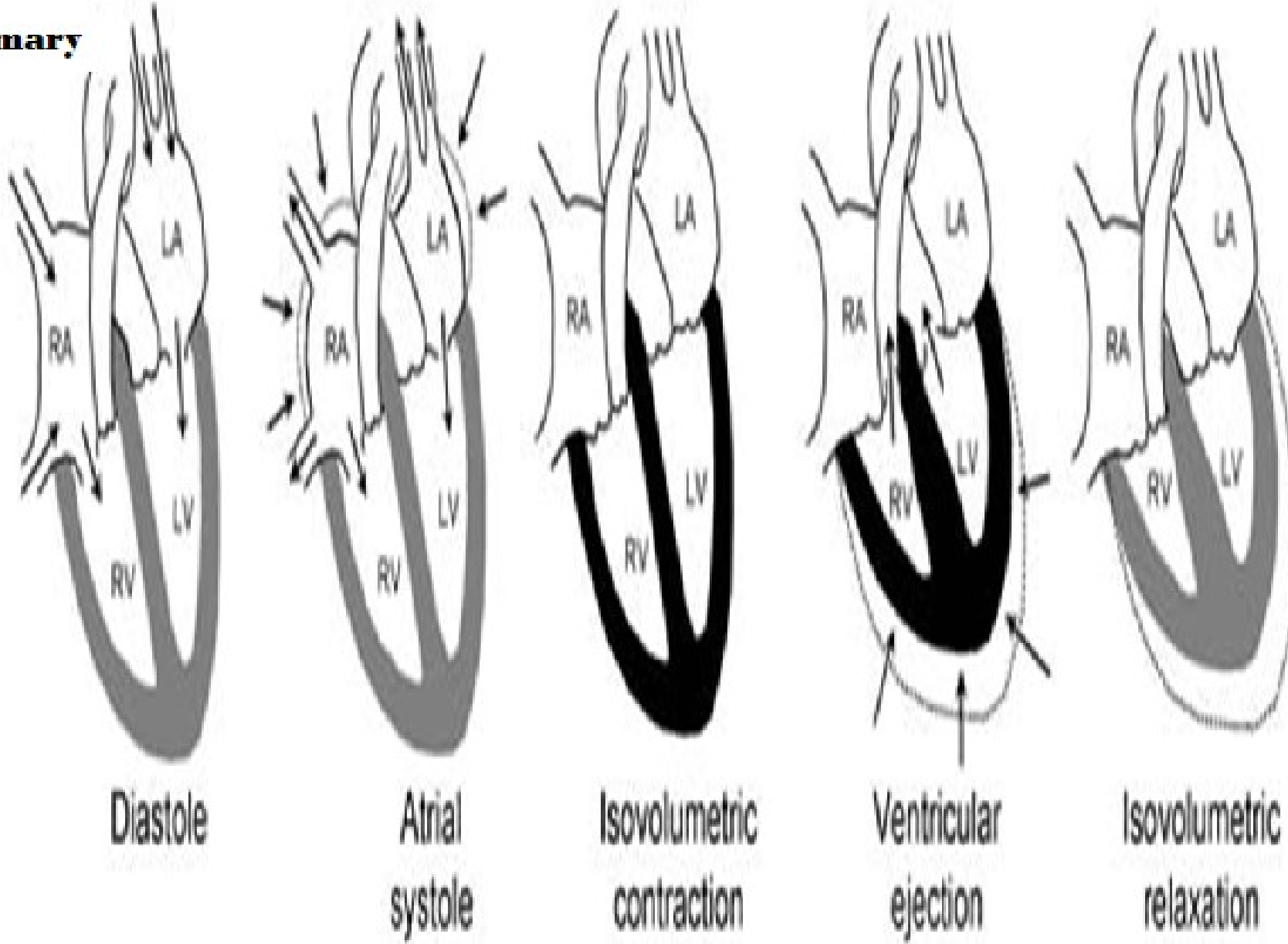
...cont'd

- $SV = 80 \text{ ml}$, i.e. ejection fraction $(80/120)$, $EF = 67\%$,
 $ESV = 120 - 80 \text{ ml} = 40 \text{ ml}$
- Ventricular volume goes down to ESV
- Most of the T-wave occurs during the ejection period
- Pressure in ventricle falls slightly below atrial pressure
- **Valves and aorta being distensible**, undergo recoil and produce a **secondary pressure wave** in aortic curve with a notch between systolic and diastolic pressure curves.
- This is referred to as **dicrotic notch - a small oscillation** on the falling phase of the pulse wave (Fig. below) formed when the **aortic valve snaps shut** following full emptying of blood.

Aortic pressure



Summary



Cardiac Output (CO)

Cardiac Output:

- It is the amount of blood pumped by each ventricle in one minute
- It is the product of heart rate (HR) and stroke volume (SV)
 - HR is the number of heart beats per minute
 - SV is the amount of blood pumped out by ventricle with each beat
 - $SV = \text{ejection fraction} \times EDV$
 - **Ejection fraction** = 2/3 of blood contained and ejected by the heart

...cont'd

Cardiac reserve:

- It is the difference between maximal exercise and resting CO
 - CO = 5-6 liters per minute at rest
 - CO = 20-40 liters per minute at maximum exercise

E.g. 35L-5 L = 30L

Cardiac Index:

- It is the amount of blood pumped of the left ventricle each minute

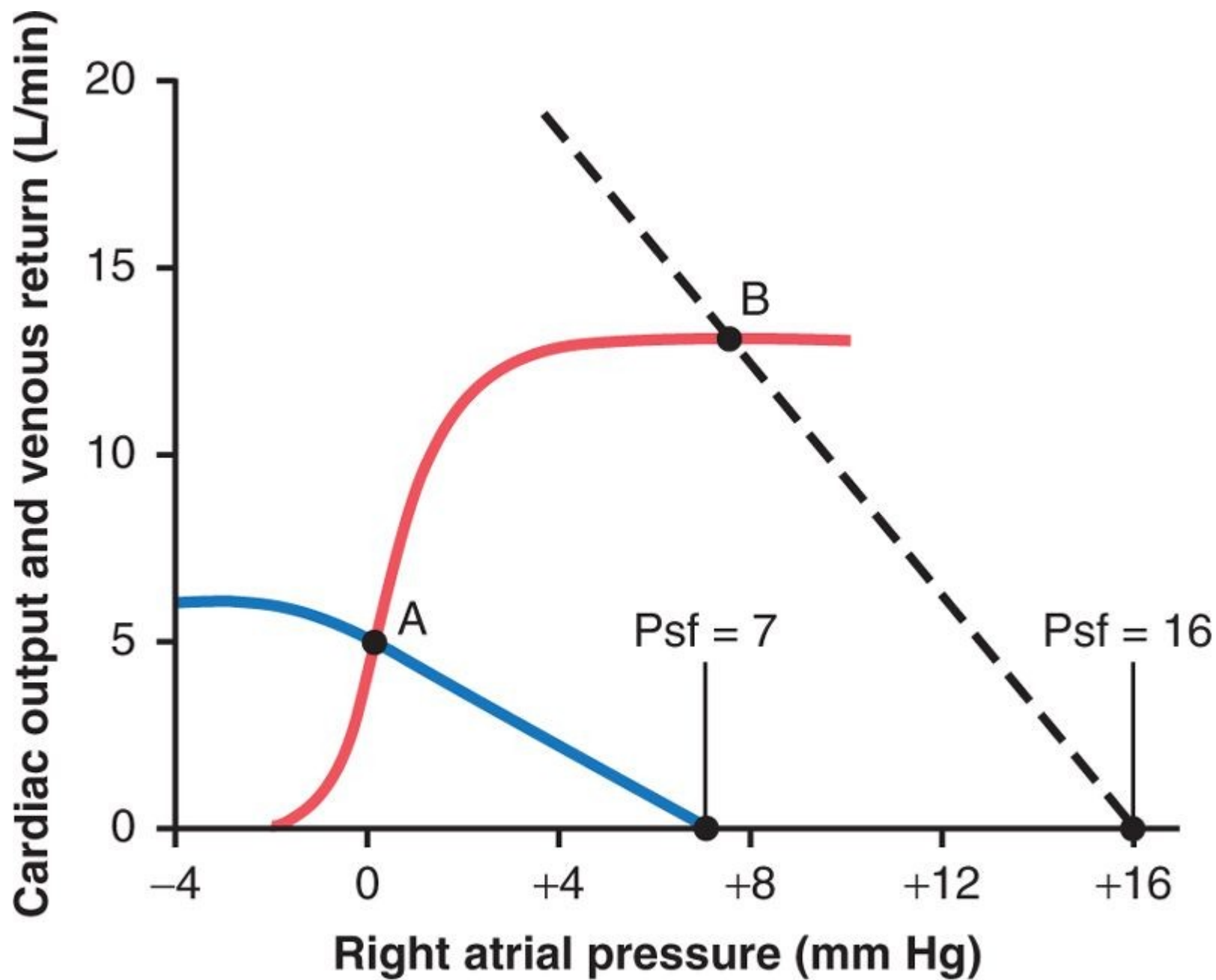
CI = Cardiac output ÷ Body surface area

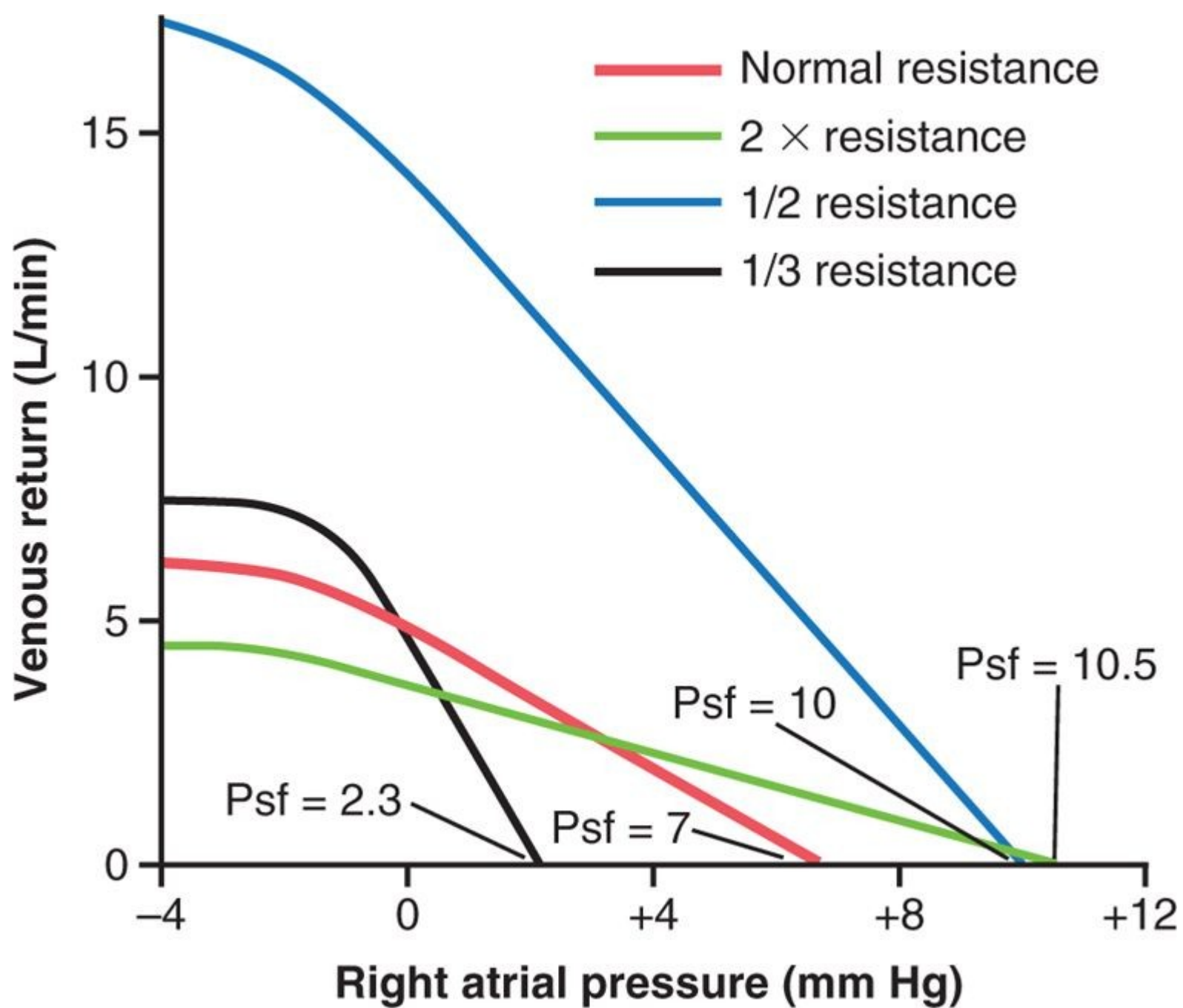
E.g. 5.5 L/min ÷ 1.7 = 3.23 L/m²

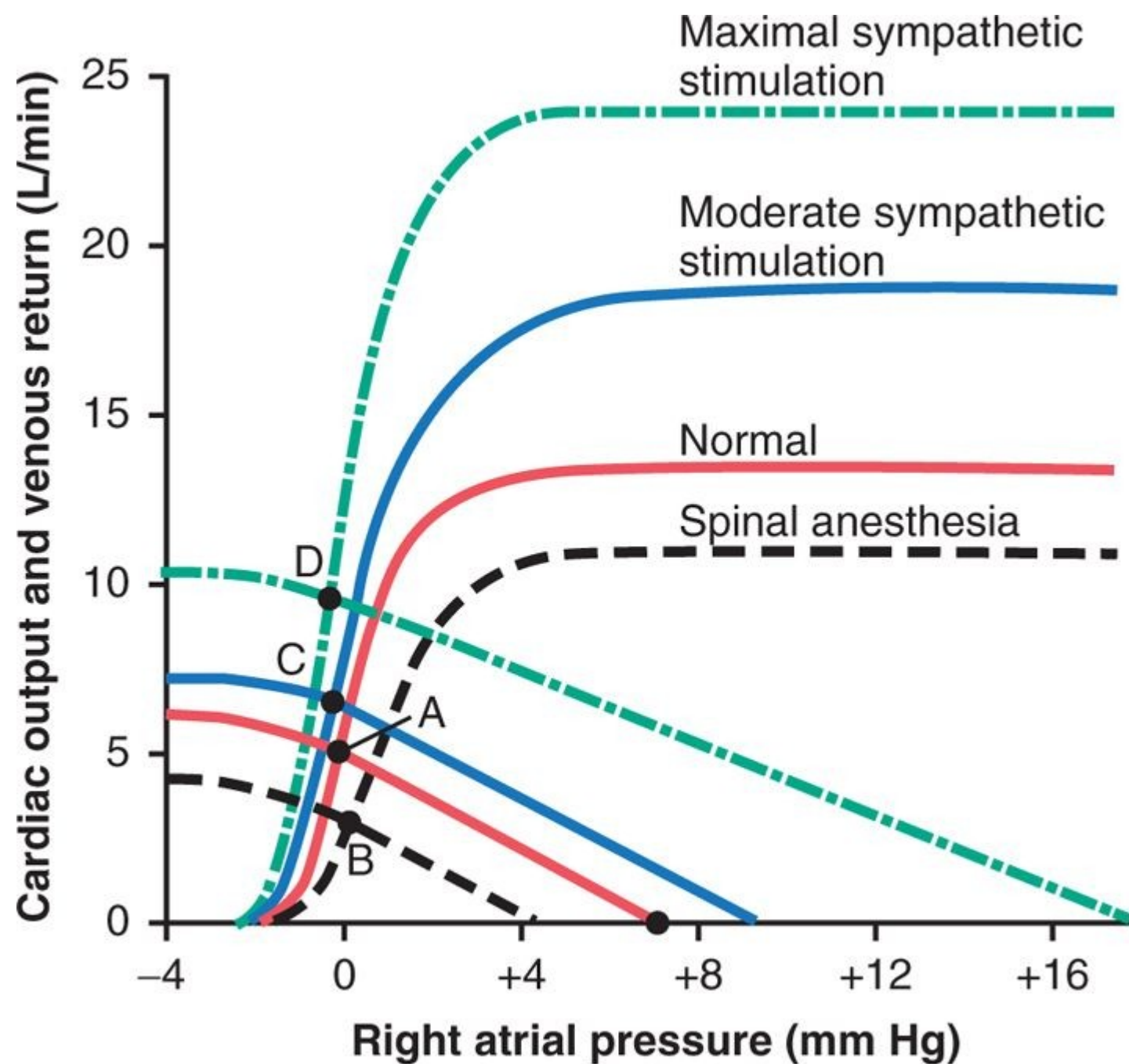
Effect of Venous Return (VR) on C.O

Venous return is:

1. volume of blood flowing from veins into RA/min
 2. 5.5L/min at rest; about 35L/min in well trained athletes during strenuous exercise
 3. influenced by RAP like C.O as shown on cardiac function curves (Fig..)
- **C.O = VR** at the “operating point” of the cardiac function curves (Fig..).
 - Any factor influencing C.O or VR shifts the “operating point” of C.O/VR intersection to a new position.







Factors affecting Venous Return

1. Mean Systemic Filling Pressure (MSFP)

- Pressure measured every where in the systemic circulation 1 min after blood flow has been stopped by clamping large blood vessels near the heart.
- It is 7 mmHg and indicates the degree of filling of the systemic circulation
- It is the driving force for VR.

2. Right Atrial Pressure(RAP)

- Mean pressure in the right atrium (Central venous pressure) = 2 mmHg
- Pressure gradient = $MSFP - RAP$
- $\uparrow RAP \rightarrow \downarrow VR$

Factors affecting ...cont'd

3. Resistance to venous return (RVR)

- Resistance blood meets (1 mmHg/L/min) during its flow from arterial side to RA
- Occurs mainly at arterioles
- $VR = \frac{MSFP - RAP}{RVR} = \frac{MAP - RAP}{TPR}$

4. Sympathetic stimulation →

- ↑VR by inducing vasoconstriction, improved cardiac pumping power and arteriolar dilatation

5. Blood volume

- ↑Blood volume → ↑VR by ↑MSFP and vice versa.

6. Respiratory movements

- VR increases with inspiration and decreases with expiration

Factors affecting ...cont'd

7. Arteriolar dilatation $\rightarrow \downarrow \text{RVR}$ and $\uparrow \text{VR}$

8. Capillary dilatation $\rightarrow \uparrow \text{vascular capacity}$, $\downarrow \text{MSFP}$ and $\downarrow \text{VR}$

9. Skeletal muscle contraction

- Squeezes veins b/n muscles $\rightarrow \uparrow \text{MSFP} \rightarrow \uparrow \text{VR}$

10. Gravity

- Standing motionless for some time $\rightarrow$ pooling of blood in lower extremities $\rightarrow \downarrow \text{MSFP} \rightarrow \downarrow \text{VR} \rightarrow \downarrow \text{C.O} \rightarrow$ hypotension $\rightarrow$ brain ischaemia $\rightarrow$ **syncope**.

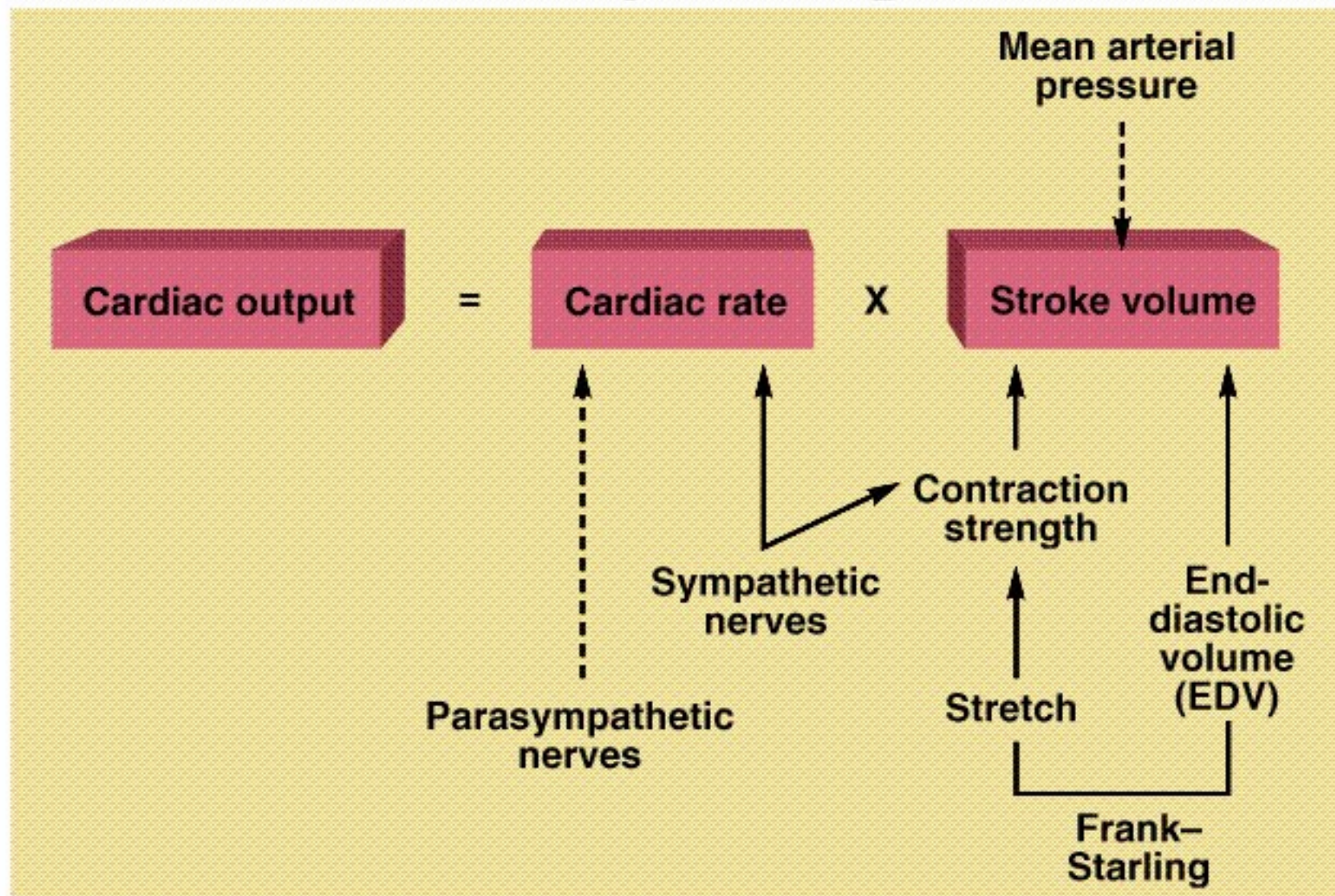
Regulation of Pumping Activity of the Heart

- There are two types of regulatory mechanisms:
 1. **The Intrinsic Auto-regulatory Mechanisms (The Frank Starling Mechanism)**
 - ☐ “Within physiological limits, the contractile force of the heart is proportional to the amount of the blood enters the heart”.
 2. **The Extrinsic Regulatory Mechanisms**
 - ☐ Sympathetic, parasympathetic stimulation
 - ☐ Catecholamine
 - ☐ Electrolytes (Ca^{2+} , K^{+})
 - ☐ Drugs: Cardiac glycosides, xanthenes
 - ☐ Hormones: AD, Glucagon, T3/T4

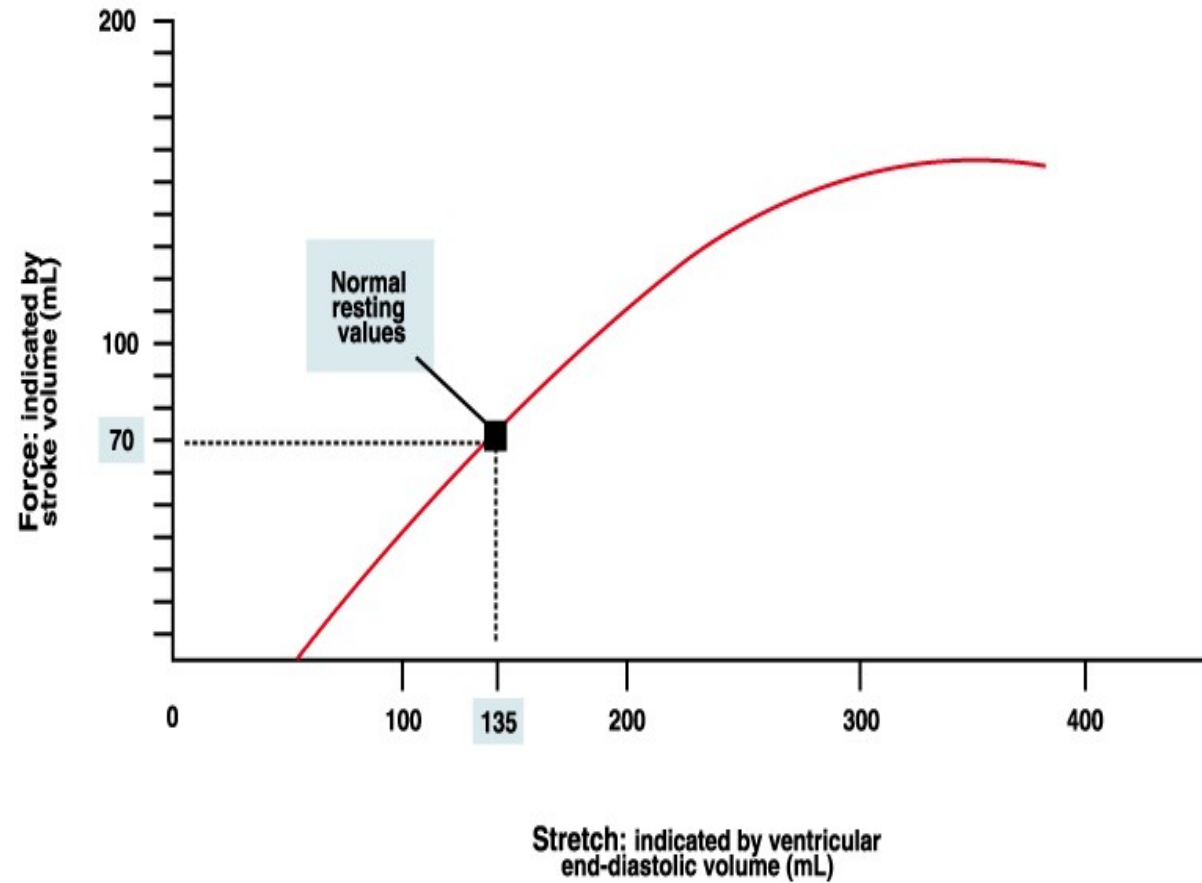
Frank Starling Law of the Heart

- Frank-Starling mechanism means the greater the heart muscle is stretched during filling $\rightarrow$ $\uparrow$ force of contraction $\rightarrow$ $\uparrow$ blood pumped to the aorta.
- Pre-load or degree of stretch of cardiac muscle cells before they contract is the critical factor controlling stroke volume
 - It depends on end diastolic volume
- Slow heart beat and exercise $\rightarrow$ $\uparrow$ VR to the heart $\rightarrow$ increasing EDV that in turn, $\uparrow$ SV
- Blood loss and extremely rapid heartbeat $\rightarrow$ decreases ventricular filling time and decreasing EDV $\rightarrow$ that decreases SV

Cardiac Output Regulation



Starling's Law of the Heart

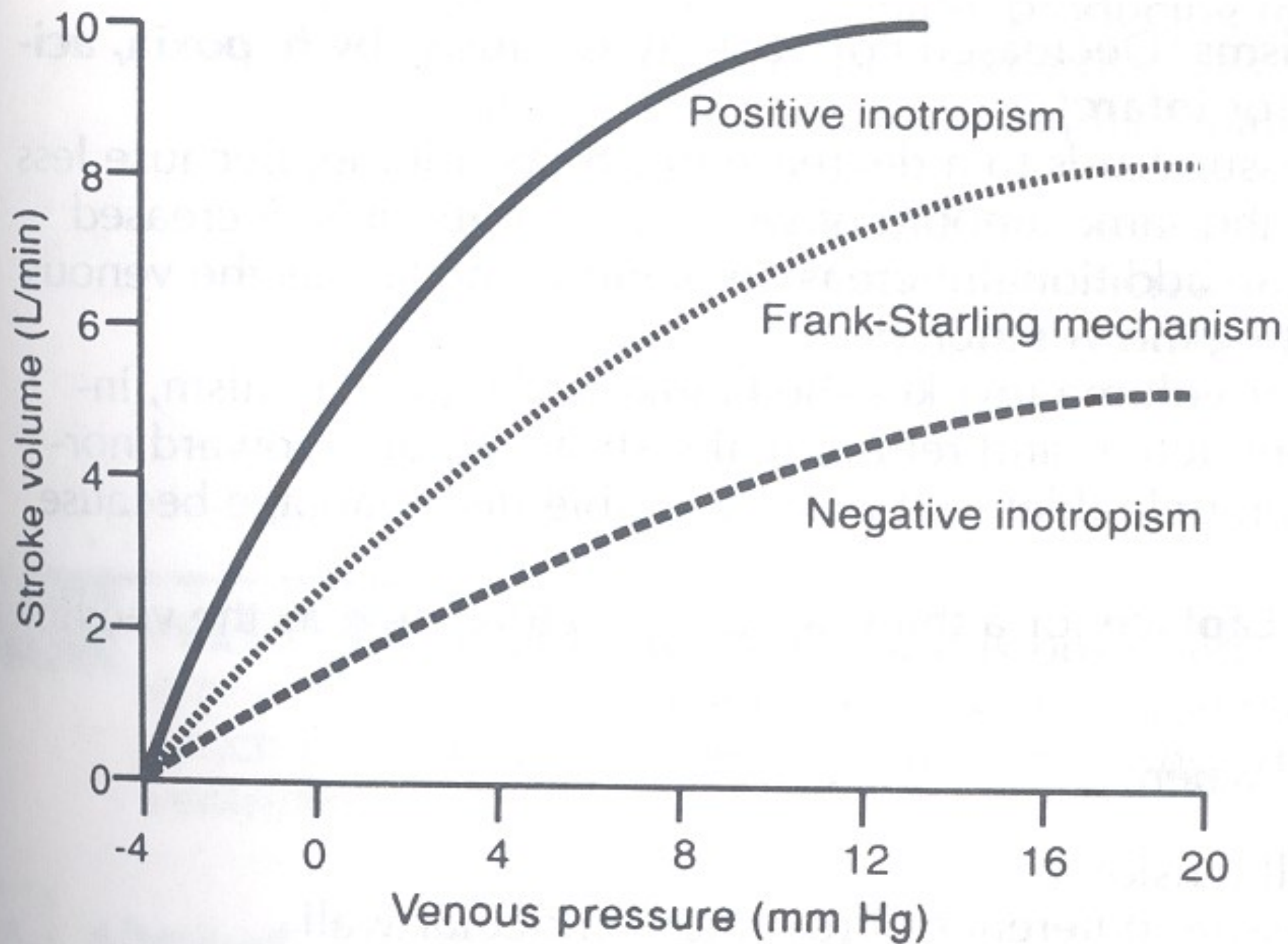


...cont'd

- A few seconds of **heterometric** autoregulation(**pre-load**)→↑ metabolic state of heart→ more forceful contraction and 30-60 sec. after, pressure elevation occurs; the heart expels more blood than is brought into it
- **↑EDV & ESV persists for 2-3 minutes.**
- This increase in contractile force without a change in EDV is referred to as **Homeometric (after-load)** regulation, a condition mediated by a change in metabolic state or contractility of the heart.

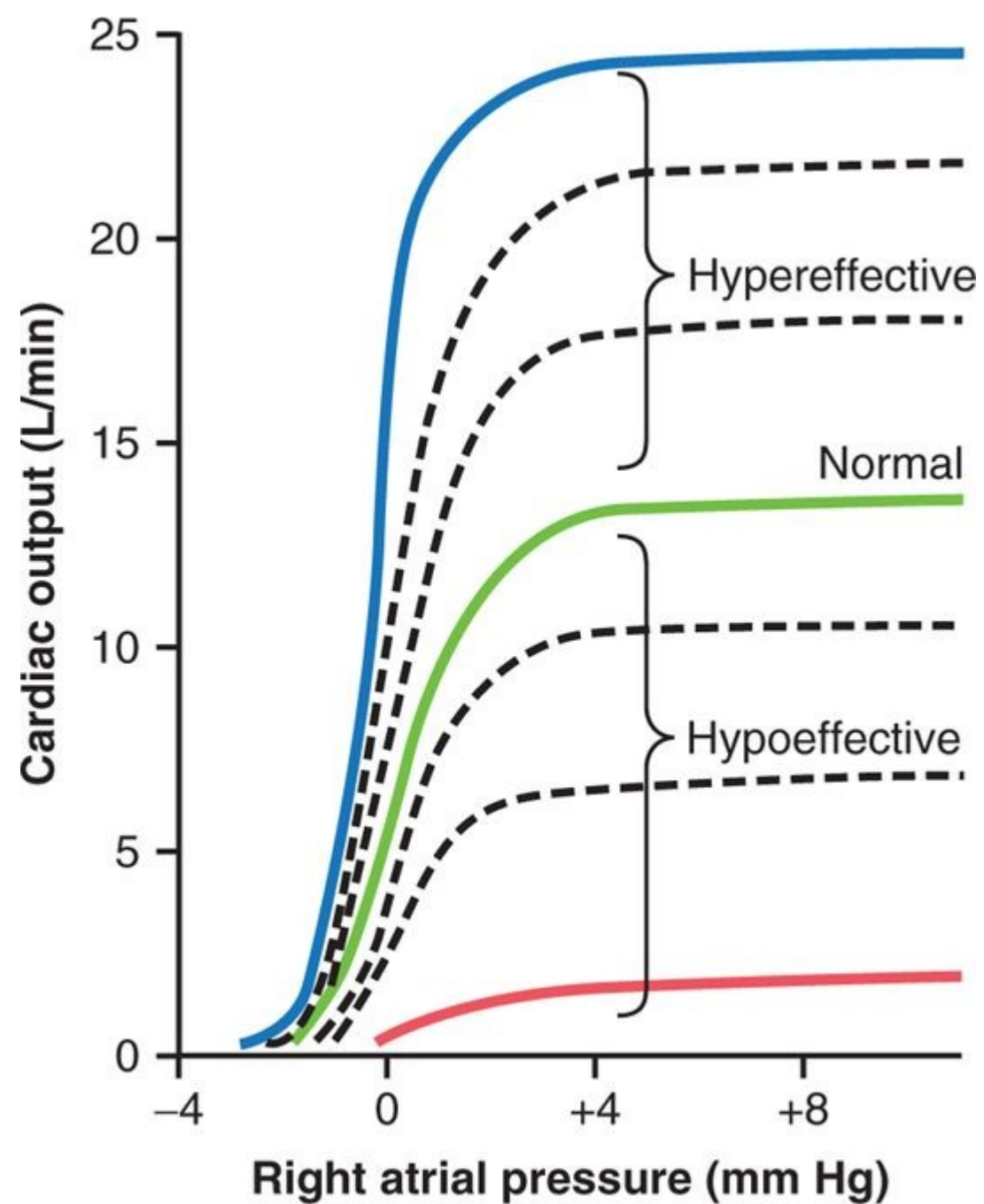
NB:

- **Heterometric autoregulation** is related to filling pressure of the heart
- while **homeometric autoregulation** is related to maximum pressure (contractile force).

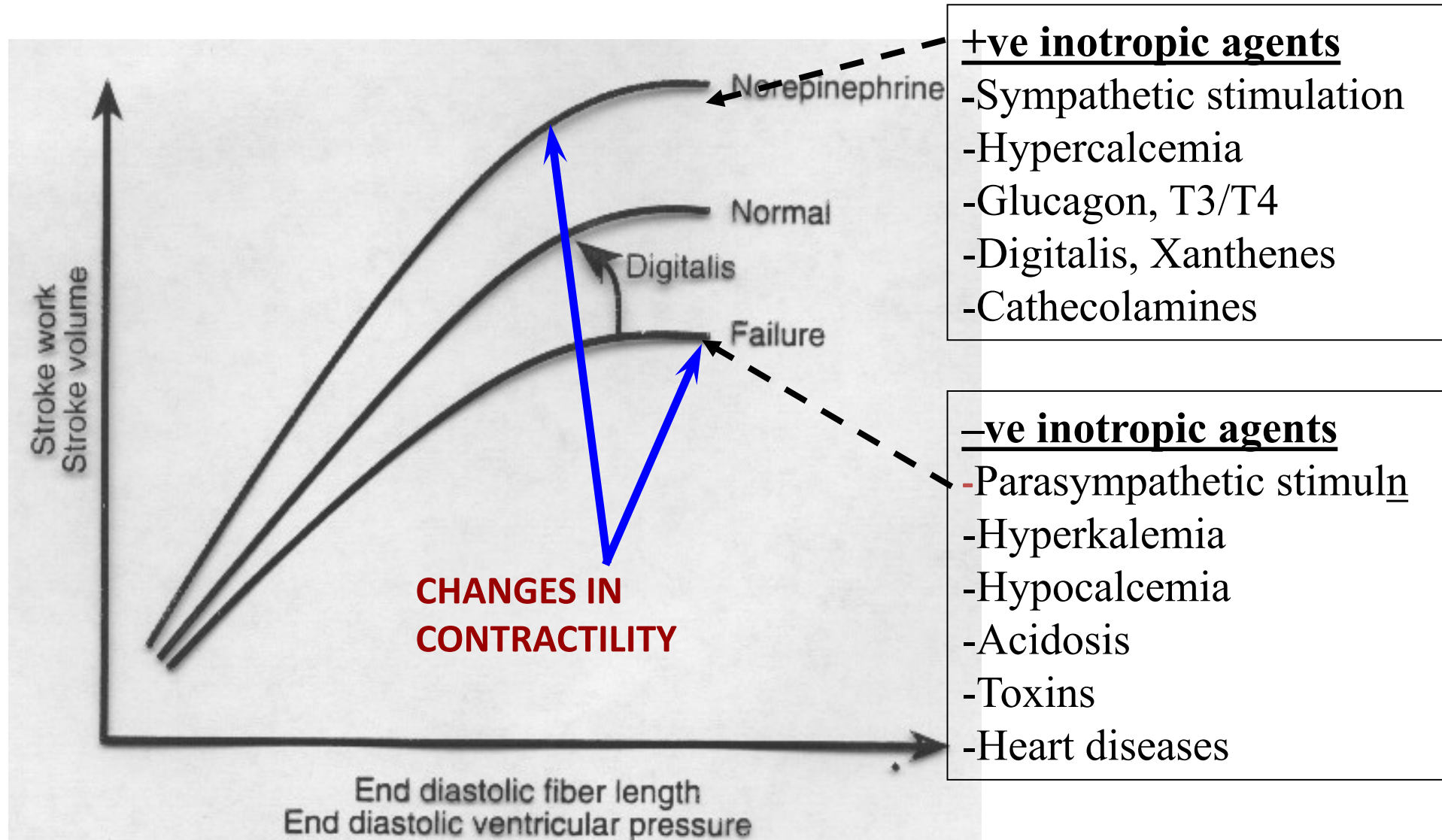


...cont'd

- A sudden increase in venous return (VR) → ↑ in heart size.
- For example, a change from erect to supine position leads to shift of blood from lower extremities to thorax → ↑ CVP (4-12cm H₂O) → ↑ in heart size.
- The heart may be:
 - **Hyper-effective** (capacity more than normal) due to sympathetic stimulation or hypertrophy.
 - **Hypo-effective** and pumping capacity is lower than normal due to:
 - Sympathetic denervation
 - Parasympathetic stimulation
 - Atrial flutter or fibrillation
 - Myocarditis
 - Cardiac hypoxia



CONTRACTILITY: THE VENTRICULAR FUNCTION CURVE



Metabolism and Energetics (work) of the Heart

Energy provided by conversion of metabolic energy into pressure and kinetic force.

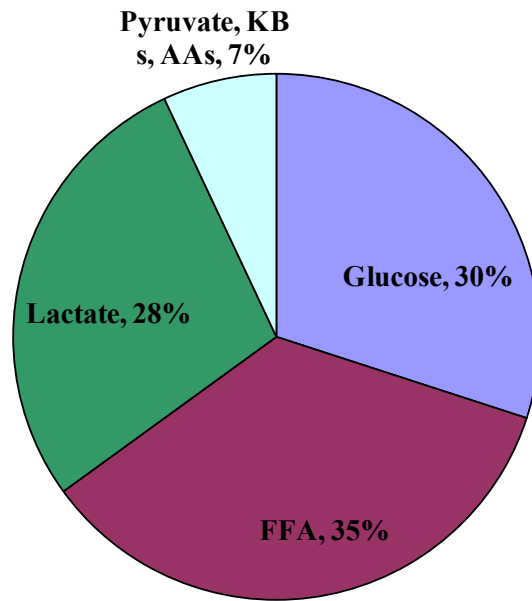
Sources of energy= glucose or glycogen, fatty acids, lactate, ketone bodies, and amino acids.

Metabolic processes are as shown in figures below

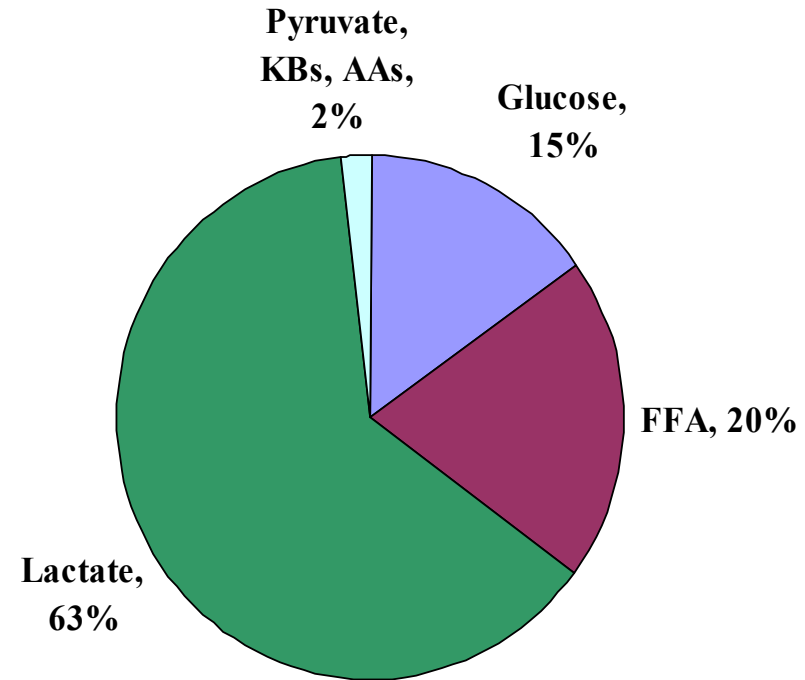
Pumping work= work done against aortic pressure= $SV \times \text{mean pressure}$ ($\Delta P \times \Delta V$) influenced by metabolic pathways

Kinetic work= energy imparted to the velocity of blood
Accounts for only some 5% of the external work of left ventricle beating at rest.

Metabolism and energetics of the cardiac muscle



At rest



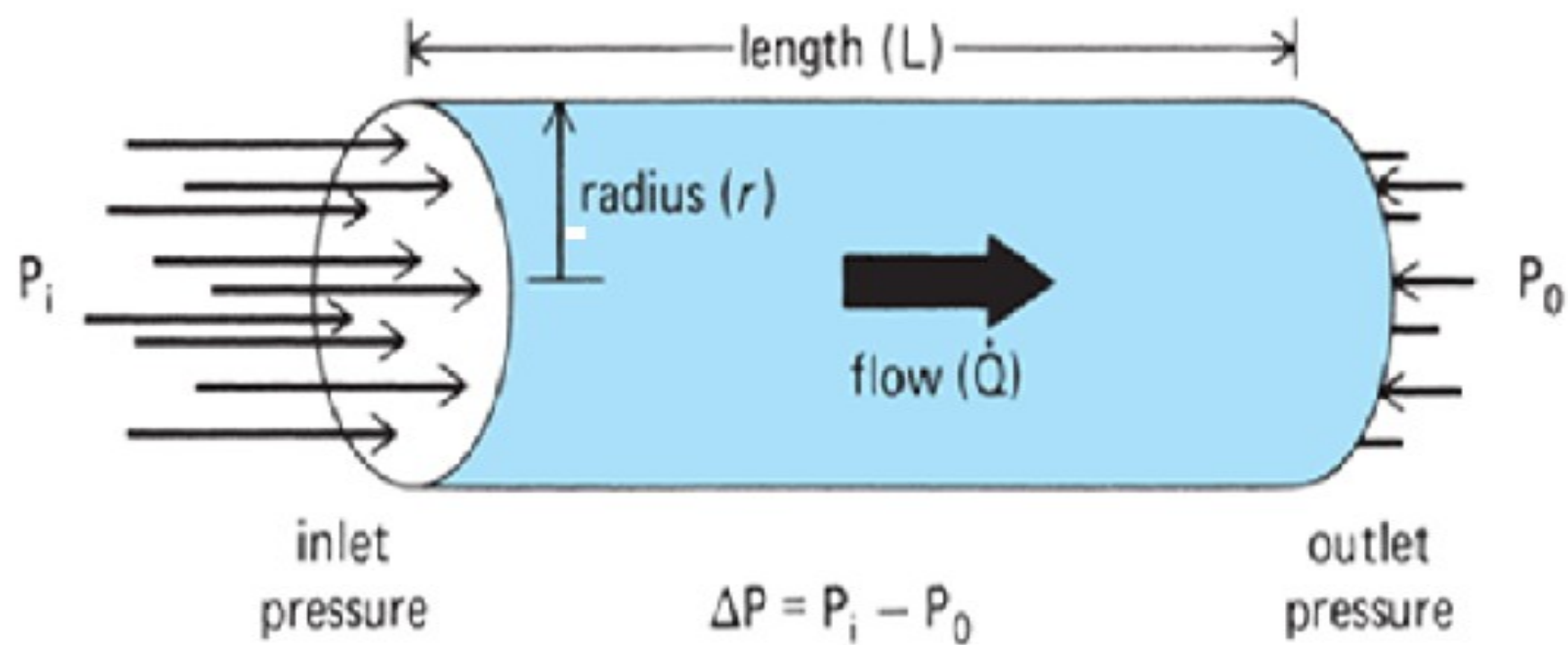
Moderate exercise

HEMODYNAMICS

- It is the study of the relationship between flow, pressure and resistance and other physical principles of blood circulation
- In 1846, Poiseuille, a French physician(1797-1869) and Hagen, a German scientist (1797-1884)
- Described factors governing non-pulsatile flow of a homogeneous (Newtonian) viscous fluid through a rigid cylindrical tube with constant circular cross-section.
- Showed that, with other things being equal, and with the inclusion of the proportionality constants $\pi/8$, the **rate of flow, Q**, through a cylindrical tube of length L, and radius r, would be:

$$Q = \Delta P \pi r^4 / 8 \eta L$$

- This is Hagen-Poiseuille's law.



Resistance to Flow(r)

- This is an impediment to flow provided by a vessel or circulatory bed.
- Vascular resistance to BF is directly proportional to the length of the vessel and viscosity of blood, but inversely proportional to the 4th power of radius of the vessel.

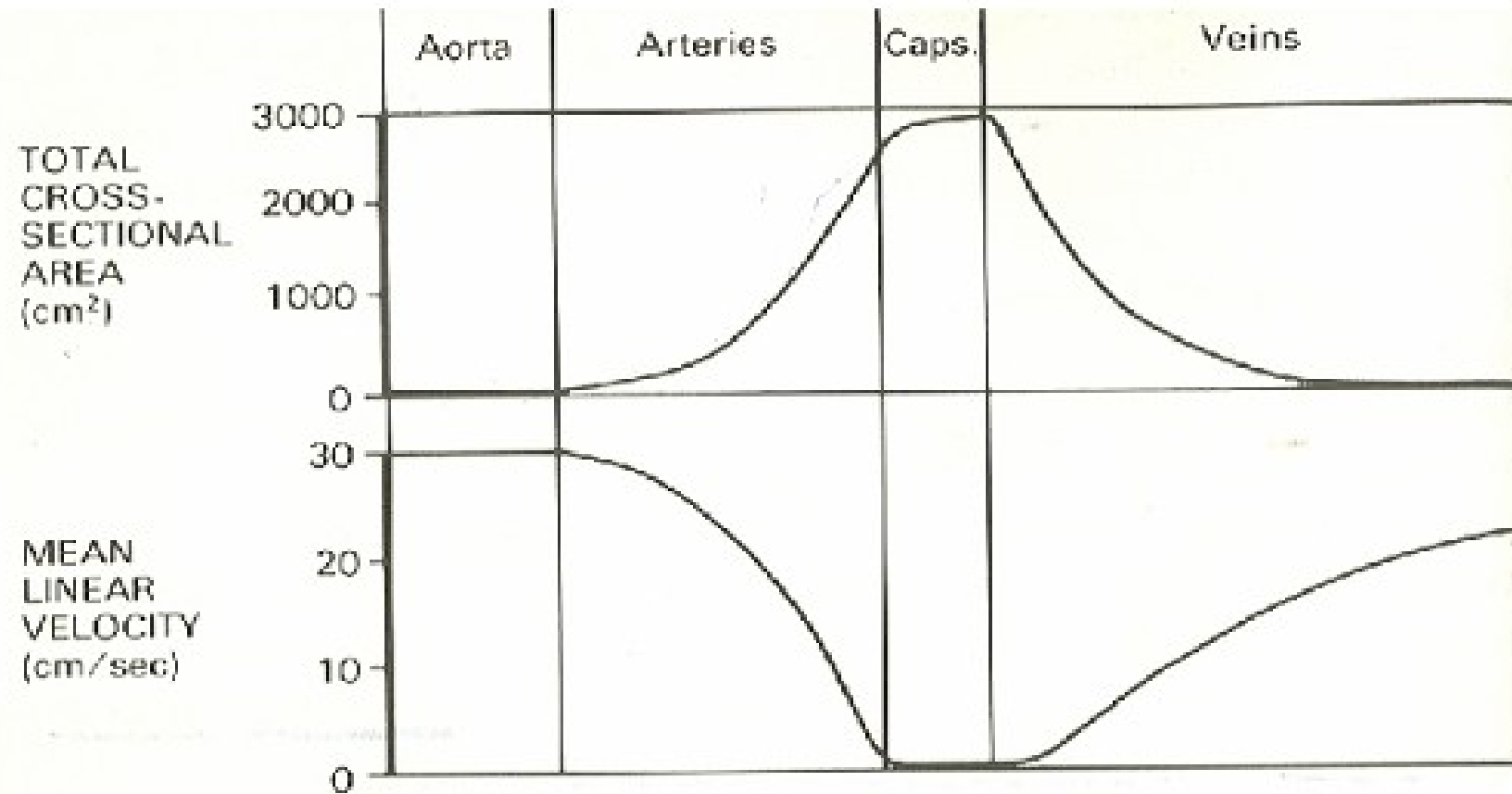
$$\Delta P = Q \times R$$

- **$R = \Delta P / Q$, analogous to Ohm's law ($R = V / \text{current}$)**
 $= \Delta P \div \Delta P \pi r^4 \div 8 \eta L$,
- Using this relationship, resistance in different vascular beds can be assessed.
- E.g. Aortic pressure=100mmHg, Ventricular output =80mL/min
- Total peripheral resistance=100/80 =1.25 PRU
- Analogous to electrical circuits, **vascular resistances** may be in **series or parallel** (fig. below).

Velocity of Blood Flow

- Velocity (v) of blood flow is directly proportional to the pressure difference (ΔP) and diameter of the vessels, but inversely related to the viscosity of blood and length of the blood vessel.
- $V = Q \div \text{Cross-sectional area } (A = \pi r^2)$
 $= (\Delta P \pi r^4 \div 8 \eta L) \div \pi r^2$
- $V = \Delta P r^2 / 8 \eta L$

Velocity profile (Fig...below)



$$\bar{V} = \frac{\text{Flow}}{\text{X-Sect. Area}}$$

Fig.. Relation of cross-sectional area and mean velocity of flow in the systemic circulation.

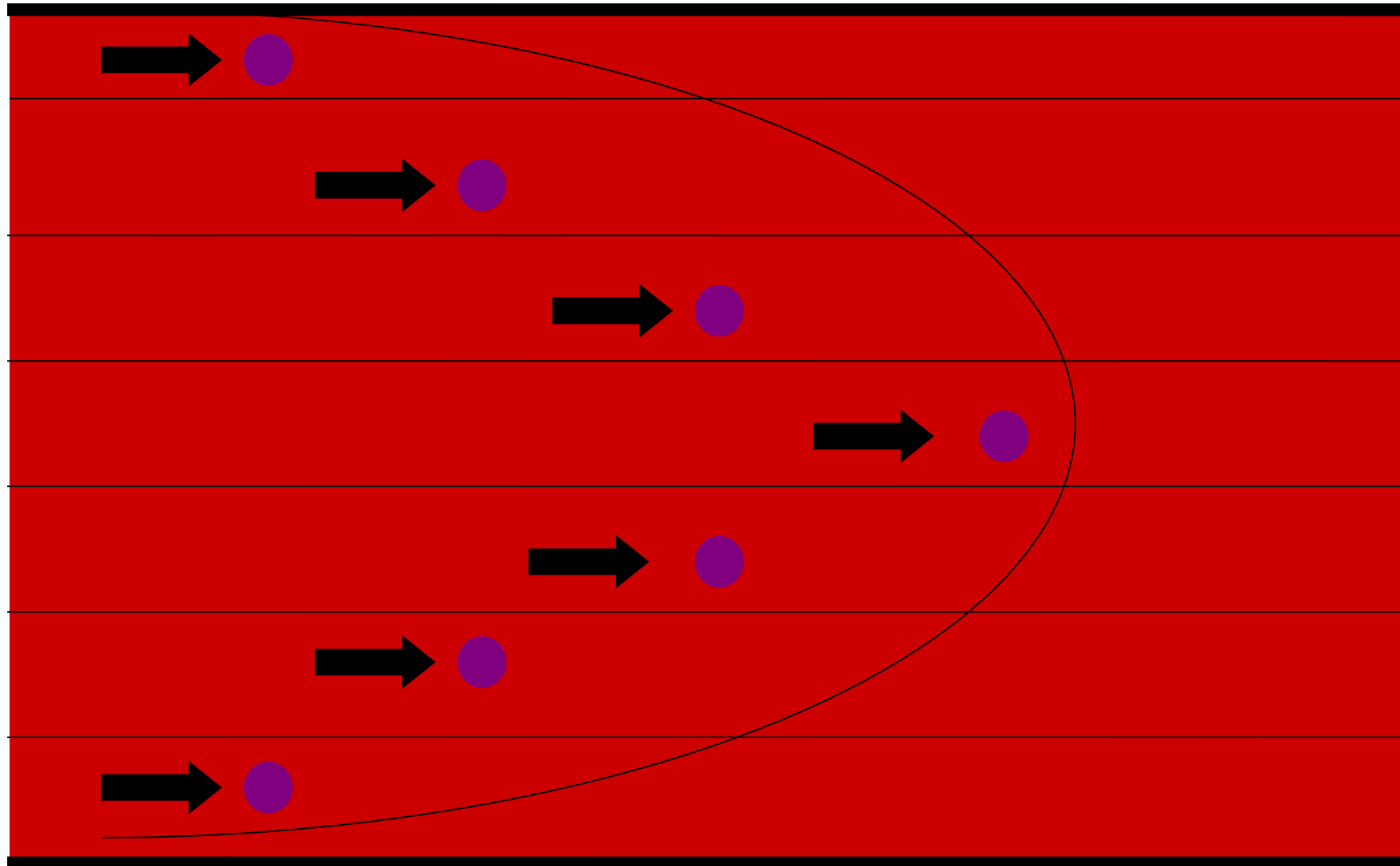
Viscosity

- This is friction of molecules in the moving stream of fluid.
- It depends on:
 1. Concentration of suspended medium (e.g. RBC in blood). Viscosity becomes high in polycythemia, and low in anaemia
 2. Velocity of flow
 3. Radius of vessel
 4. Temperature

1. Laminar or streamline blood flow

- Pulsatile (laminar) flow of fluid along a uniform tube (>2 or 3 RBC) at a fixed rate is laminar, i.e. infinite number of concentric laminae or layers sliding upon one another are seen.
- Innermost or axial layer has the highest velocity and the outermost layer touching the lining of the tube remains stationary.

LAMINAR or STREAMLINE FLOW



2. Steady (non-pulsatile) Blood Flow

- In steady blood flow, there is no variation of flow in time, i.e. flow is non-pulsatile.
- Blood flow is laminar (streamline) in all the smaller vessels of the resistance channels and steady or non-pulsatile in veins.

3. Turbulent Blood Flow

- Flow continues to be laminar (streamline) as long as the product of velocity and diameter does not exceed a critical value.
- It becomes turbulent when the critical value is exceeded.
- This is determined by the **Reynolds number Re (a dimensionless number)**.
- It represents the **ratio of inertia to viscous forces**.

Types of blood flow

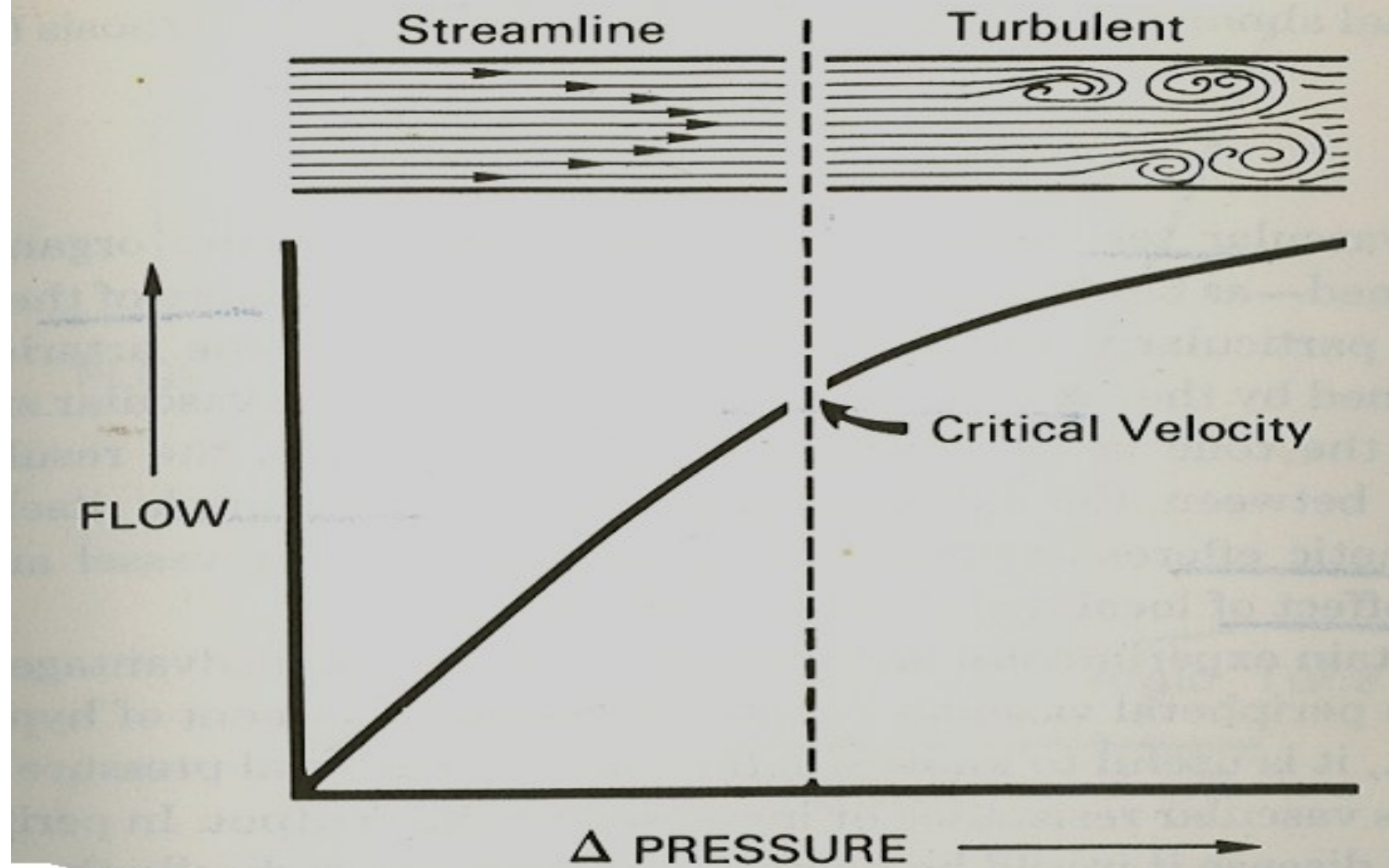


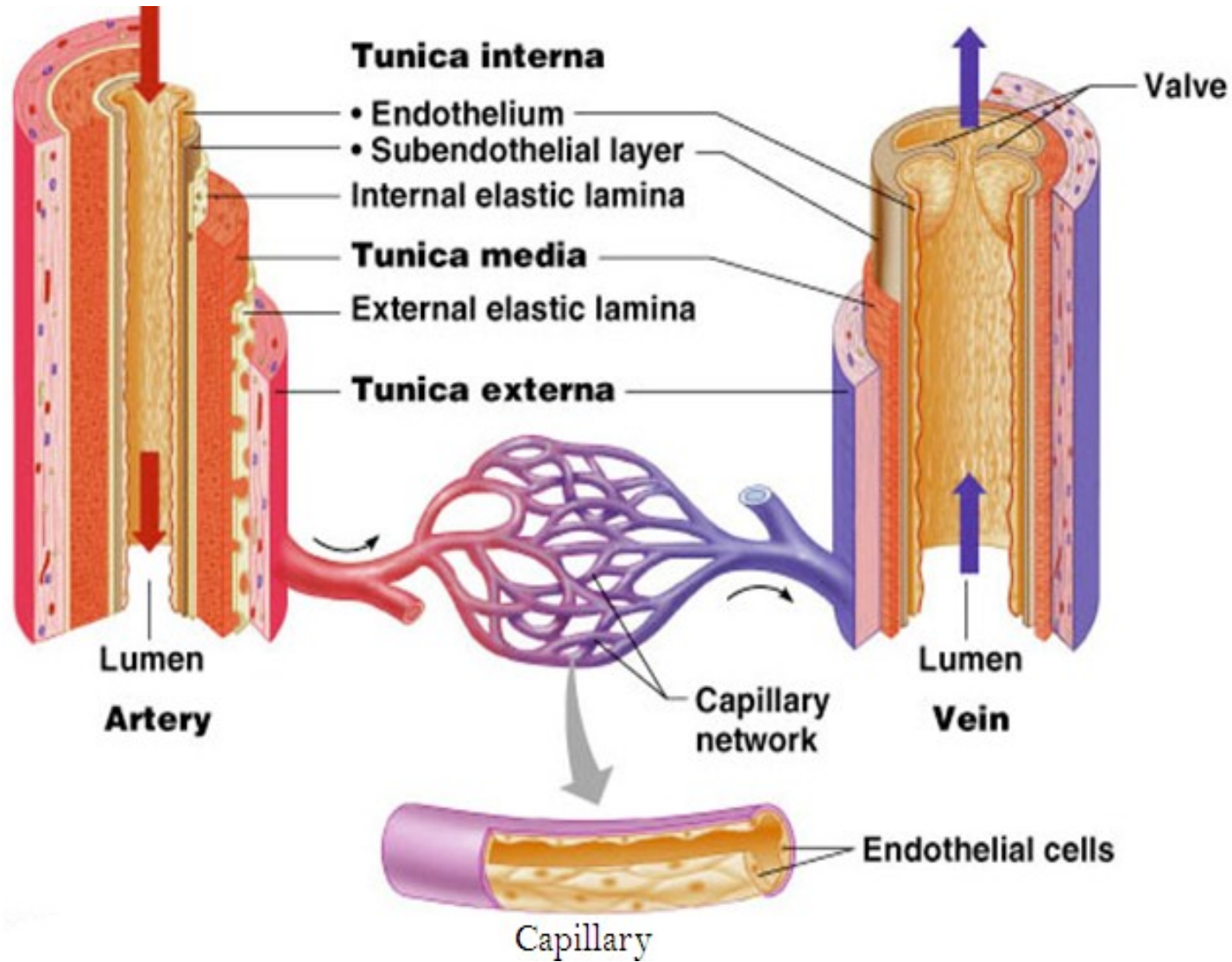
Fig.. Relationship between velocity of flow and turbulence.

- Arteries and veins resemble each other in that their walls contain three coats.
- However, the vessels adapt to their different circulatory tasks by differing in the structure of these coats.
- The **inner coat** (vascular endothelium, tunica intima) consists of a single layer of endothelial cells applied to a thin connective tissue layer, the basement membrane.
- The **middle coat** (tunica media) contains primarily smooth muscle and elastic tissue fibers.
- The **outer coat** (tunica adventitia, adventitious coat) embeds the vessel in its surroundings and consists mainly of connective tissue.
- In addition, the arteries have an elastic, fenestrated membrane.

....cont'd

- Arteries are distinguished by an especially well developed muscle coat, which contains a varying amount of elastic fiber according to its site (predominantly elastic and predominantly muscular arteries).
- This layer is the driving force of the blood vessels by **dilating (vasodilatation)** and **constricting (vasoconstriction)** the diameter of the blood vessels, it regulates **blood flow** and **blood pressure**.
- The arteries near the heart contain a high proportion of elastic fibers and this creates an elastic recoil.
- In the smallest blood vessels, the capillaries, the coats are reduced to one, the tunica intima; this **facilitates the exchange of fluids and gases**.

STRUCTURAL PATTERN OF THE VASCULAR SYSTEM



Lymph Vessels

- The lymphatic system runs parallel to the venous side of the circulation
- It begins near the capillaries as “blind” lymphatic capillaries that reabsorb fluid that has not been taken up from the tissues by the blood vessels (lymphatic fluid).
- Small and large lymph vessels then return the lymph to the venous blood.
- The wall of the lymph vessels consists of an endothelium, and a thin layer of rhythmically contracting smooth muscle cells.
- Similarly to the situation in the veins, numerous valves further the transport of the lymph.
- The course of the lymph vessels is interrupted by **lymph node** stations, which represent a kind of biological filter and fulfill important functions in immune defense.

Functional Classification of Blood Vessels

1. Elastic vessels (Windkessel vessels):

- Example: Aorta, big arteries
- Pressure storing components
- High ability of recoiling

2. Resistance vessels (Stop Cock Vessels)

- Example: small arteries and arterioles
- High muscular component
- Develop high resistance
- Regulate blood flow

Blood vessels: Classification...cont'd

3. **Exchange vessels:**

- Example: capillaries
- Made up of endothelium and basement membrane.
- Thin enough for exchange
- 3 types: continuous, discontinuous and fenestrated capillaries

4. **Capacitance vessels** (big to small veins)

- Very high capacity of distension
- Can accommodate large volume of blood (65% of blood volume)

Arteries

- **AORTA** branches → peripheral vascular bed → 1000 fold increase in cross-sectional area at capillaries
- Large amount of elastin → ↑ expansion and recoil → ↑ pressure storage.
- A velocity of 40 cm/sec at aorta → <2mm/sec at capillaries, i.e. $V=Q/A$.
- Beyond capillaries, the situation is reversed and blood flow accelerates as total cross-sectional area of veins decreases.

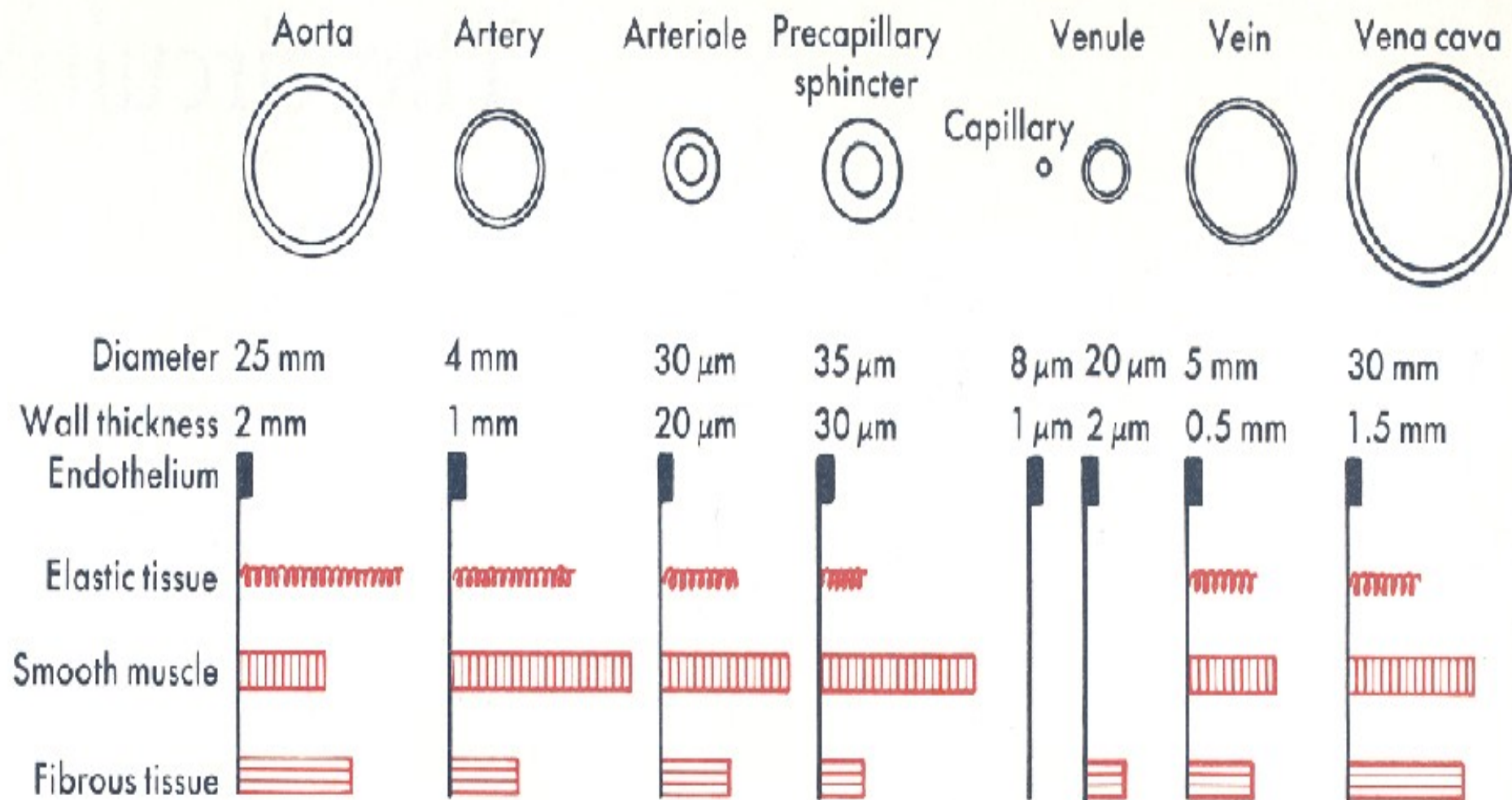
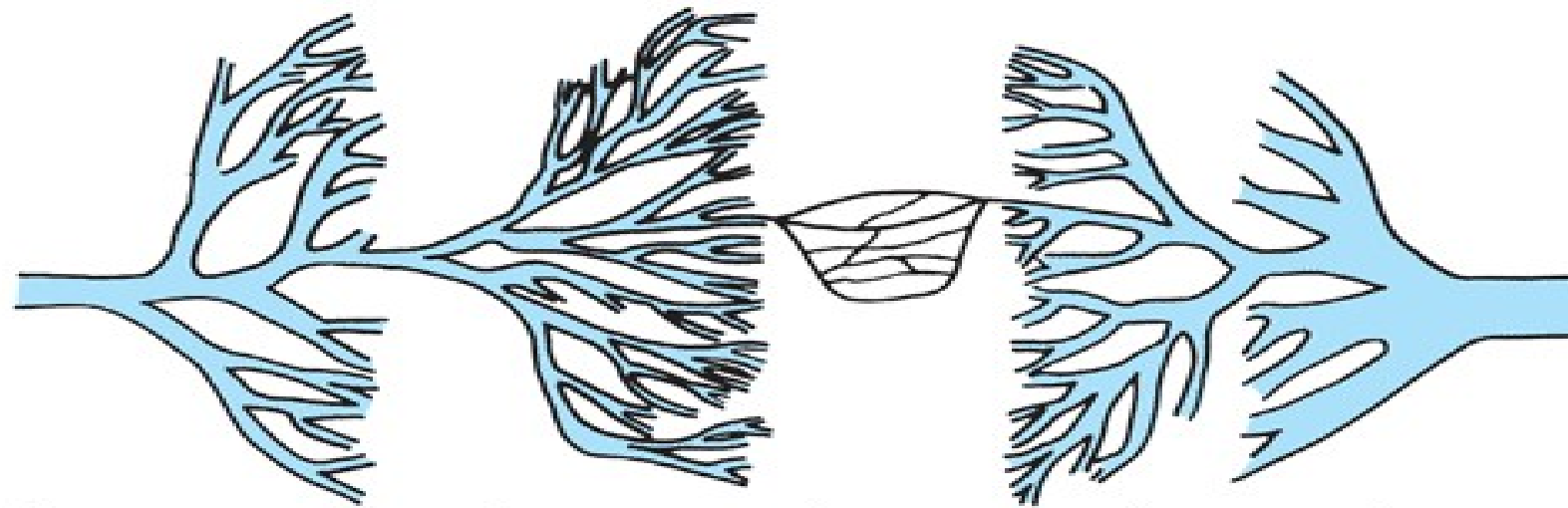


Fig. Internal diameter, wall thickness, and relative amounts of the principal components of the vessel walls of the various blood vessels that compose the circulatory system. Cross sections of the vessels are not drawn to scale because of the huge range from aorta and venae cavae to capillary.



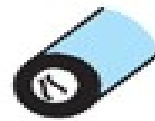
ARTERIES

ARTERIOLES

CAPILLARIES

VENULES

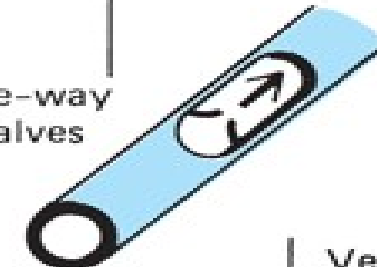
VEINS



Aorta



one-way
valves



Venae
cavae

	ARTERIES		ARTERIOLES	CAPILLARIES	VENULES	VEINS	
internal diameter	2.5 cm	0.4 cm	30 μm	5 μm	70 μm	0.5 cm	3 cm
wall thickness	2 mm	1 mm	20 μm	1 μm	7 μm	0.5 mm	1.5 mm
number	1	160	5×10^7	10^{10}	10^8	200	2
total cross-sectional area	4.5 cm^2	20 cm^2	400 cm^2	4500 cm^2	4000 cm^2	40 cm^2	18 cm^2

...cont'd

- Resistance (R), pressure (P) and flow (F) patterns change as we go from arteries to arterioles, small branches of arterial system measuring about 0.10mm in diameter.
- R increases as we go from the aorta to the arterioles and hence arterioles are referred to as resistance vessels and “stop cocks” of the arterial tree.
- Pressure decreases from aorta to capillaries and blood flow becomes steady as it passes the arterioles.
- Decrease in O_{ce} or in transmural pressure → extreme narrowing → active or passive closure in resistance vessels → obliteration of lumen (“critical closure”) → flow ceases.

...cont'd

- “Critical closure” occurs most readily in such vessels as the thick walled **cutaneous A-V shunts** or in the **ductus arteriosus** and in strongly constricted arterioles- **met-arterioles sphincters**.
- One **arteriole** branches to **10 capillaries** leading to the formation of **10 billion capillaries** in the peripheral tissues.
- **Capillaries**, in spite of their thinness and weakness, are capable of withstanding ABP without damage because they have **narrow lumen** and the **tension developed is small**.
- The protective effect of small size is an example of the operation of Laplace's law.
- According to this law, tension in the wall of a cylinder (T) is equal to the product of transmural pressure (p) and radius divided by the wall thickness (w).

$$T = p \times r / w$$

Arterial Blood Pressure (BP)

- The overall strategy of the cardiovascular system is to :
 1. provide all organs with a constant perfusion pressure and
 2. allow each individual organ to regulate its blood flow in accordance with the local needs of the tissue.
- **ABP** includes pressure in the aorta and other large arteries.

NB:

- BP is conventionally written as systolic (peak, ejection pressure)/ diastolic (minimum, filling pressure) blood pressure.

Normal SBP: 90 – 130 mm Hg (120 mm Hg)

DBP: 60 – 90 mm Hg (80 mm Hg)

PP: SBP – DBP

MAP: $P_d + \frac{1}{3}(P_s - P_d)$

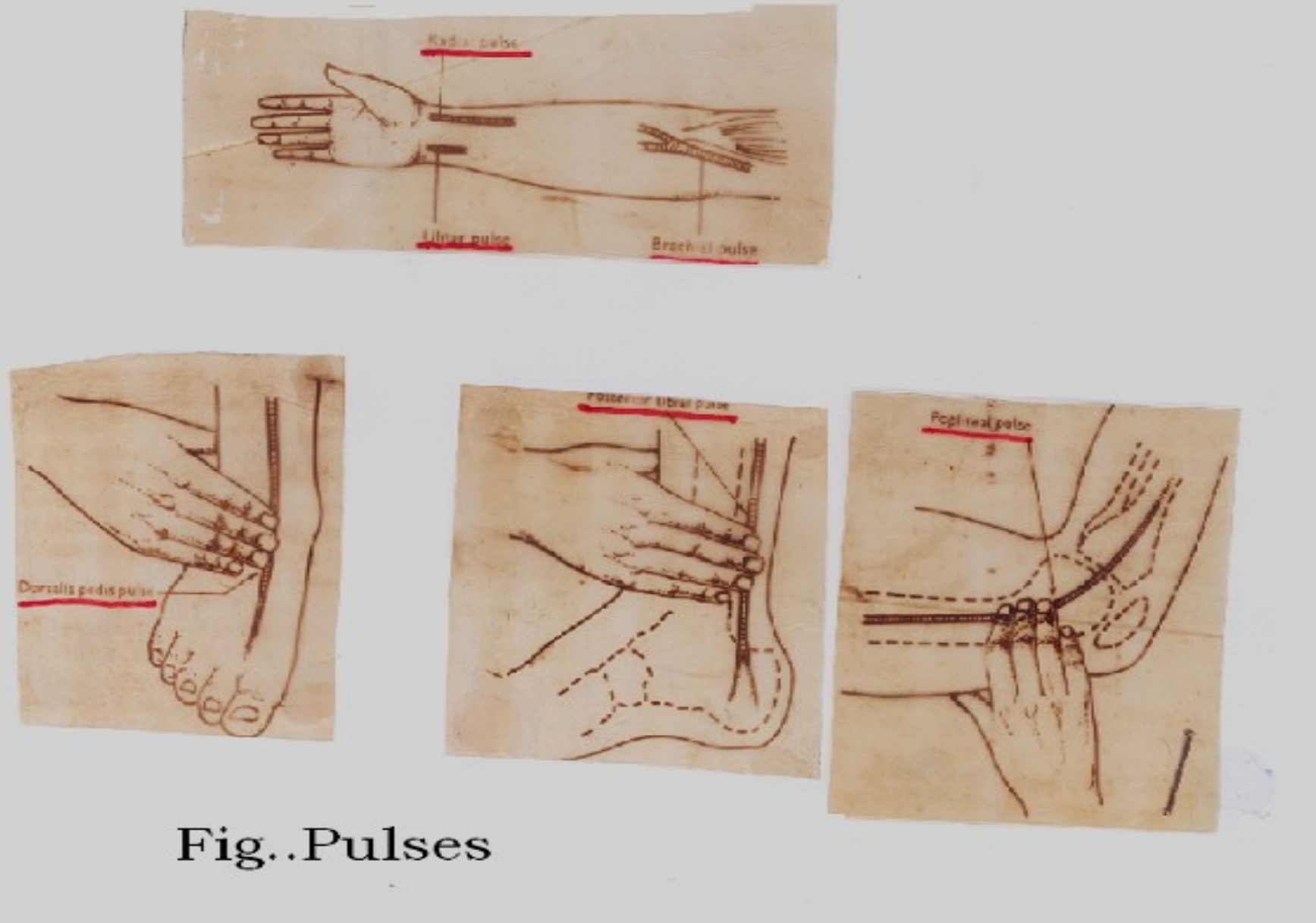


Fig..Pulses

Sites of arterial blood pressure pulsations

Measurement of ABP

1. Invasive (direct) measurement:

- A cannula is inserted into a large artery and pressure measured directly with a manometer or a strain gauge connected to a recorder.
- The record is made as shown below.

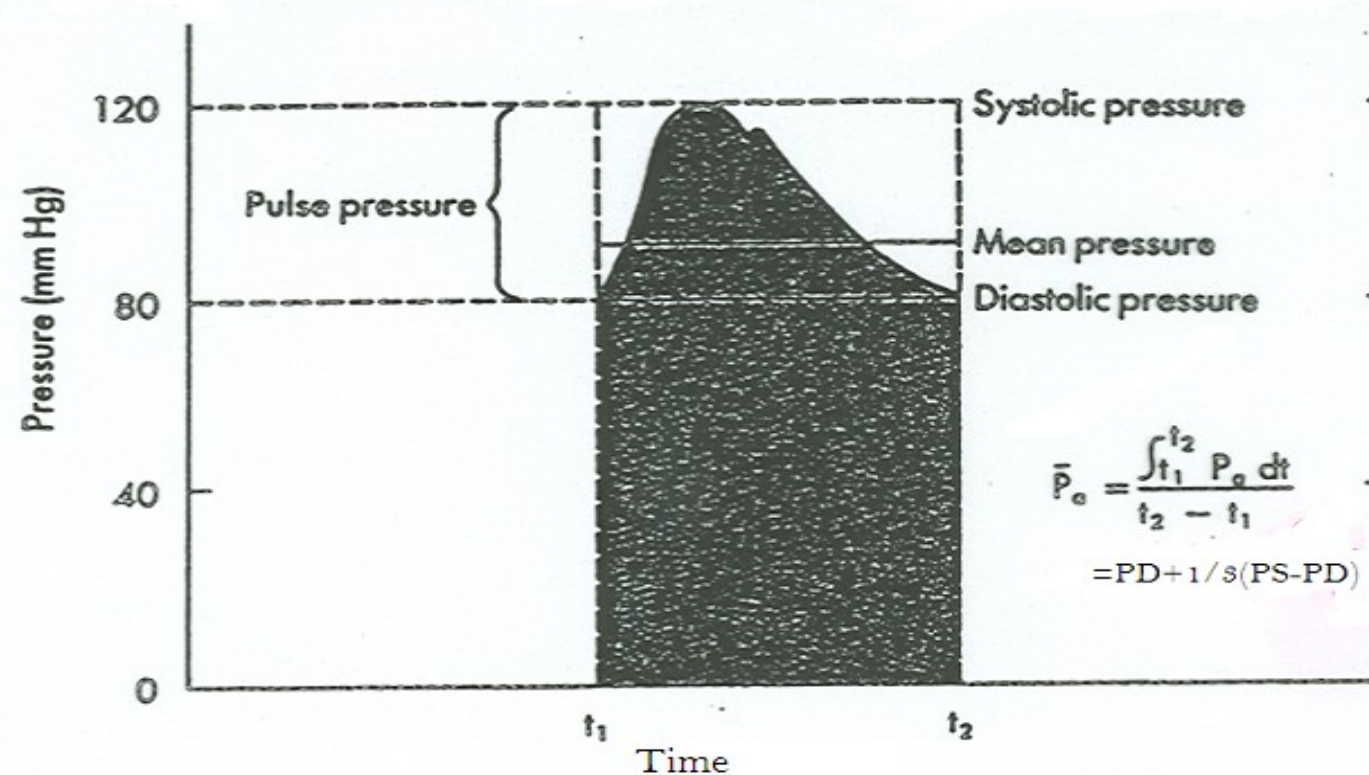


Fig.. ABP: systolic, diastolic, pulse and mean

2. Indirect routine measurement

- This method uses Riva Rocci cuff attached to a sphygmomanometer

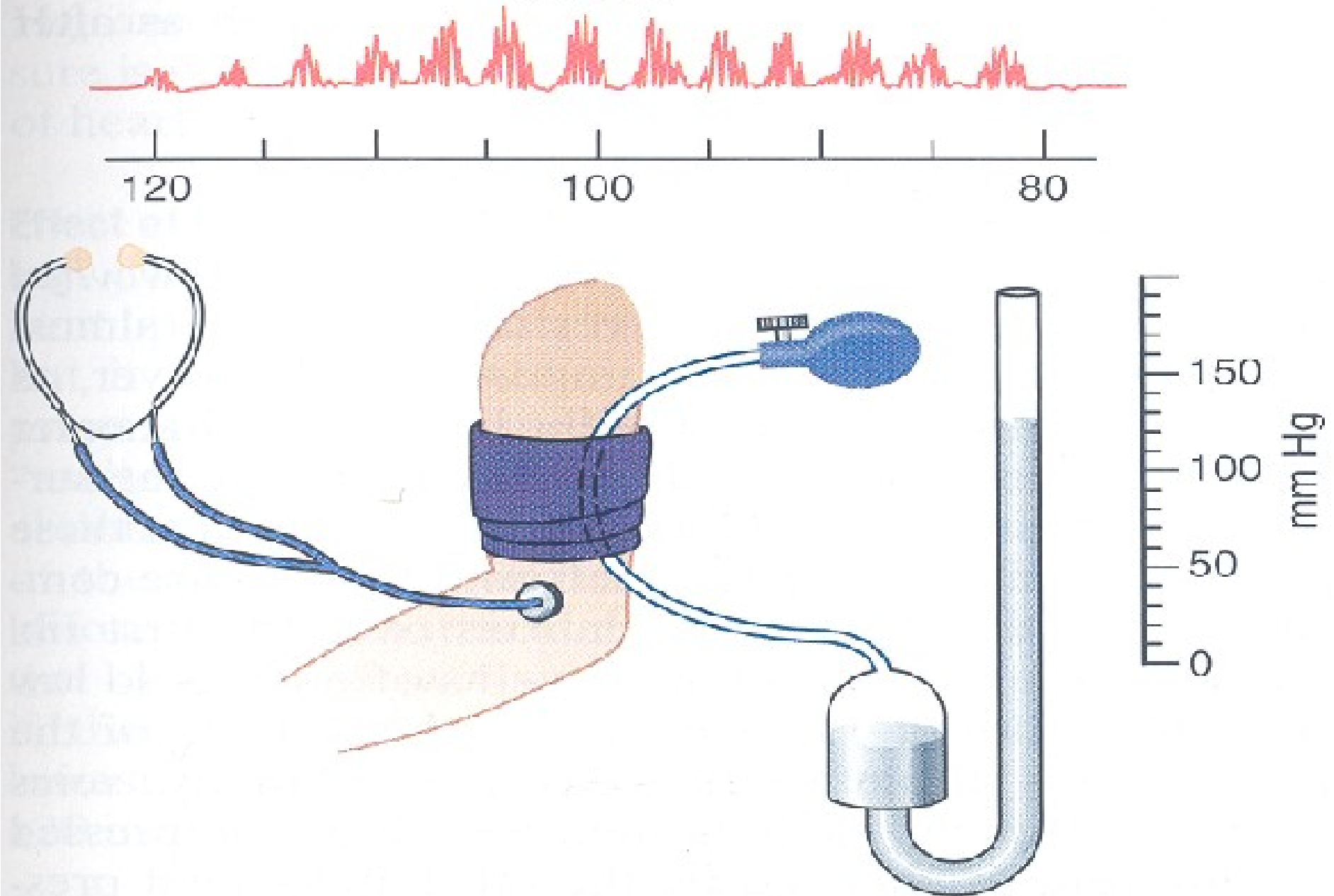
a. Auscultatory method: BP(D/S) is read using Korotkoff's sound.

- BP reading in:
 - **Children:** muffling of Korotkoff's sound for DB
 - **Adults:** complete cessation of Korotkoff's sound for DB

b. Palpation method: used to estimate systolic BP

- Approximation of mean BP

Sounds



Recommended cuff sizes for estimating BP

1. Under one yr of age =2.5cm
 2. 1-3 yrs =5-6cm
 3. 4-8 yrs- 9-10cm
 4. Average adults =12.5cm
 5. Obese adults =14cm
- **Venous pressure measurement:** use a hypodermic needle to be inserted into a vein and connect it to a sensitive aneroid manometer.

General Factors that affect arterial BP

1. **Posture:** In the upright position any vessel below heart level has increased BP,
 - e.g. BP in dorsalis pedis = 200 mmHg
 - Gravitational effect on BP = 0.77 mmHg/cm
2. **Age:** Newborn: 70/50, infants: 90/60, children: 100/70
 - SBP increases by ~ 1 mmHg/yr, secondary to atherosclerosis.
 - DBP increases by ~ 0.4 mmHg/yr secondary to increase in TPR.
3. **Sex:** ♀ sex ≤ 40 -50 yrs $\rightarrow$ low BP, ♀ ≥ 50 yrs $\rightarrow$ high BP.
 - This Bp could be higher in women than in men.
 - probably due to hormonal changes during menopause.
4. **Body weight:** both SBP and DBP are directly related to body wt.
 - ABP is higher by 10-15 mm Hg in obese persons.

- 5. **Race and socioeconomic status:** Both SBP & DBP are higher in blacks than those in whites at all ages and for both sexes.
 - This difference is attributable to genetic factors and/ or economic status.
- 5. **Exercise:** SBP and DBP increase with exercise.
 - $\uparrow$ SBP is secondary to $\uparrow$ contractility;
 - DBP may decrease initially secondary to vasodilatation in skeletal muscle vasculature but $\uparrow$ heart rate limits runoff time $\rightarrow$ $\uparrow$ DBP.
 - Exercise increases ABP by 40 – 50 mm Hg
- 5. **Emotion:** increases ABP by 10– 30 mm Hg
- 5. **Deep sleep:** decreases ABP by 20 mm Hg
- 5. **Time of the day:** $\downarrow$ BP in the morning, $\uparrow$ BP in the evening
- 5. **Thermal stress** decreases ABP

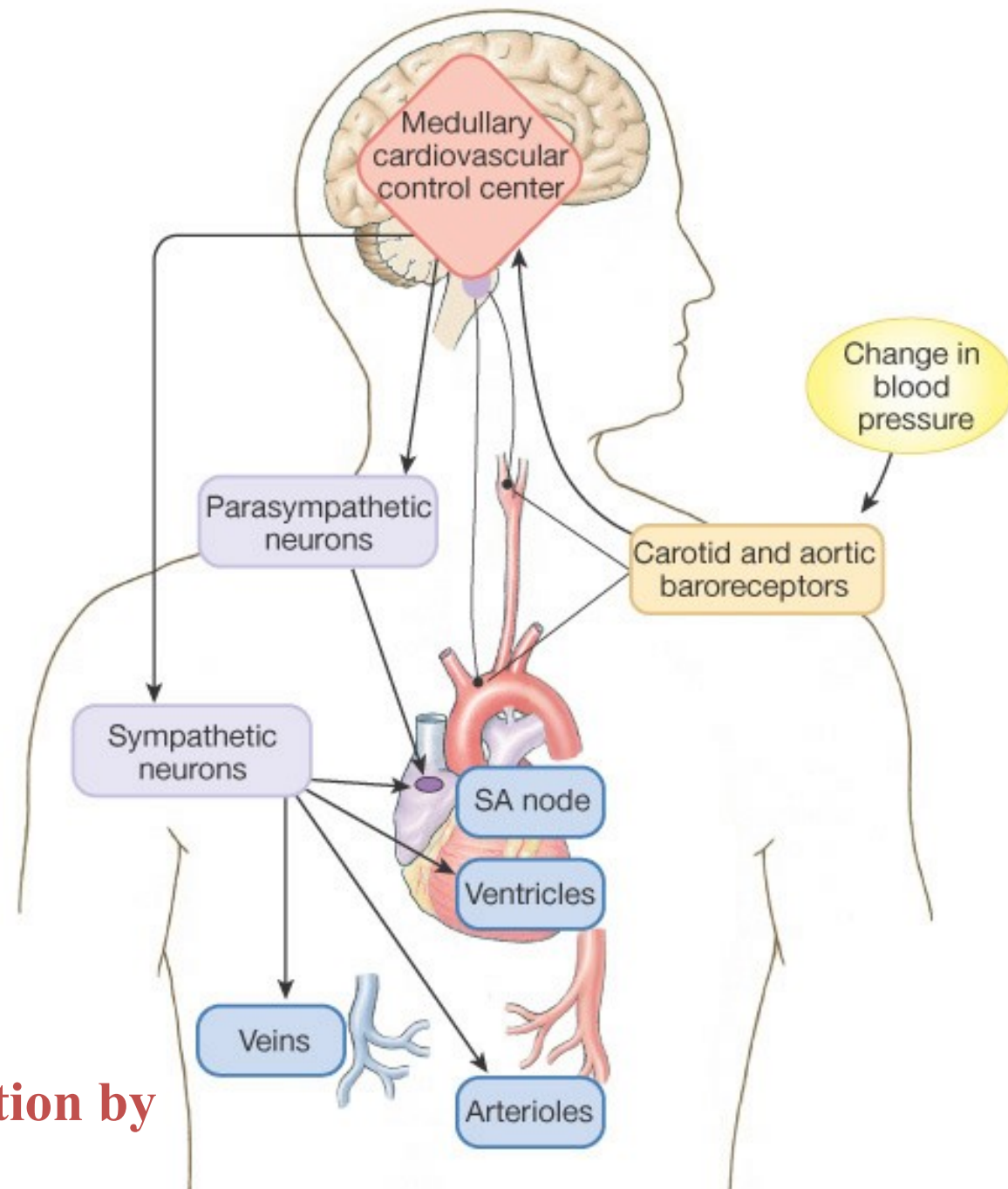
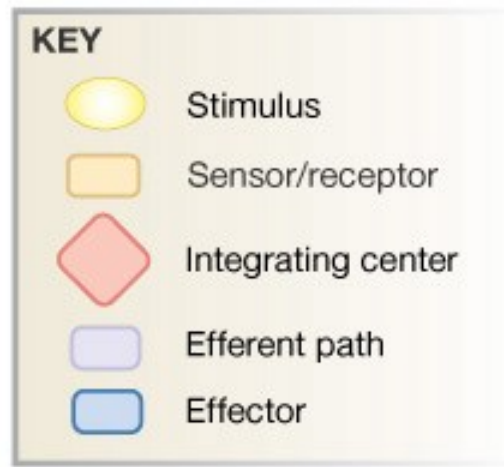
Regulation of ABP

A. Short-term controlling mechanisms

- CNS ischemic response
- Baroreceptor reflex
- Chemoreceptor reflex
- Atrial stretch reflex
- Stress relaxation and reverse stress relaxation
- Capillary fluid shift
- Hormonal mechanisms: AD, NAD, Vasopressin (ADH)

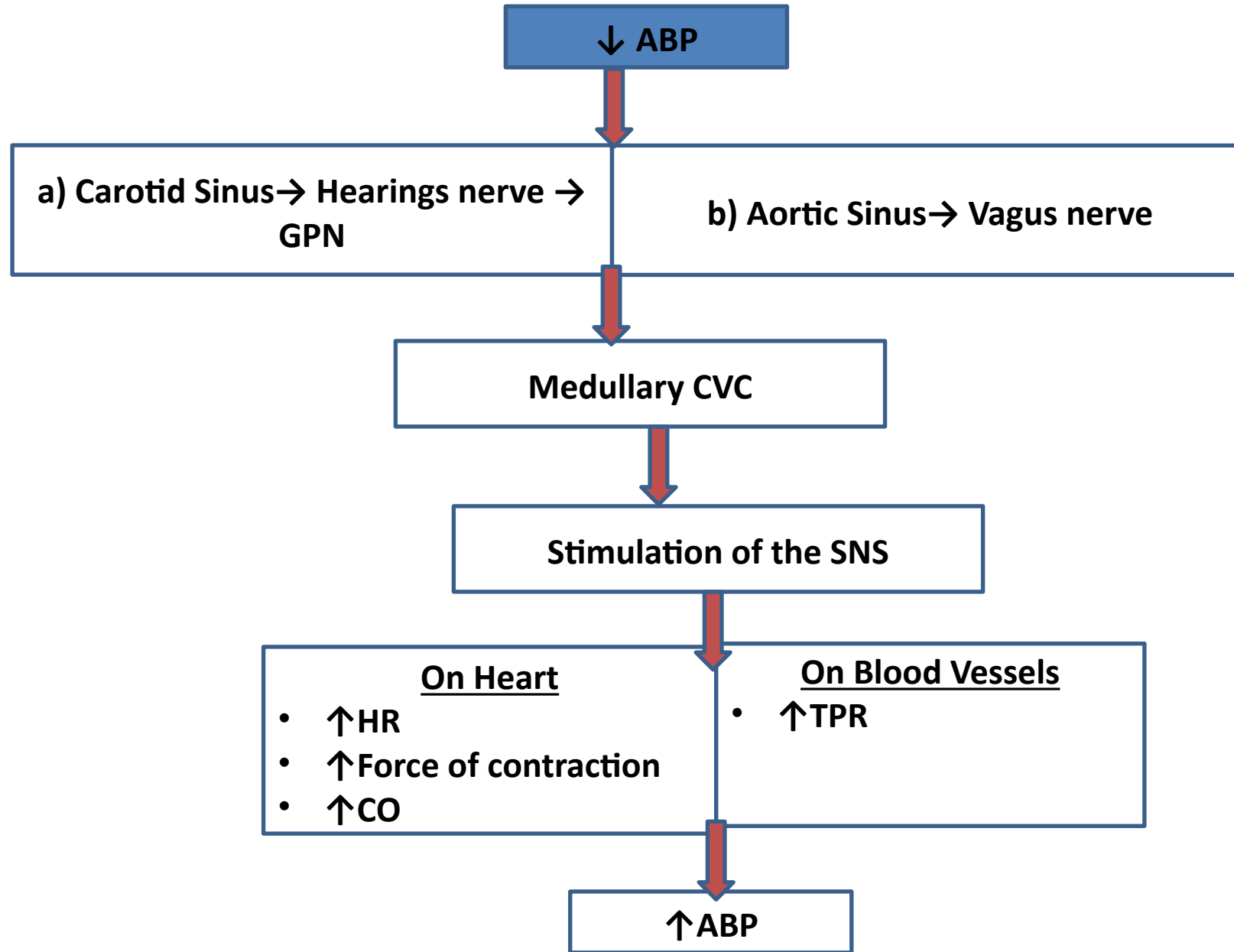
B. Long-term controlling mechanisms

- The renal mechanism
- The Renin-angiotensin-aldosterone system

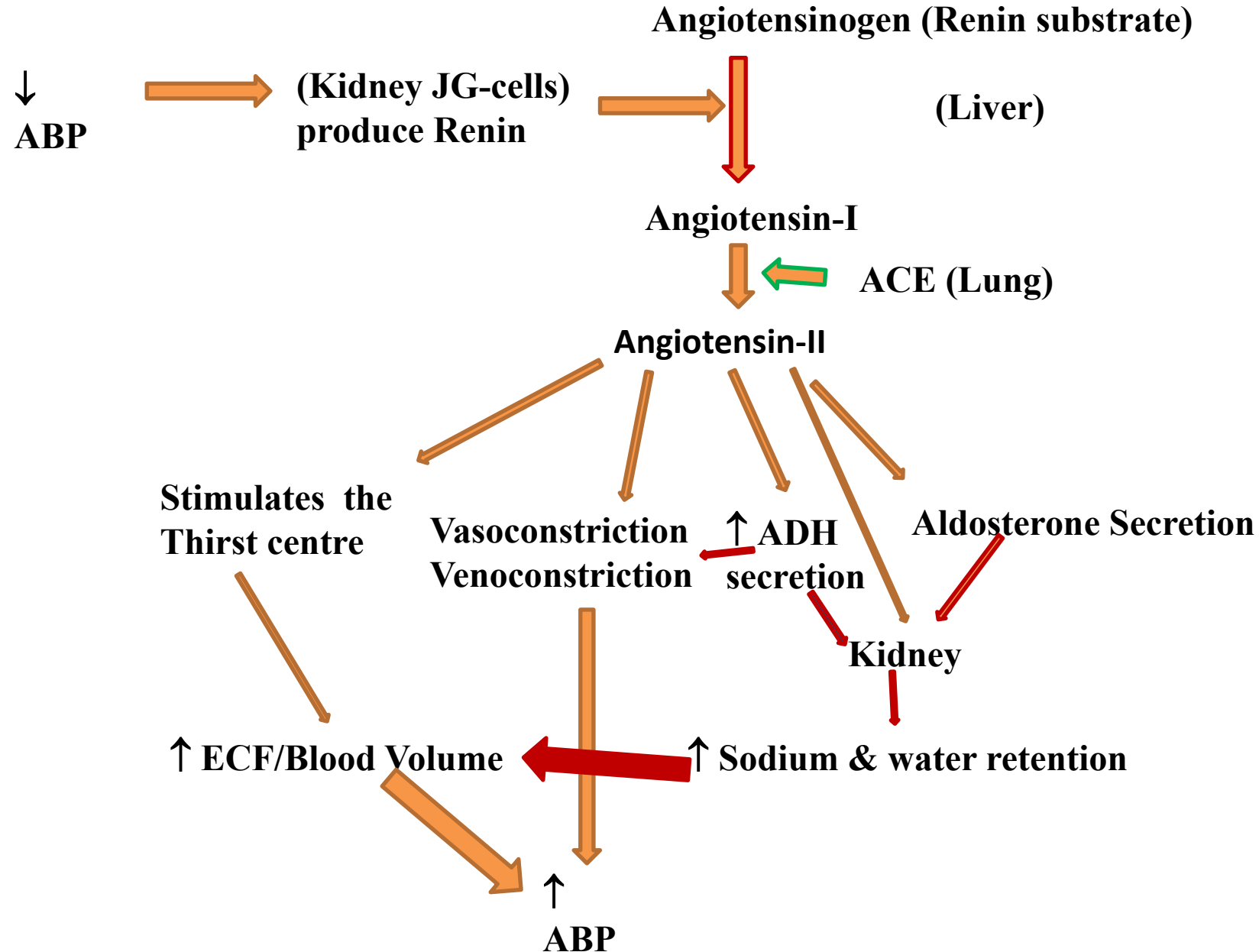


Blood Pressure Regulation by Baroreceptors

Baroreceptor Reflex



RENIN-ANGIOTENSIN-ALDOSTERONE MECHANISM



Blood Pressure Abnormalities

Hypertension:

- Hypertension is persistent elevation of ABP $> 160/95$ mmHg
- Upper limit of ABP in different age groups is different

Age (yrs).....Upper limits of normal

- 10-20..... 140/90 mmHg
- 20-50..... 150/95 mmHg
- >50 160/100 mmHg
- Any levels higher than these limits are considered as hypertension.

Types of Hypertension

A. Primary (Essential) Hypertension (90% of the cases)

- ↑↑ ABP with no clear cause, It may be benign or malignant.

B. Secondary (Symptomatic) Hypertension (10%)

Causes:

1. Renal HTN: renal ischemia, excessive renin
2. Endocrine disturbance
 - Pheochromocytoma (↑↑ AD)
 - Cushing's disease (↑↑ Cortisol)
 - Hyperaldosteronism
 - Oral contraceptives
3. Cardiovascular diseases due to changes in CO, PR, arterial compliance
 - Aortic coarctation: congenital constriction of the aorta (↑BP in upper parts of body)
 - Atherosclerosis
4. Drug induced HTN

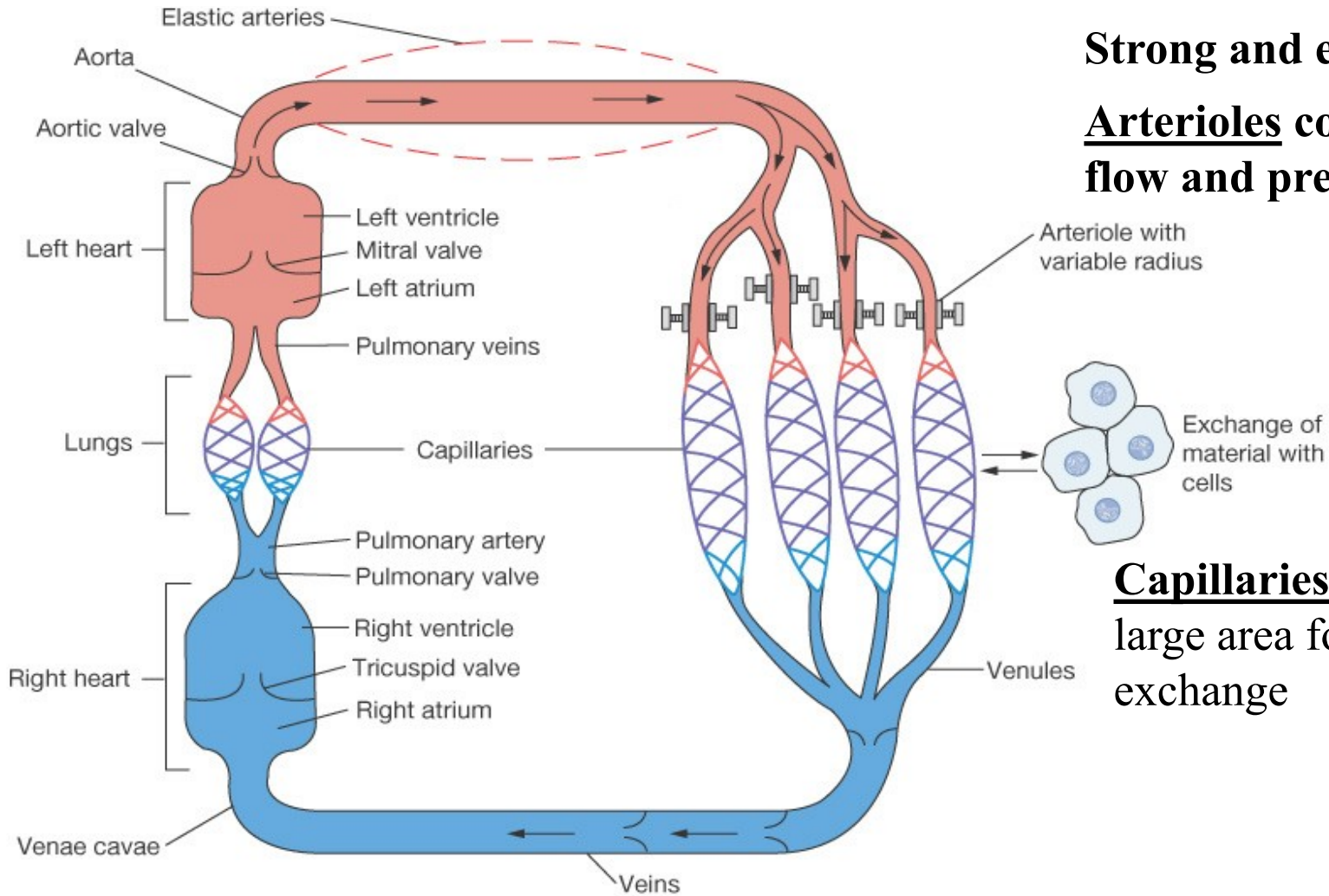
Hypotension:

- SBP= $\leq$ 90 to 110 mmHg
 - a. Essential hypotension
 - b. 2^o hypotension
 - Acute fever
 - Myocardial infarction
 - TB
 - Adrenal or pituitary insufficient.
 - c. Orthostatic or postural hypotension
- Reflex mechanisms greatly depressed.

General therapeutic measures of HTN

- Wt. reduction
- Salt-intake restriction
- Mild exercise
- Psychotherapy, relaxation, yoga, transcendental meditation, music therapy, etc.
- Avoid:- Smoking
 - Excessive alcoholic intake
- Patient education
- Drug treatment
 - Diuretics
 - Drugs acting on ANS

Blood vessel functions: overview



Strong and elastic arteries

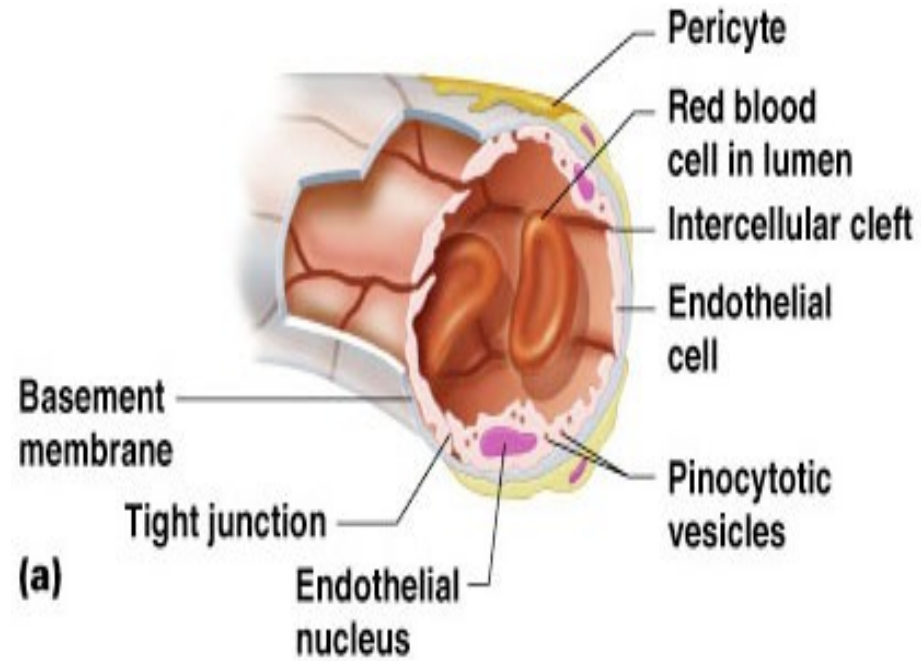
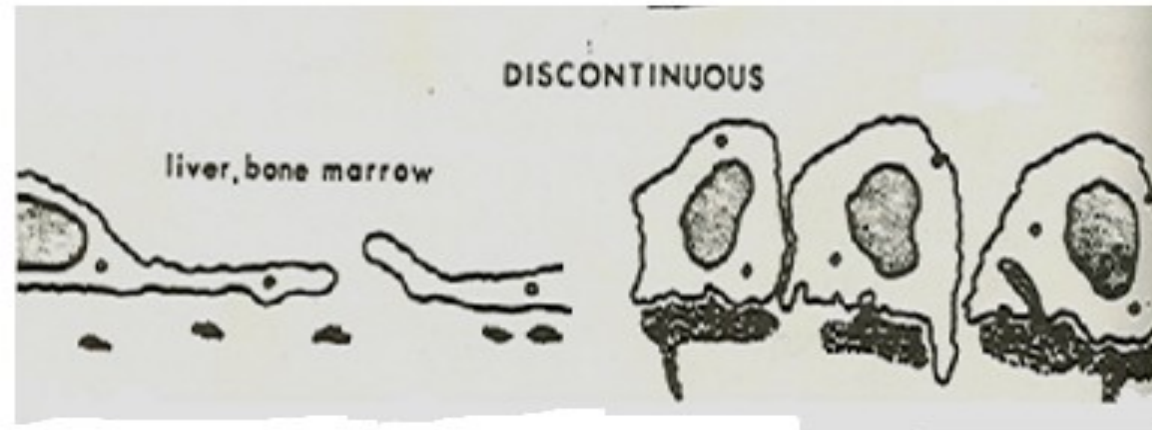
Arterioles control blood flow and pressure

Capillaries: thin and with large area for diffusional exchange

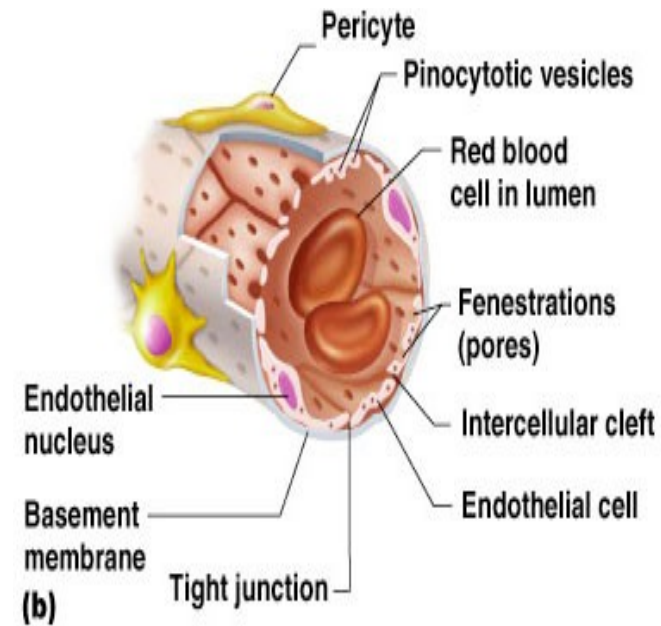
Veins: compliant, large, low resistance
veins have valves & assure blood return to the heart

CAPILLARIES

- Pressure inside is 15 to 35 mmHg
- 5% of the blood is in capillaries
- Necessary for exchange of gases, nutrients, and wastes
- Flow is slow and continuous
- Blood flow controlled by pre-capillary sphincter muscles *via* metabolic regulation
- No smooth muscle, no innervations
- Largest total cross sectional area
- Based on the continuity of their filtration barriers, capillaries have been labelled as **continuous, fenestrated and discontinuous**



Continuous Capillaries



Fenestrated capillaries

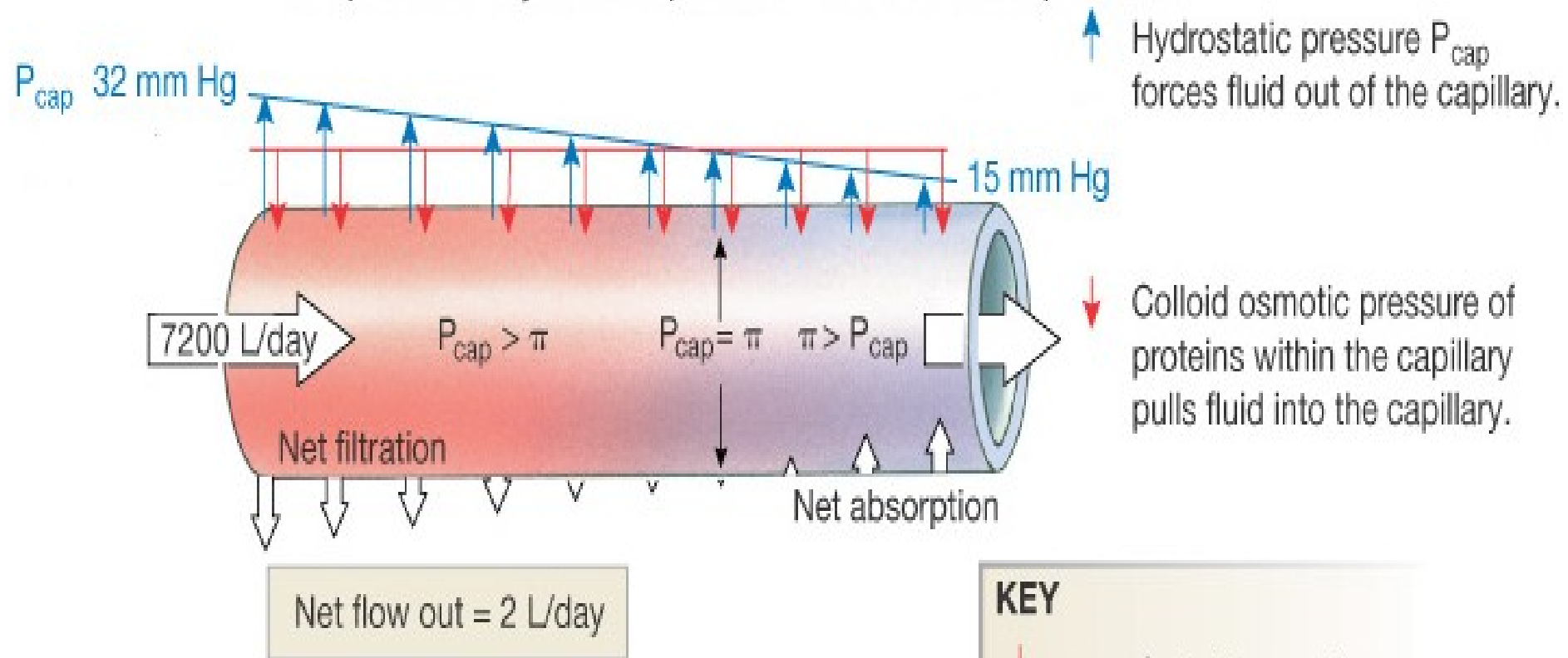
Capillaries Exchange

- There are two forces that determine the direction of flow of fluid through the capillaries
 - **The capillary colloid osmotic pressure (25 mm Hg)**
 - **The capillary hydrostatic pressure (32mmHg)**
- Fluid that is leak out of the capillaries is taken up by the lymphatic vessels to prevent edema
- **Net outflow into ECF**
 - Net filtration – net absorption = net outflow
 - About 2 L/day collected by lymph vessels

Capillary Exchange:

(a) Filtration in systemic capillaries

Net pressure = hydrostatic pressure - colloid osmotic pressure

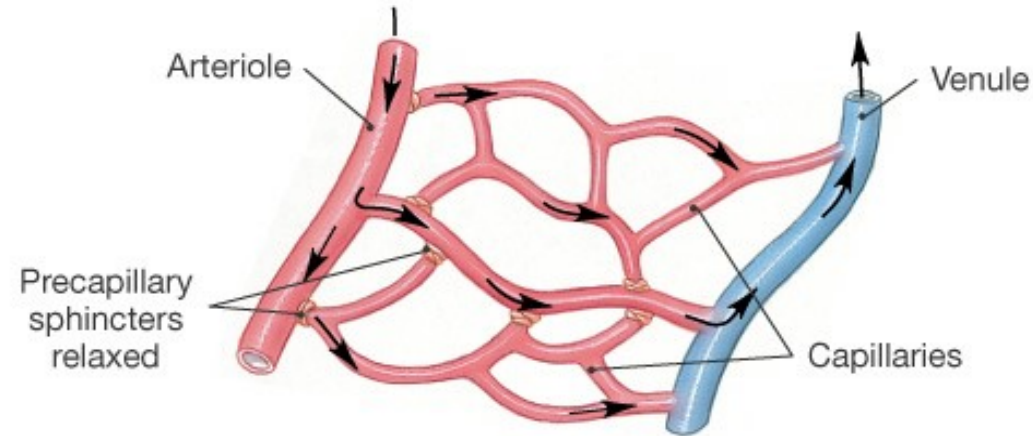


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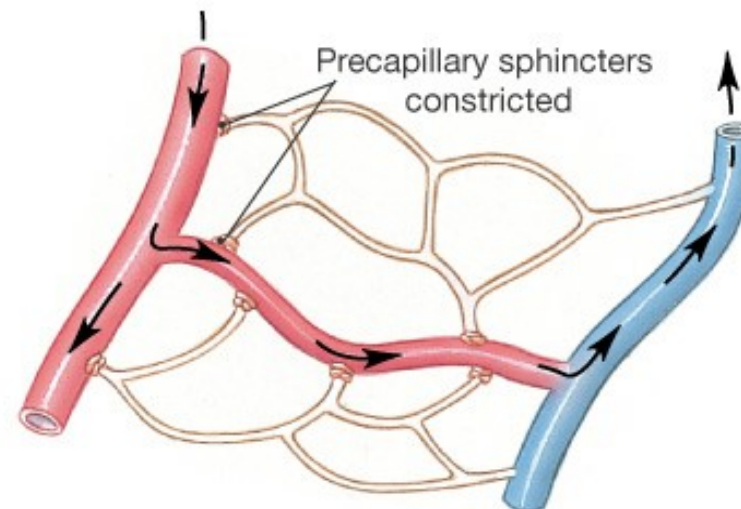
- ↓ π = Colloid osmotic pressure
- ↑ P_{cap} = Capillary hydrostatic pressure

Figure: Capillary fluid exchange

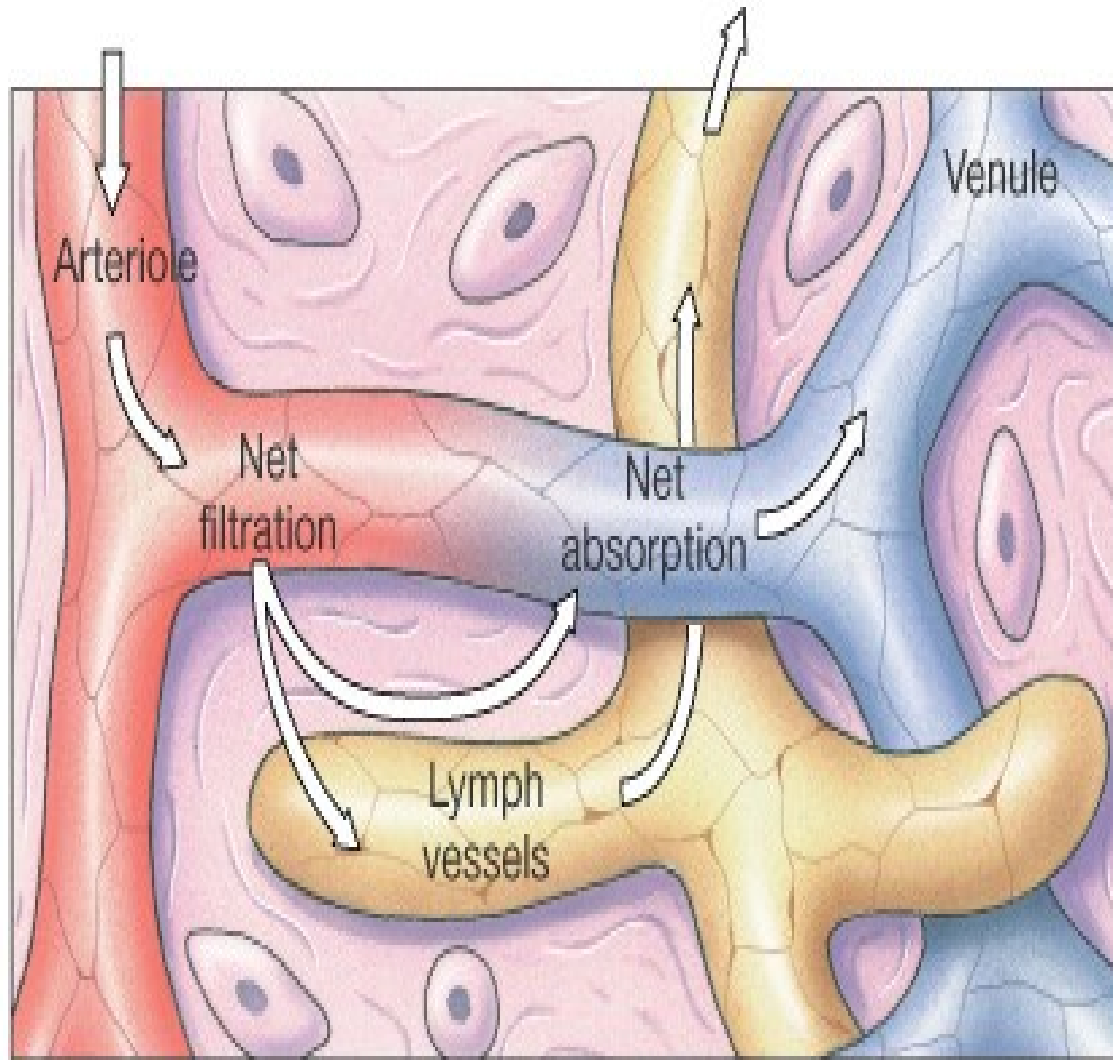
(a) When precapillary sphincters are relaxed, blood flows through all capillaries in the bed.



(b) If precapillary sphincters constrict, blood flow bypasses capillaries completely and flows through metarterioles.



(b) Relationship between capillaries and lymph vessels



The excess water and solutes that filter out of the capillary are picked up by the lymph vessels and returned to the circulation.

Figure: Fluid exchange at the capillary

Capillary Blood Flow-Velocity

- Lowest velocity
- Largest total cross sectional area
- Hydrostatic pressure drops slightly

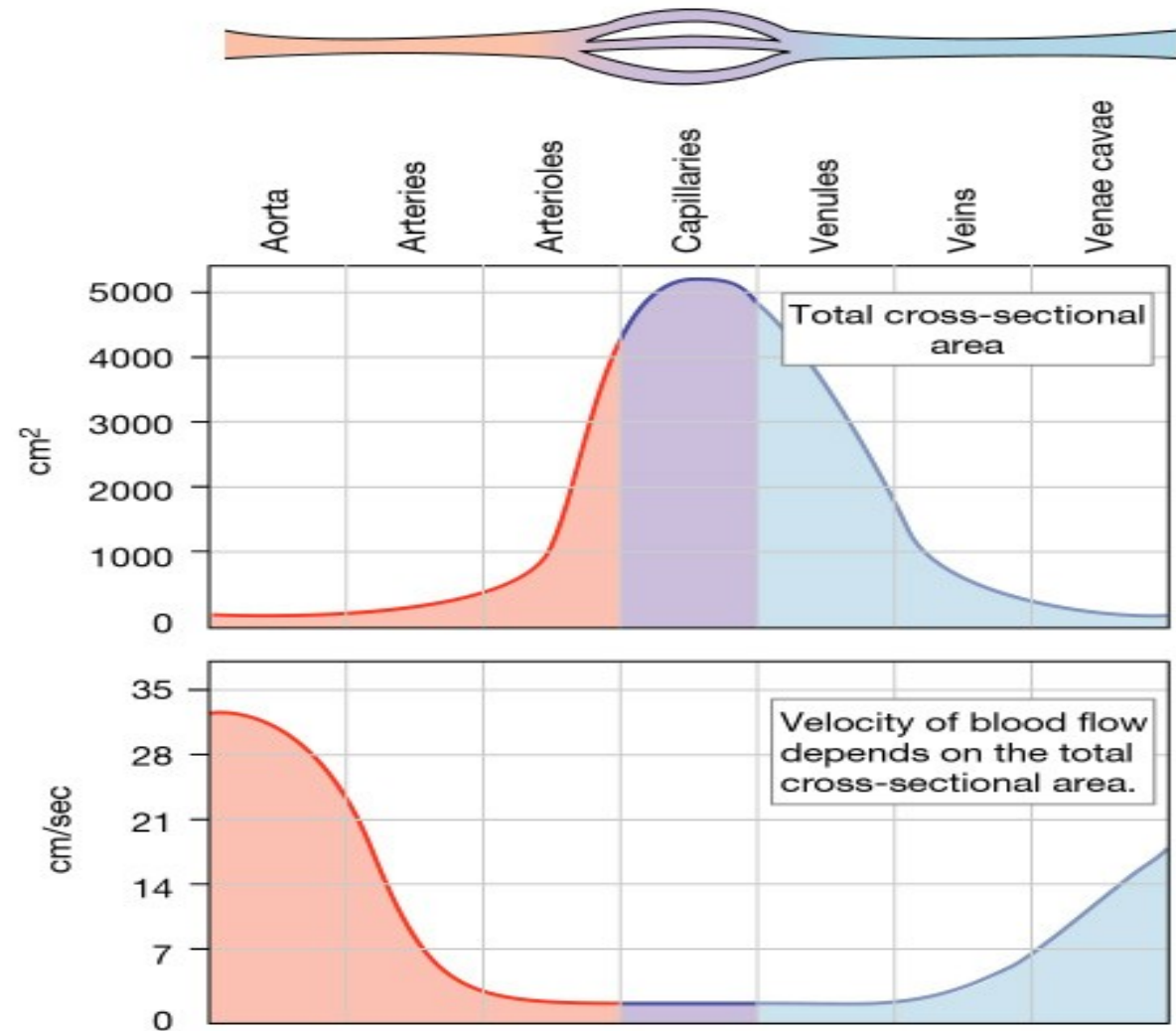


Figure. The velocity of flow depends on the total cross-sectional area

Blood flow

- Blood flow (BF) is the amount of blood that moves to a particular organ in a given time
- Total blood flow is equal to CO; 5-6 L/min
- BF is determined by 2 factors
 1. Pressure difference b/n 2 ends of the vessel
i.e., the force that pushes blood through vessels
 2. Resistance of flow, hindrance to flow through vessels

Where **Q = Blood flow**

$$Q = \Delta P / R$$

ΔP = Change in pressure

R = Resistance

- **Ohms Law:** states that BF is directly proportional to the ΔP but inversely proportional to resistance (R).

Vascular resistance

- **Poiseuille's Law**: Vascular resistance to BF is directly proportional to the length of the vessel and viscosity of blood, but inversely proportional to the 4th power of radius of the vessel.

Where, R = Resistance

$$Q = \frac{\Delta P \pi r^4}{8 \eta l}$$

$$R = 8 \eta l / \pi r^4$$

l = Length

η = Viscosity

π = Circle constant

Q=Blood flow

(3.14)

r = Radius

Factors affecting viscosity of blood

- Velocity of BF: $\uparrow$ Velocity = $\downarrow$ Resistance
- Hematocrit: polycythemia = $\uparrow$ Viscosity = $\uparrow$ PR
Anemia = $\downarrow$ Viscosity = $\downarrow$ PR
- Diameter of blood vessels
- Plasma protein concentration

PR=Peripheral resistance

Resistance to blood flow

- It is the hindrance to blood flow
- Has no direct means of measurement, but can be calculated from values of BF and ΔP in vessels
- If the ΔP is 1 mm Hg and BF 1 ml/sec, then vascular resistance is said to be 1 peripheral resistance unit (PRU)

Total peripheral resistance and total pulmonary resistance

- The rate of BF through the circulatory system at rest is 100 ml/sec or 5 L/min.
- The ΔP between systemic arteries and veins is 100 mm Hg
- Resistance of the entire systemic circulation, called TPR is $100/100 = 1$ PRU.
- TPR ranges between 0.2 and 4 PRU during vascular dilation and constriction

TOTAL PERIPHERAL RESISTANCE

SYSTEMIC CIRCULATION:

RAP=Right atrial
pressure

$$\text{TPR} = \frac{\text{Aortic Pressure} - \text{RAP}}{\text{FLOW}}$$

$$\text{TPR} = \frac{100 - 0 \text{ mmHg}}{83.3 \text{ ml/sec (5 L/min)}} = 1.2 \text{ PRUs}$$

PULMONARY CIRCULATION:

$$\text{Pul. R.} = \frac{\text{Pul. Art. P.} - \text{LAP}}{\text{FLOW}}$$

$$\text{Pul. R.} = \frac{15 - 5 \text{ mmHg}}{83.3 \text{ ml/sec}} = 0.12 \text{ PRUs}$$

Oedema

- Water and solutes accumulate in IF compartment→ oedema,
- It is pitting if firm pressure on swollen area leaves an impression- occurring in dependent parts- extracellular oedema.
- If the oedema is non-pitting, it is cellular.
- Oedema in general occurs when the increase in tissue fluid volume is 30% or more of the basal volume or when filtration is >reabsorption + lymphatic drainage

Oedema may be evident from:

- Puffy face (eyes) in the morning
- Puffy feet
- Sacral oedema when patient is lying in bed.

Generalized Oedema (Anasarca):

- Expanded IF volume
- ↑Total body sodium

Possible causes of pitting oedema

1. Increased **hydrostatic pressure** in capillaries secondary to heart failure
2. Decreased **osmotic pressure** due to low plasma proteins secondary to conditions like nephrotic syndrome
3. Increased **capillary permeability**
4. **Lymphatic obstruction** (non-pitting edema). . Localized

Cellular or non-pitting oedema:

- may be caused by:
 1. Depression of cellular metabolism $\rightarrow$ $\downarrow$ Na^+ - K^+ pump $\rightarrow$ $\downarrow$ Na^+ influx into cell $\rightarrow$ Water following Na^+ $\rightarrow$ Cellular oedema
 2. Inflammation $\rightarrow$ $\uparrow$ membrane permeability to sodium and other ions that diffuse into cell with water $\rightarrow$ oedema.
Accumulation of water inside cells $\rightarrow$ **Water intoxication.**

Water intoxication: occurs

- when excessive volumes of water are absorbed too quickly leading to nausea, vomiting and shock--conditions caused by undue drop in plasma osmolality before adequate inhibition of ADH secretion occurs.

Other related problems

- **Depletion of ECF volume:** low plasma vol. → low BP, rapid pulse; low IF vol. → poor skin turgor, dry tongue, sunken eyes.
- **Depletion of ICF Volume:** Brain goes mad with thirst due to intense stimulation of hypothalamic osmoreceptors

Characteristics of the Venous System

Veins:

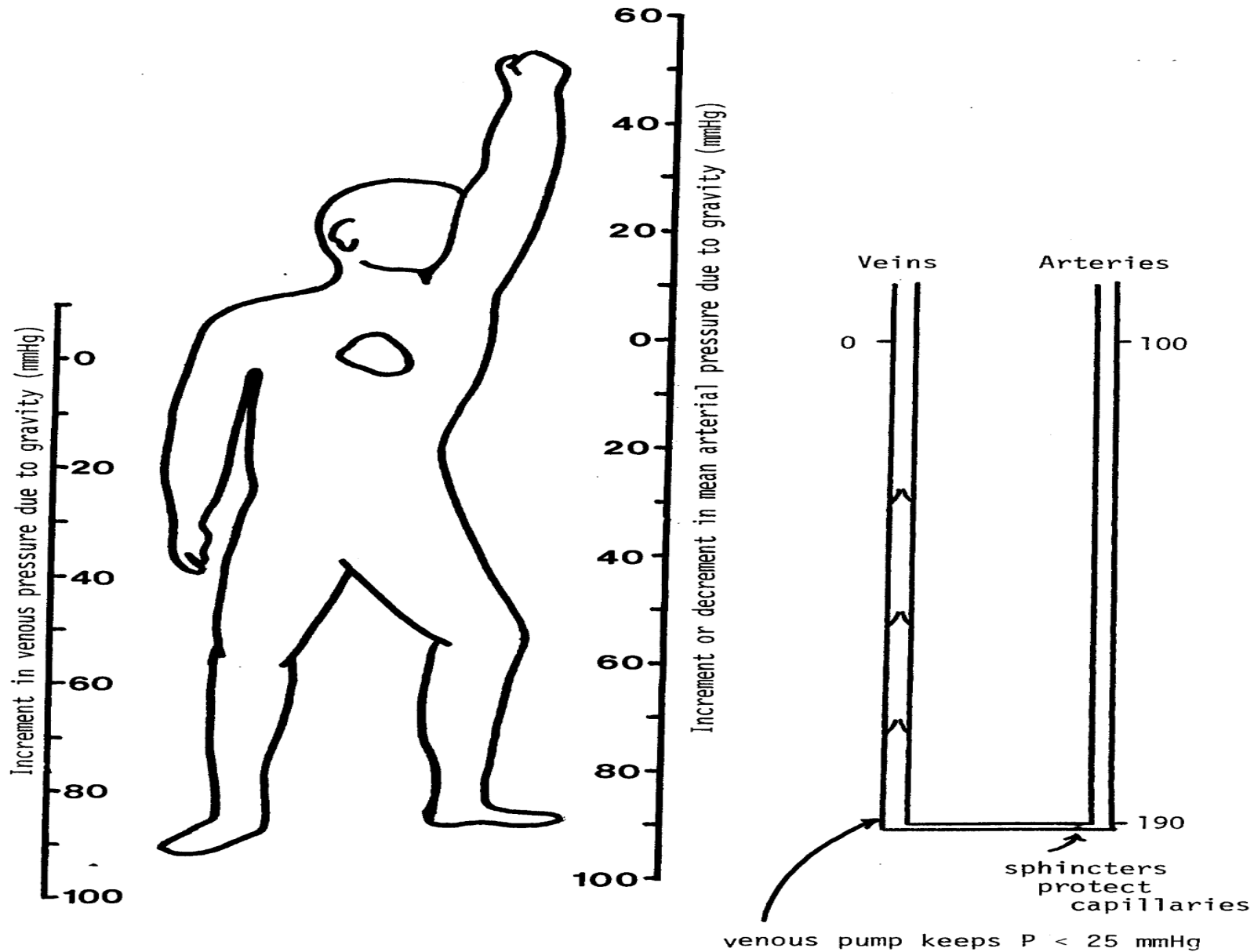
- Constitute the low pressure system and extend b/n smallest venules ($>10\text{mm Hg}$) and largest terminal veins (venae cavae = 0 mmHg).
- Are **wide-bored** and **thin-walled**
- Are **volume storer** (high capacitance) of the circulation
- Are low pressure vessels
- Have **great distensibility**
- Have 5-6 times greater distensibility and 2-3 times greater capacity than the arteries.
- Have **one-way valve** to break up columns of blood and to prevent excessive venous distension.
- Faulty valves may lead to development of varicose veins.
- Use **sympathetic supply to facilitate circulatory homeostasis**.

...cont'd

Venous Pressure

- 12-18 mmHg in venules
- 5.5 mmHg in large veins
- Mean pressure in ante-cubital vein is 7.1mmHg
- At entrance of large veins into RA (CVP), the average pressure is 4.6 mmHg
- Like ABP, peripheral venous BP is affected by gravity; it is increased by 0.77mmHg for each cm below RA and decreased by a similar amount for each cm above RA (Fig..).

Effect of gravity on arterial and venous pressures



Measurement of venous pressure

- Pressures, either central (CVP, low) or peripheral (relatively high), can be estimated by inserting a catheter into the thoracic great veins.
- The pressure in peripheral veins is greater than central venous pressure by an amount proportional to the distance away from the right side of the heart.
- CVP can be estimated simply by noting the height to which the external jugular veins are distended when the subject lies with the head slightly above the heart.
- The vertical distance b/n RA and the place the vein collapses (pressure=0) is the venous pressure in mm of blood.
- CVP is decreased during –ve pressure breathing and shock. (inspiration)
- It is increased by +ve pressure breathing(expiratorion), straining, expansion of blood volume and heart failure.

CENTRAL VENOUS PRESSURE (CVP)

- Normal venous pulsations are weak and usually do not appear at levels higher than the sternal angle.
- To make them visible in the neck veins, the subject has to recline on his back with the head supported on pillows (Fig..below). In this position, pulsations in internal jugular vein can be seen and examined.
- Maximum jugular pulsations are observed when the trunk is inclined by less than 30°.
- The vertical distance b/n the right atrium (sternal angle) and the place the vein collapses (the place where the pressure in it is zero) is the venous pressure in mm of blood.

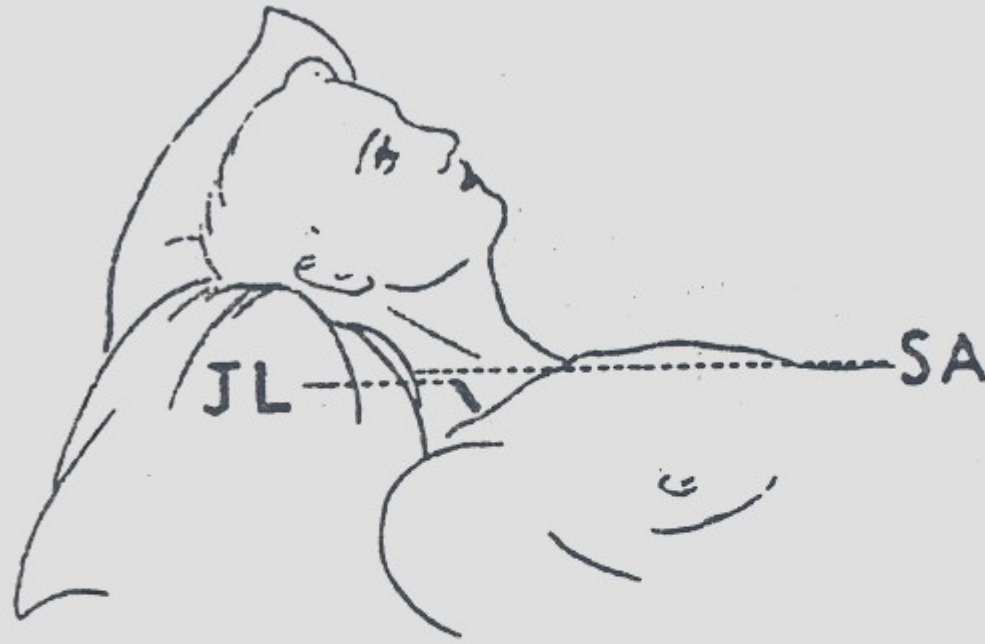


Fig.. The position to examine normal jugular venous pulsations. SA = sternal angle level; JL = upper level where jugular pulsations appear (Jugular level).